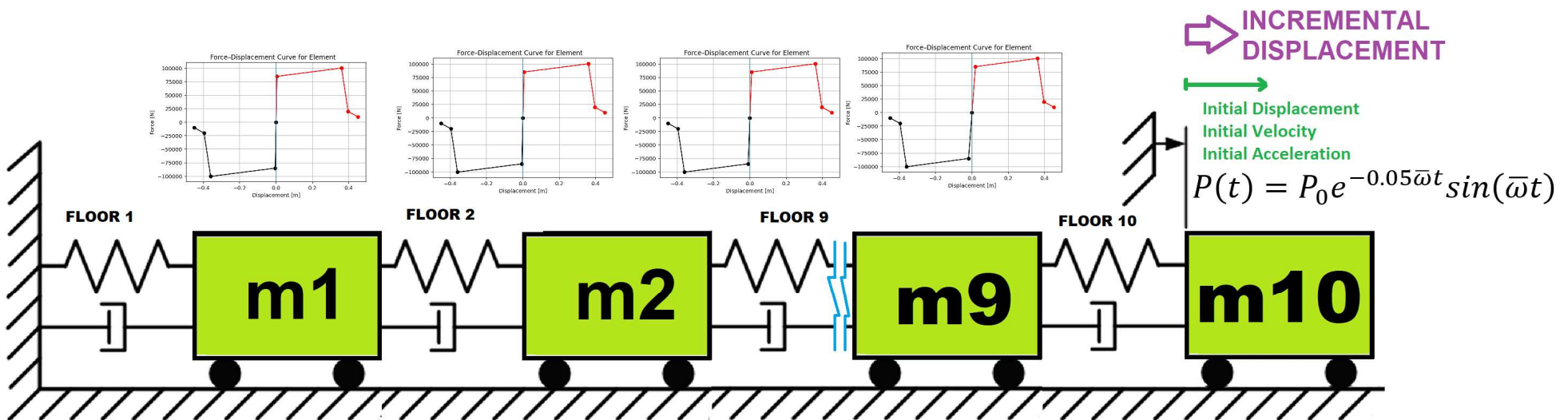


>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

**COMPREHENSIVE NONLINEAR
SEISMIC ASSESSMENT OF A MULTI-
DEGREE-FREEDOM (MDOF)
STRUCTURE: AN OPENSEES
FRAMEWORK FOR STATIC
PUSHOVER, CYCLIC DEGRADATION,
STATIC TIME-HISTORY AND
DYNAMIC TIME-HISTORY ANALYSIS**

THIS PROGRAM IS WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)



$$\text{Structural Ductility Damage Index} = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

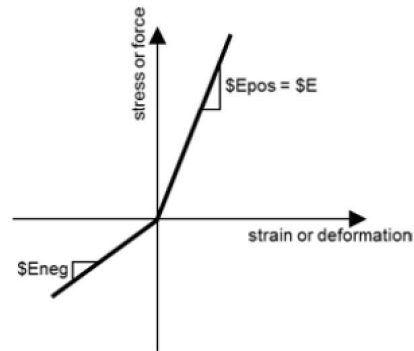
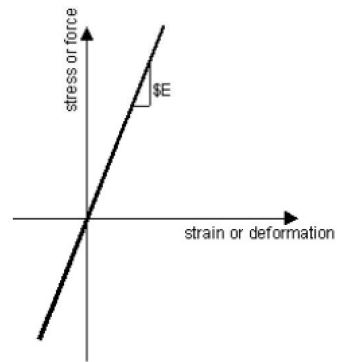
Δ_d = Lateral Displacement from Dynamic Analysis

Δ_y = Lateral Yield Displacement from Pushover Analysis

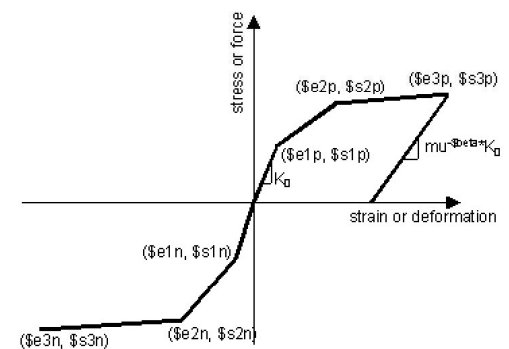
Δ_u = Lateral Ultimate Displacement from Pushover Analysis

$$P(t) = P_0 e^{-0.05\bar{\omega}t} \sin(\bar{\omega}t)$$

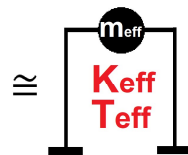
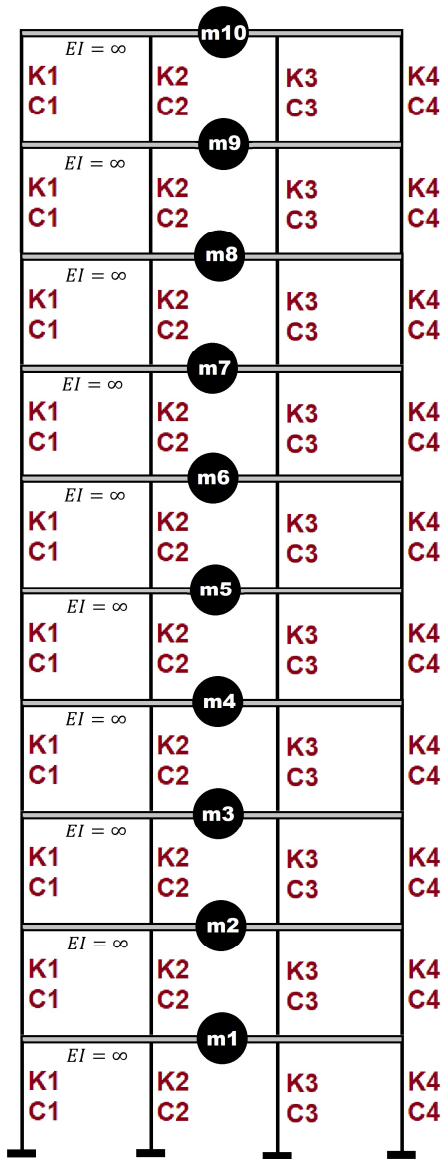
$$m\ddot{u} + c\dot{u} + ku = P(t)$$



ELASTIC MATERIAL



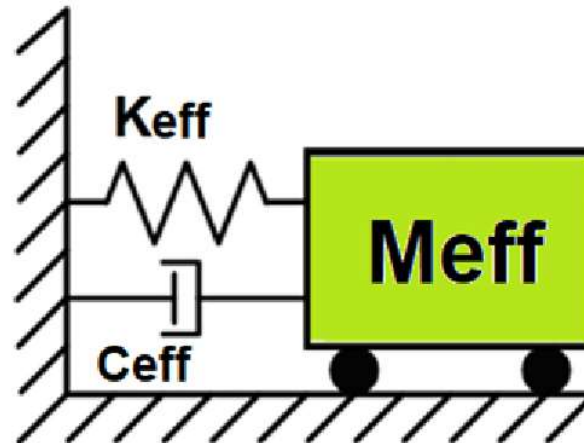
INELASTIC MATERIAL

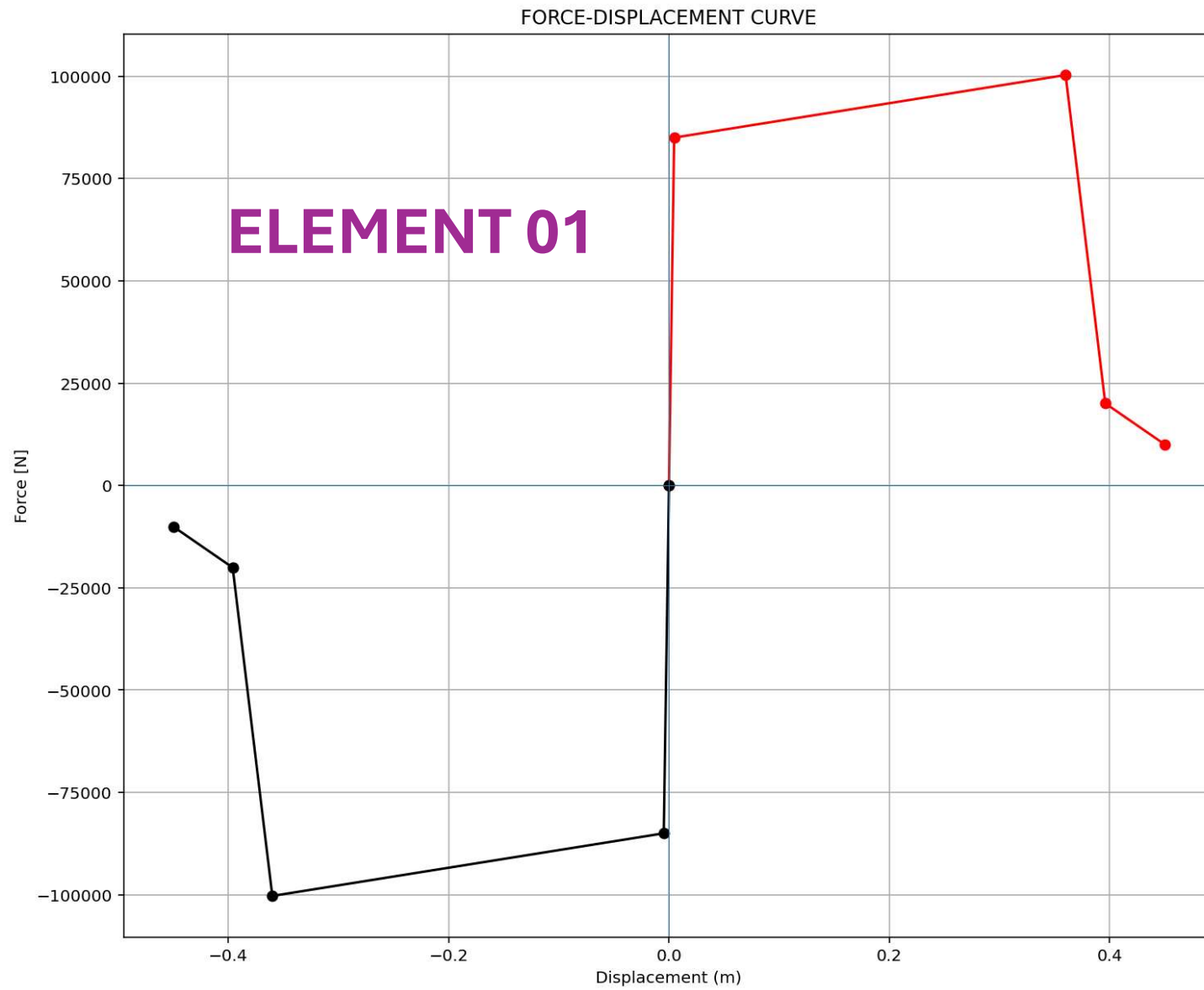


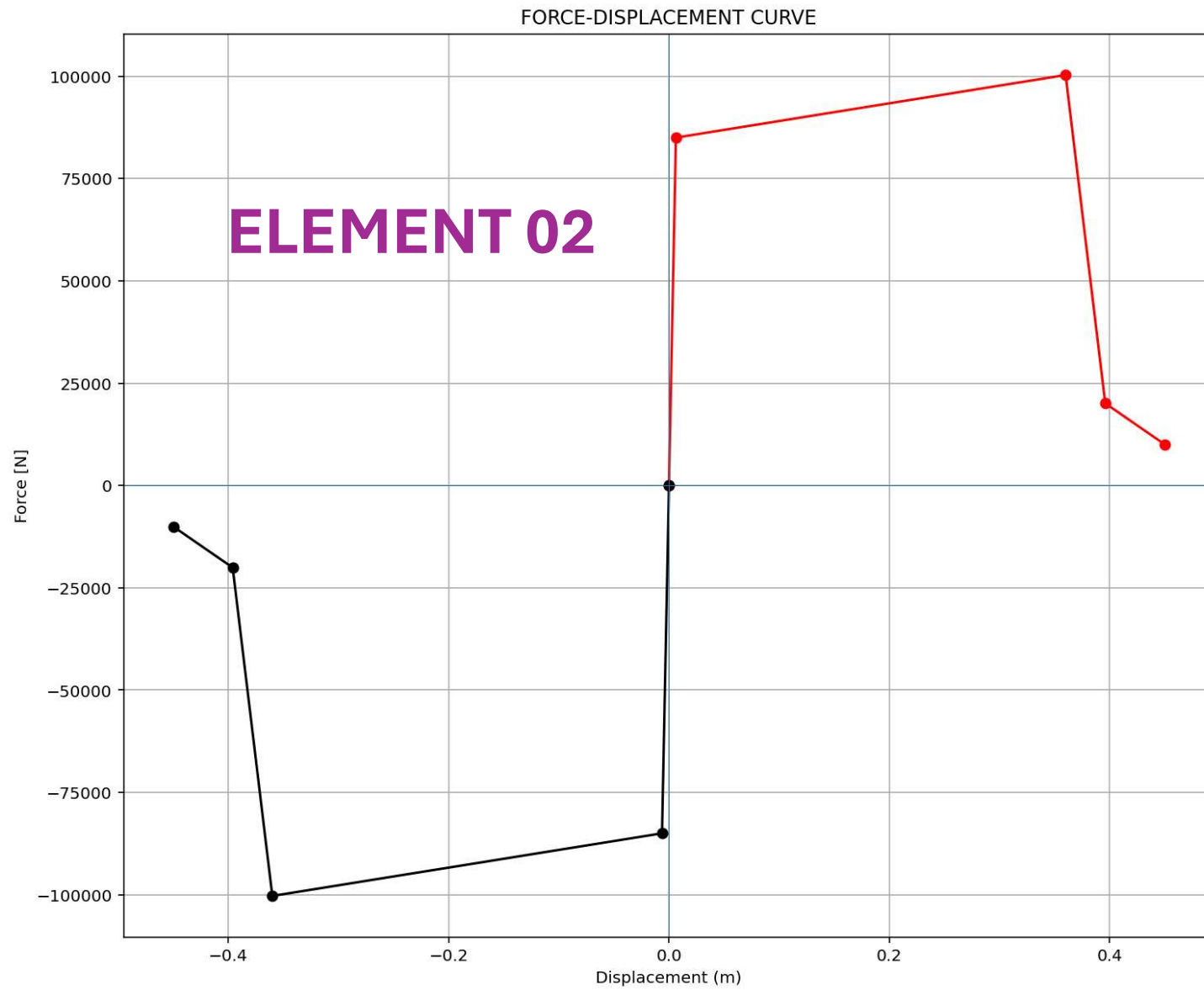
Displacement-based seismic analysis transformation,
converting a multi-degree-of-freedom (MDOF) system
into an equivalent single-degree-of-freedom (SDOF)
system for seismic assessment

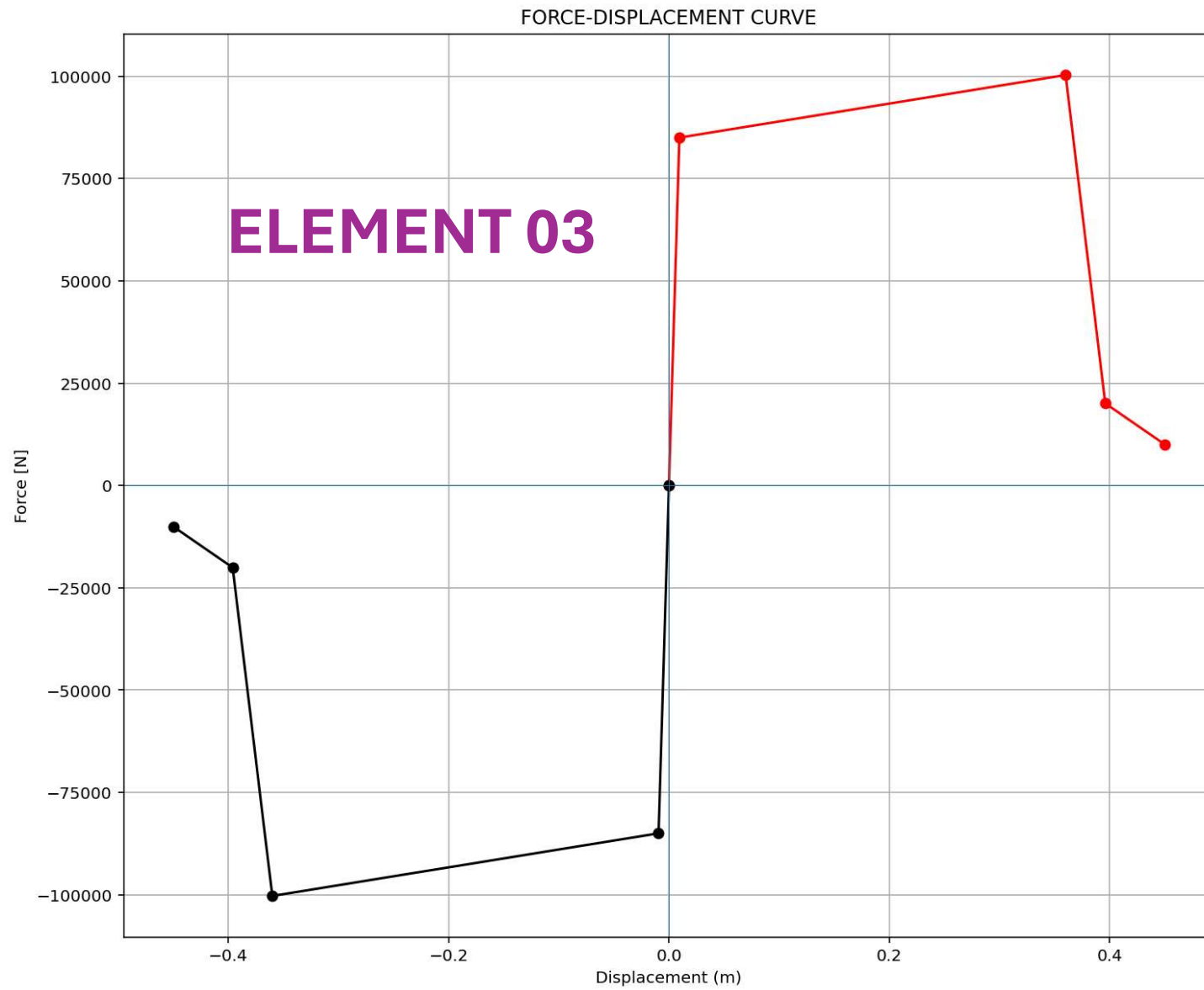
$$\Delta_d = \frac{\sum_{i=1}^n m_i \Delta_i^2}{\sum_{i=1}^n m_i \Delta_i} \quad m_{eff} = \frac{\sum_{i=1}^n m_i \Delta_i}{\Delta_d}$$

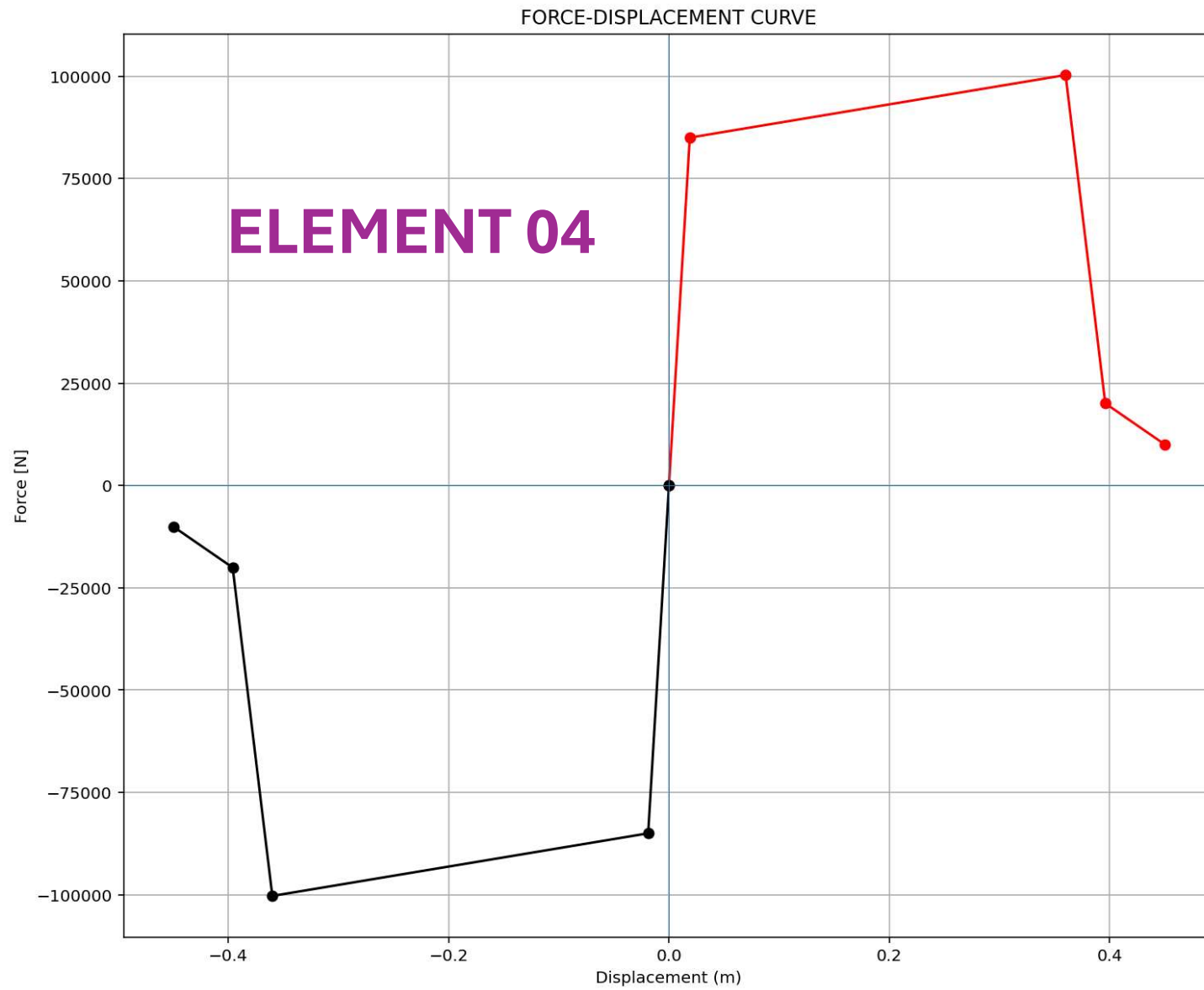
$$k_{eff} = \frac{\sum_{i=1}^n k_i \Delta_i}{\Delta_d} \quad T_{eff} = \frac{2\pi}{\sqrt{\frac{k_{eff}}{m_{eff}}}}$$











STATIC ANALYSIS

Spyder (Python 3.12)

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C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...MNS_8_ANA\p(t)_MDOF_10_FLOORS_04_COLUMNS_8_ANA.py

P(t)_MDOF_10_FLOORS_04_COLUMNS_8_ANA.py X

```
1006
1007
1008 #%%-----
1009 # STATIC ANALYSIS
1010 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
1011 ANAL_TYPE = 'STATIC'
1012
1013 DATA = MDOF(MAT_TYPE, TOTAL_MASS, ANAL_TYPE)
1014 (reaction, disp, DI) = DATA
1015
1016 #%%-----
1017 # PUSHOVER ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
1018 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
1019 ANAL_TYPE = 'PUSHOVER'
1020
1021 DATA = MDOF(MAT_TYPE, TOTAL_MASS, ANAL_TYPE)
1022 (reaction_PUSH, disp_PUSH, DI_PUSH,
1023  ele_force_PUSH, node_displacements_PUSH,
1024  PERIOD_MIN_PUSH, PERIOD_MAX_PUSH) = DATA
1025
1026
1027 XDATA = disp_PUSH
1028 YDATA = reaction_PUSH
1029 XLABEL = 'Displacement [m]'
1030 YLABEL = 'Base Reaction [N]'
1031 TITLE = 'Base Reaction and Displacement of Structure During Pushover Analysis'
1032 COLOR = 'black'
1033 SEMILOGY = False
1034 PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
1035
1036 DATA = S07.BILINEAR_CURVE(np.abs(disp_PUSH), np.abs(reaction_PUSH), SLOPE_NODE=10)
1037 (X_PUSH, Y_PUSH, Elastic_ST, Plastic_ST, Tangent_ST, Ductility_Rito, Over_Strength_Factor) = DATA
1038
1039 # PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
```

Effective Displacement - Median: 0.15446

Effective Mass - Median: 482523.93674

Effective Stiffness - Median: 4928770.20397

Effective Period - Median: 1.96595

IPython Console Plots Variable Explorer Help Debugger History Files

Inline Conda: anaconda3 (Python 3.12.7) ✓ LSP: Python Line 1, Col 1 UTF-8 CRLF RW Mem 43%

PUSHOVER ANALYSIS

Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...MNS_8_ANA\P(t)_MDOF_10_FLOORS_04_COLUMNS_8_ANA.py

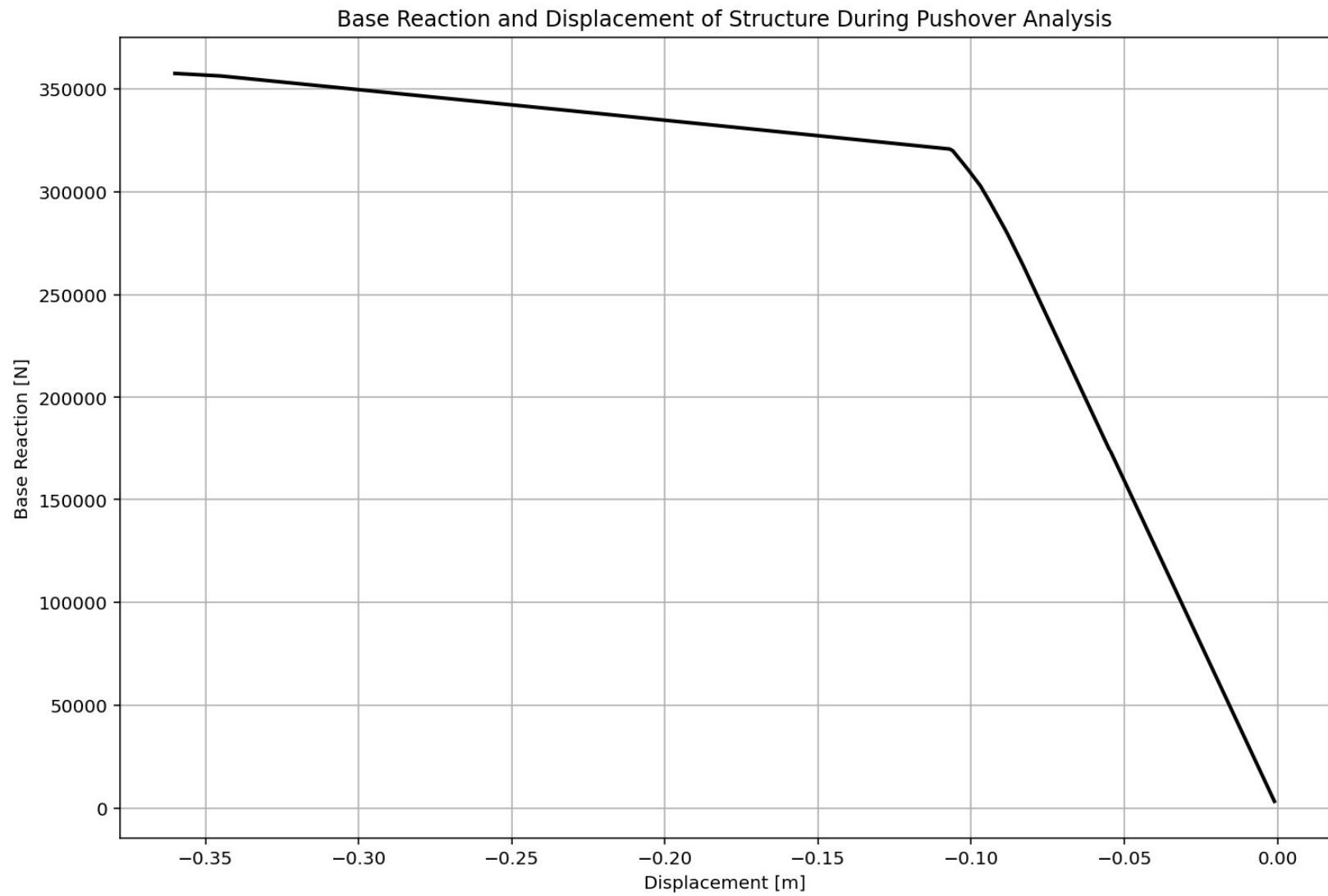
P(t)_MDOF_10_FLOORS_04_COLUMNS_8_ANA.py X

```
1017 # PUSHOVER ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
1018 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
1019 ANAL_TYPE = 'PUSHOVER'
1020
1021 DATA = MDOF(MAT_TYPE, TOTAL_MASS, ANAL_TYPE)
1022 (reaction_PUSH, disp_PUSH, DI_PUSH,
1023  ele_force_PUSH, node_displacements_PUSH,
1024  PERIOD_MIN_PUSH, PERIOD_MAX_PUSH) = DATA
1025
1026
1027 XDATA = disp_PUSH
1028 YDATA = reaction_PUSH
1029 XLABEL = 'Displacement [m]'
1030 YLABEL = 'Base Reaction [N]'
1031 TITLE = 'Base Reaction and Displacement of Structure During Pushover Analysis'
1032 COLOR = 'black'
1033 SEMIOLOGY = False
1034 PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMIOLOGY)
1035
1036 DATA = S07.BILINEAR_CURVE(np.abs(disp_PUSH), np.abs(reaction_PUSH), SLOPE_NODE=10)
1037 (X_PUSH, Y_PUSH, Elastic_ST, Plastic_ST, Tangent_ST, Ductility_Rito, Over_Strength_Factor) = DATA
1038
1039 # PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
1040 plt.figure(0, figsize=(12, 8))
1041 plt.plot(disp_PUSH, PERIOD_MIN_PUSH, linewidth=3)
1042 plt.plot(disp_PUSH, PERIOD_MAX_PUSH, linewidth=3)
1043 plt.title('Period of Structure During Pushover Analysis')
1044 plt.ylabel('Structural Period [s]')
1045 plt.xlabel('Displacement [m]')
1046 #plt.semilogy()
1047 plt.grid()
1048 plt.legend([f'PERIOD - MIN VALUES: Min: {np.min(PERIOD_MIN_PUSH):.3f} (s) - Mean: {np.mean(PERIOD_
1049 f'PERIOD - MAX VALUES: Min: {np.min(PERIOD_MAX_PUSH):.3f} (s) - Mean: {np.mean(PERIOD
1050 ]])
```

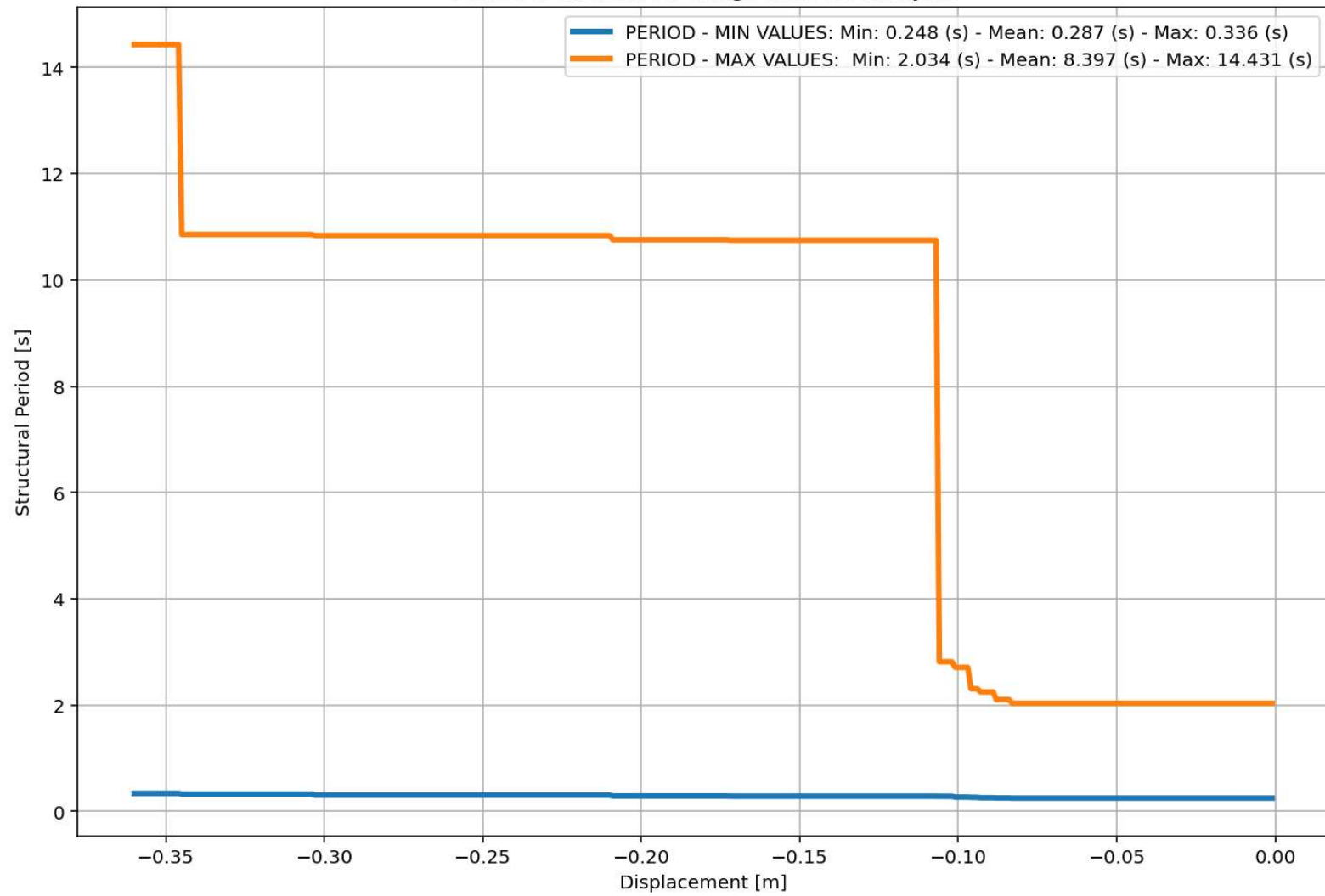
Base Reaction and Displacement of Structure During Pushover Analysis

IPython Console Plots Variable Explorer Help Debugger History Files

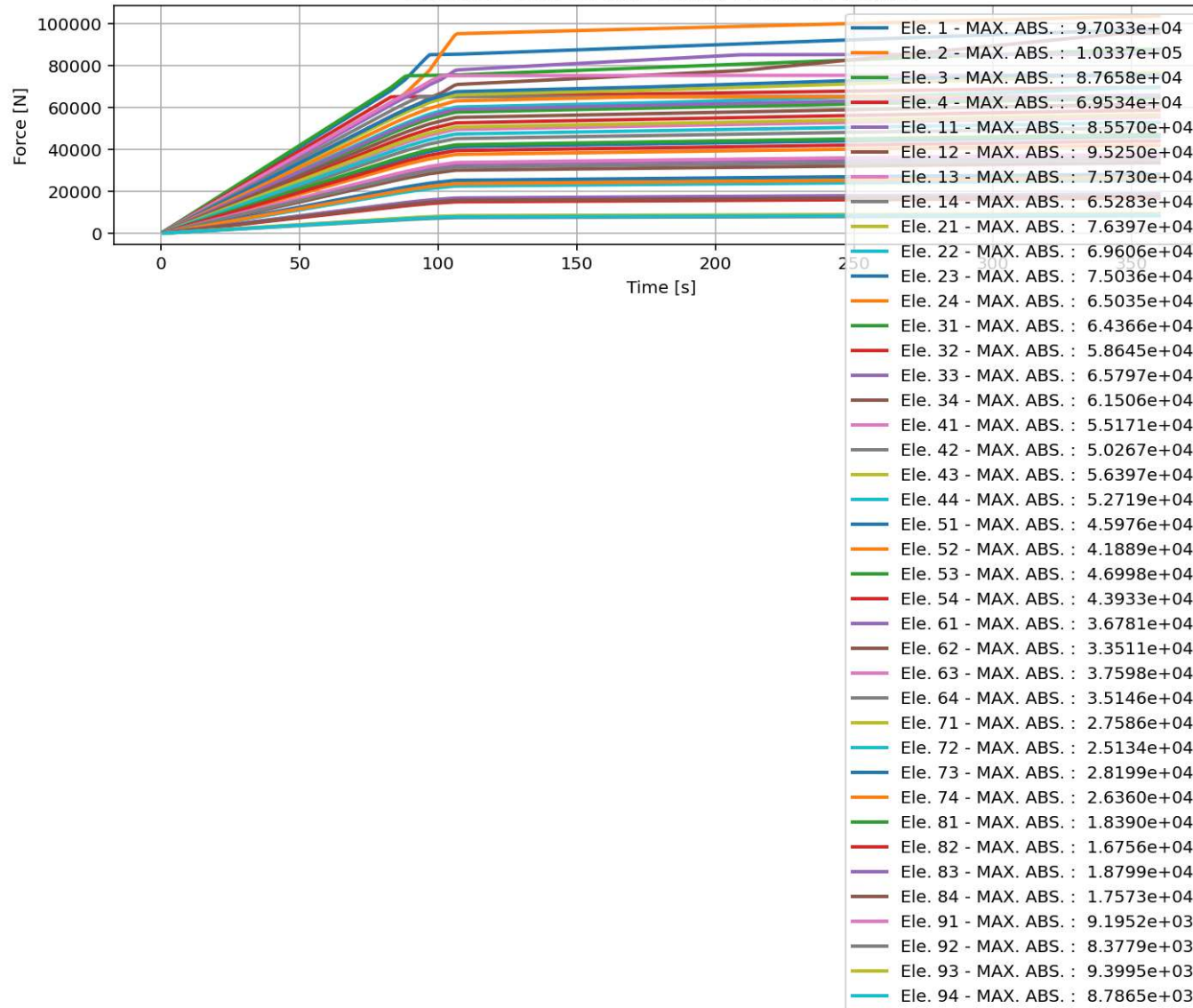
Inline Conda: anaconda3 (Python 3.12.7) ✓ LSP: Python Line 1, Col 1 UTF-8 CRLF RW Mem 42%



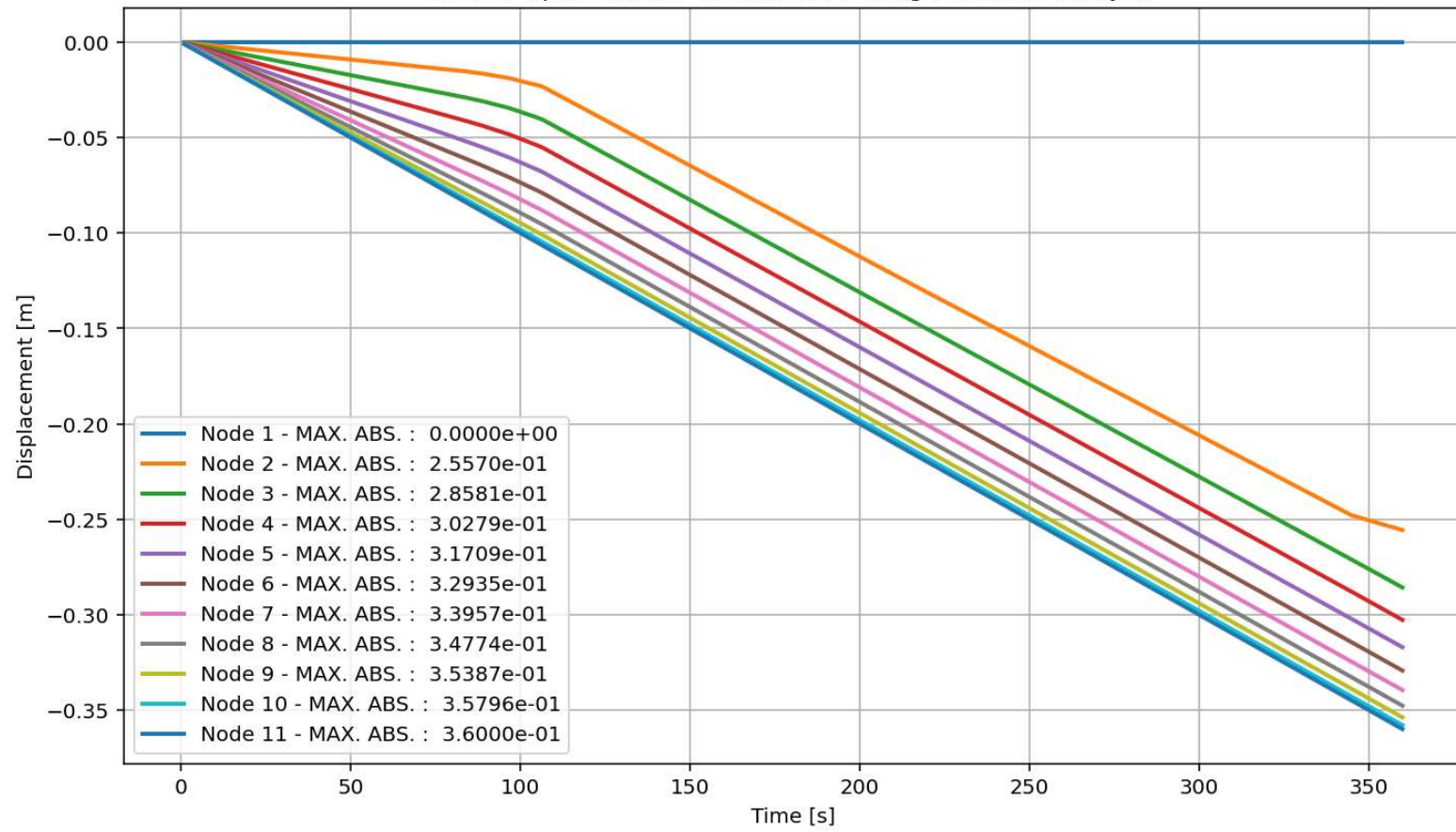
Period of Structure During Pushover Analysis

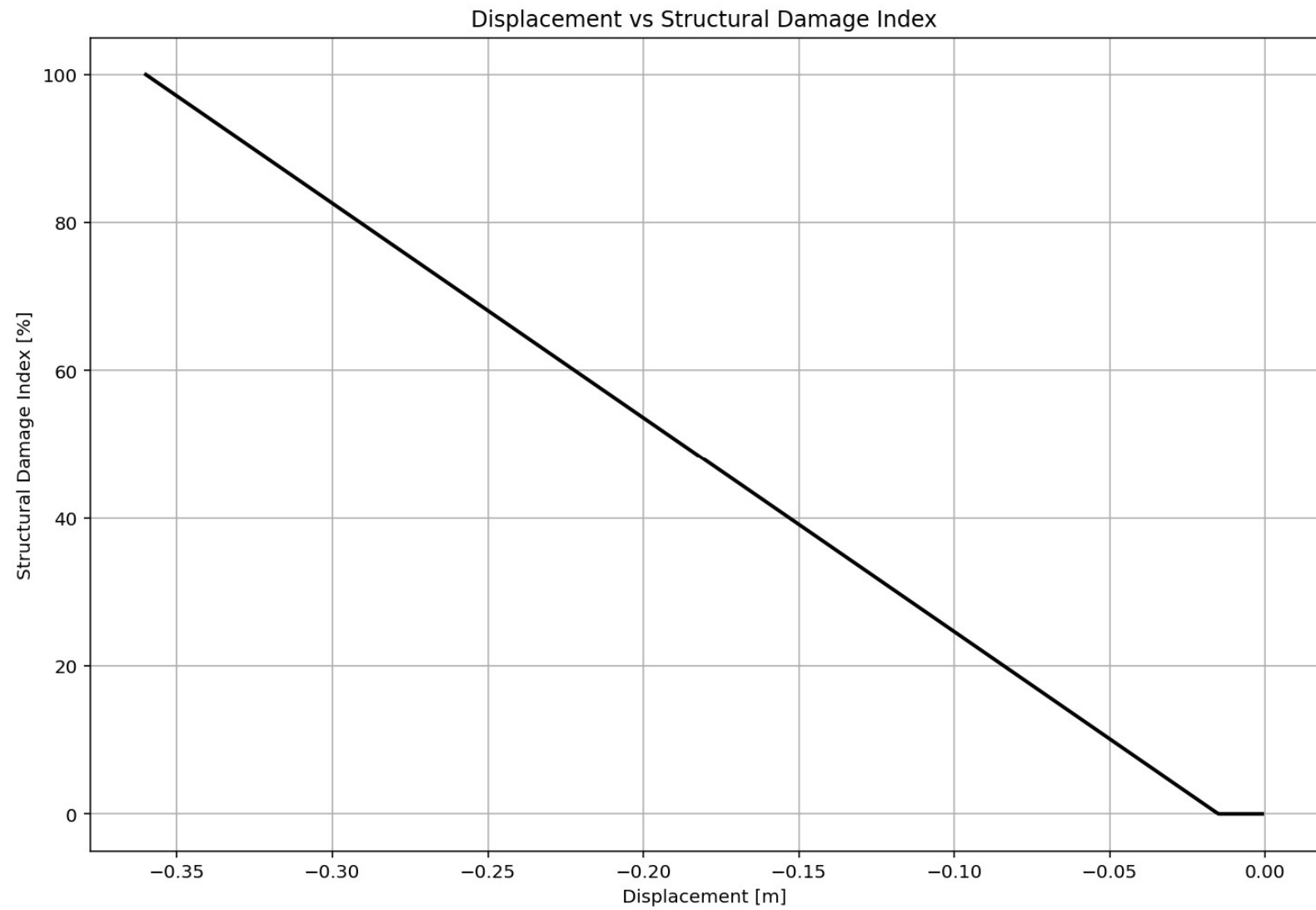


elements force vs Time During Pushover Analysis



Node Displacements vs Time for During Pushover Analysis





Spyder (Python 3.12)

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C:\Users\ DELL\Desktop\OPENSEES_FILES\+EIGHT_ANALY...MNS_8_ANA\p(t)_MDOF_10_FLOORS_04_COLUMNS_8_ANA.py

P(t)_MDOF_10_FLOORS_04_COLUMNS_8_ANA.py X

```
1017 # PUSHOVER ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
1018 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
1019 ANAL_TYPE = 'PUSHOVER'
1020
1021 DATA = MDOF(MAT_TYPE, TOTAL_MASS, ANAL_TYPE)
1022 (reaction_PUSH, disp_PUSH, DI_PUSH,
1023  ele_force_PUSH, node_displacements_PUSH,
1024  PERIOD_MIN_PUSH, PERIOD_MAX_PUSH) = DATA
1025
1026
1027 XDATA = disp_PUSH
1028 YDATA = reaction_PUSH
1029 XLABEL = 'Displacement [m]'
1030 YLABEL = 'Base Reaction [N]'
1031 TITLE = 'Base Reaction and Displacement of Structure During Pushover Analysis'
1032 COLOR = 'black'
1033 SEMIOLOGY = False
1034 PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMIOLOGY)
1035
1036 DATA = S07.BILINEAR_CURVE(np.abs(disp_PUSH), np.abs(reaction_PUSH), SLOPE_NODE=10)
1037 (X_PUSH, Y_PUSH, Elastic_ST, Plastic_ST, Tangent_ST, Ductility_Rito, Over_Strength_Factor) = DATA
1038
1039 # PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
1040 plt.figure(0, figsize=(12, 8))
1041 plt.plot(disp_PUSH, PERIOD_MIN_PUSH, linewidth=3)
1042 plt.plot(disp_PUSH, PERIOD_MAX_PUSH, linewidth=3)
1043 plt.title('Period of Structure During Pushover Analysis')
1044 plt.ylabel('Structural Period [s]')
1045 plt.xlabel('Displacement [m]')
1046 #plt.semilogy()
1047 plt.grid()
1048 plt.legend([f'PERIOD - MIN VALUES: Min: {np.min(PERIOD_MIN_PUSH):.3f} (s) - Mean: {np.mean(PERIOD_MIN_PUSH):.3f} (s)',
1049            f'PERIOD - MAX VALUES: Min: {np.min(PERIOD_MAX_PUSH):.3f} (s) - Mean: {np.mean(PERIOD_MAX_PUSH):.3f} (s)'])
1050
```

This is a string

...S_04_COLUMNS_8_ANA+\+60_MDOF_10_FLOORS_04_COLUMNS_8_ANA

23 %

Effective Displacement - Median: 0.15446

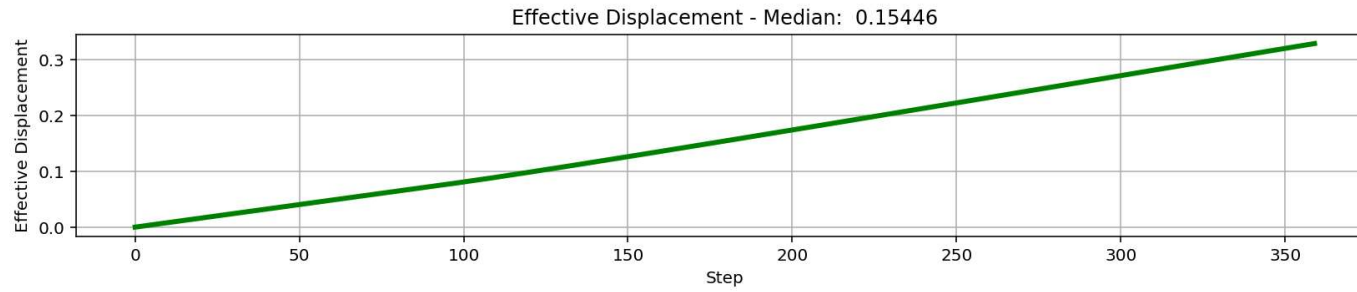
Effective Mass - Median: 482523.93674

Effective Stiffness - Median: 4828770.20397

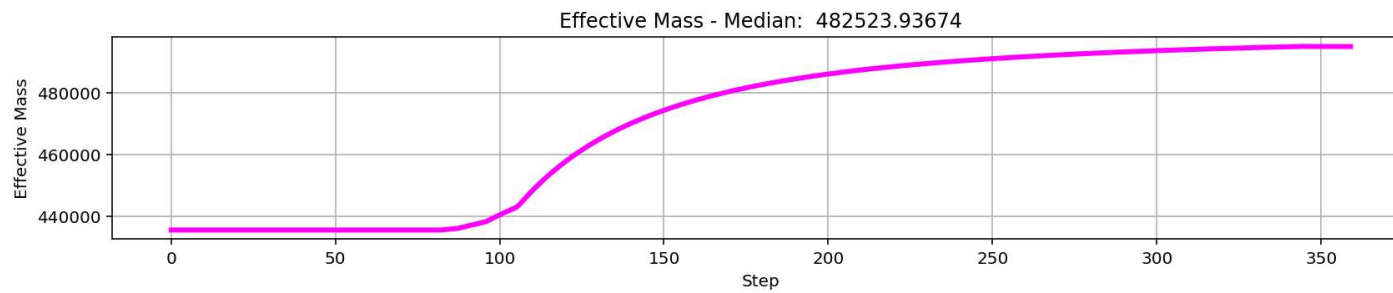
Effective Period - Median: 1.96595

IPython Console Plots Variable Explorer Help Debugger History Files

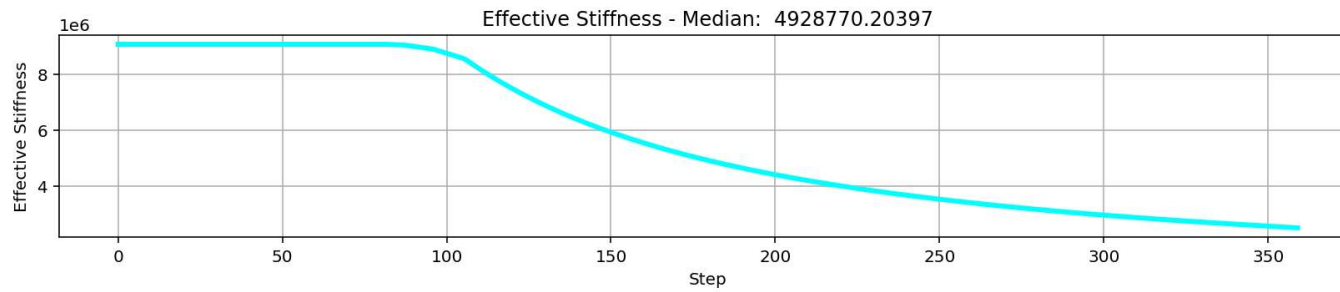
Inline Conda: anaconda3 (Python 3.12.7) ✓ LSP: Python Line 1, Col 1 UTF-8 CRLF RW Mem 43%



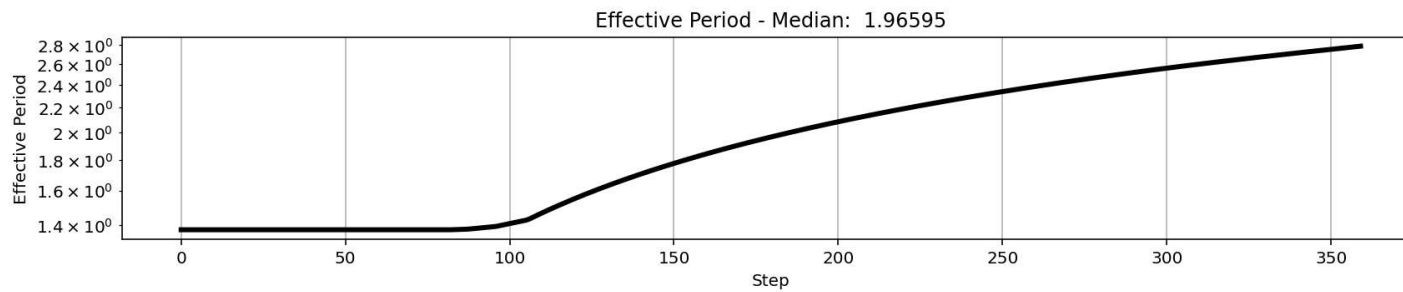
$$\Delta_d = \frac{\sum_{i=1}^n m_i \Delta_i^2}{\sum_{i=1}^n m_i \Delta_i}$$



$$m_{eff} = \frac{\sum_{i=1}^n m_i \Delta_i}{\Delta_d}$$

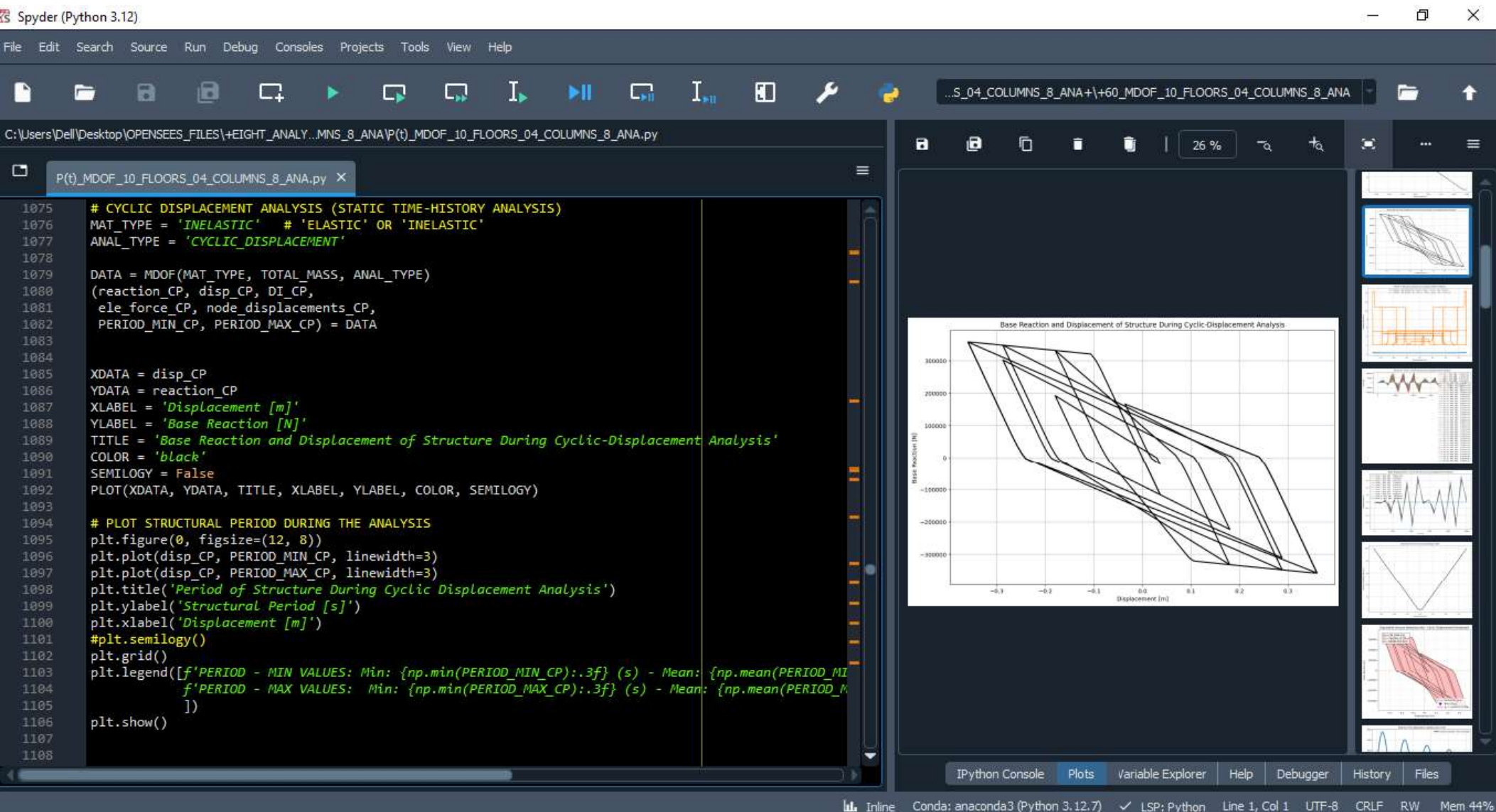


$$k_{eff} = \frac{\sum_{i=1}^n k_i \Delta_i}{\Delta_d}$$

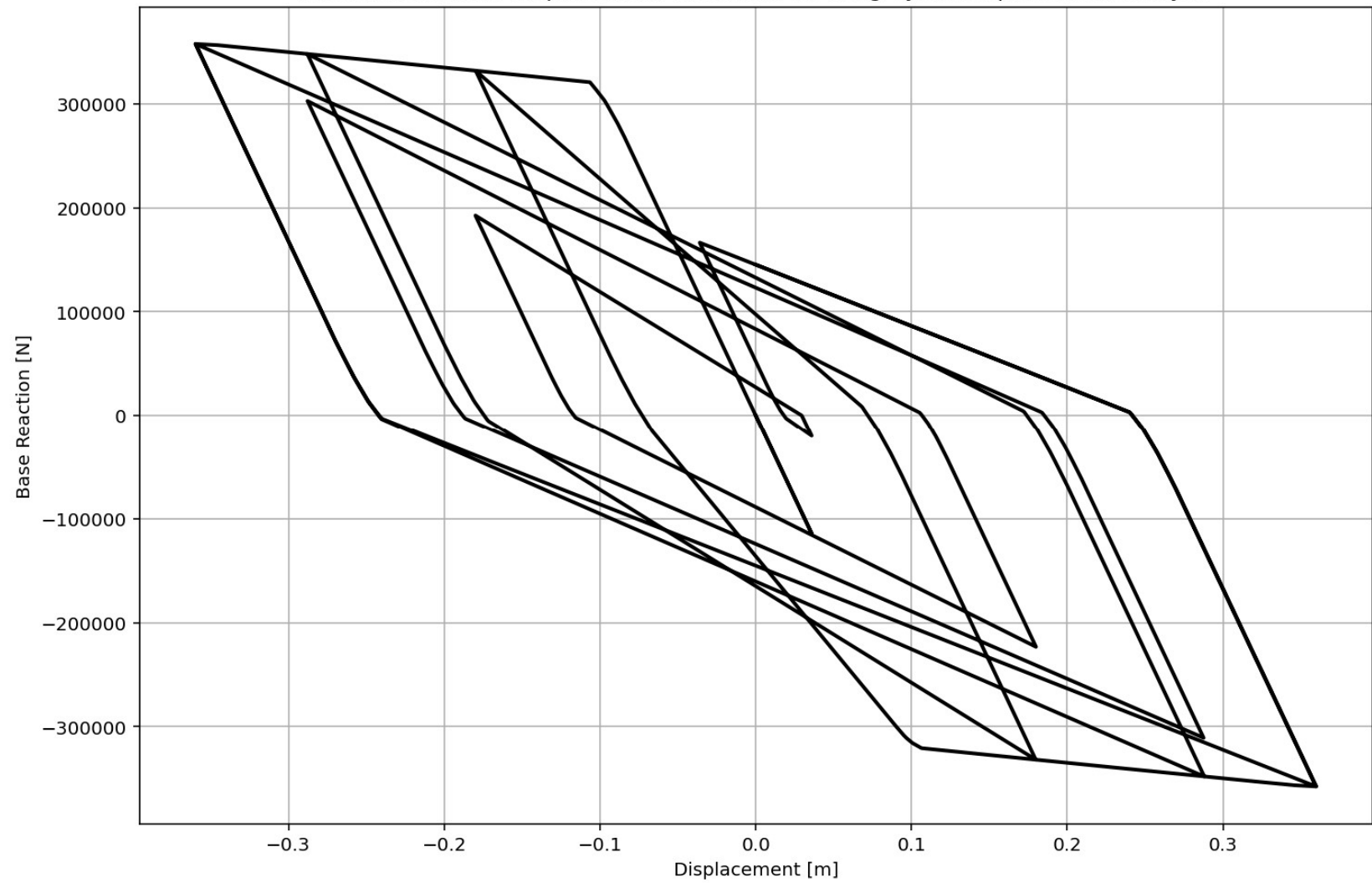


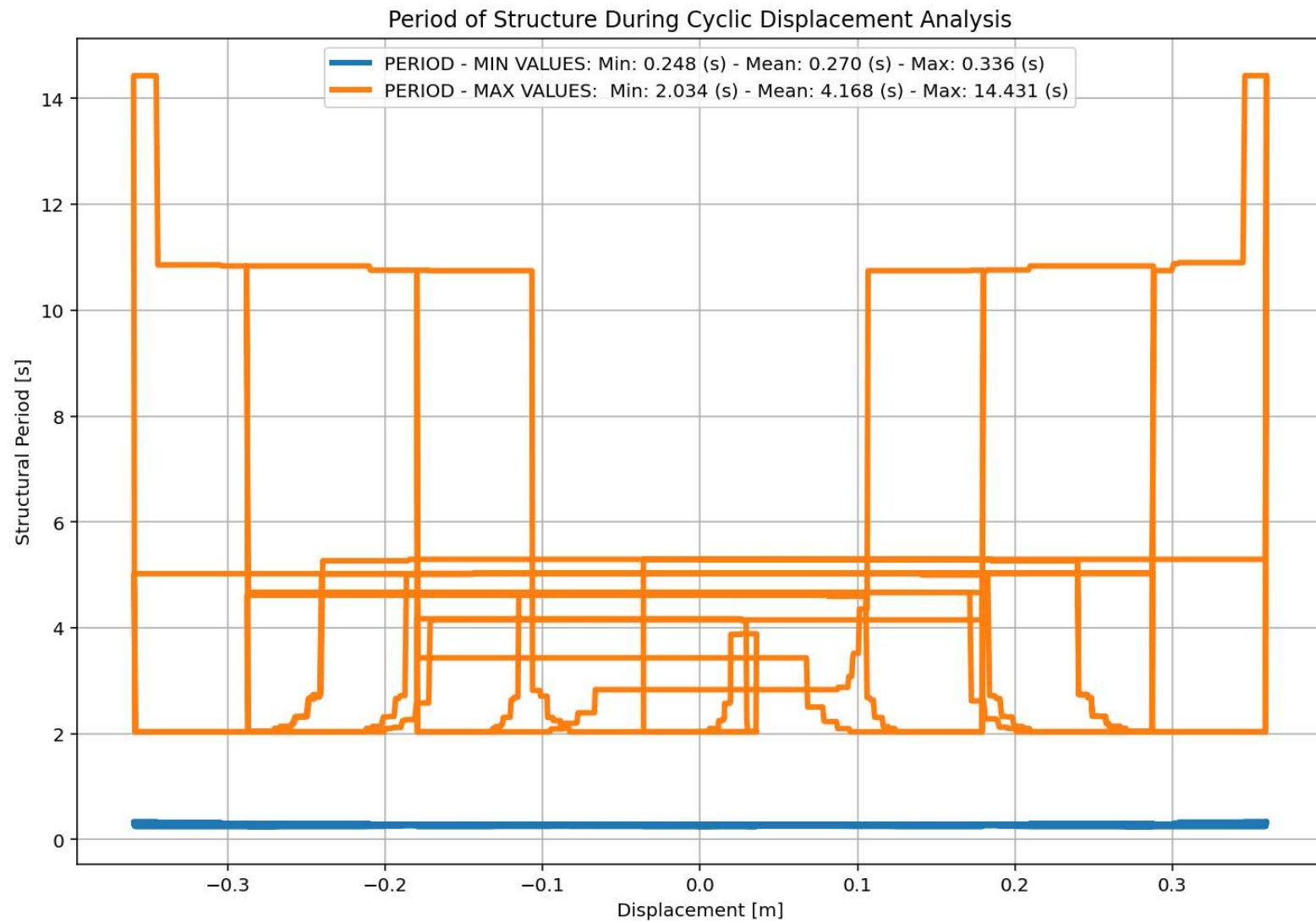
$$T_{eff} = \frac{2\pi}{\sqrt{\frac{k_{eff}}{m_{eff}}}}$$

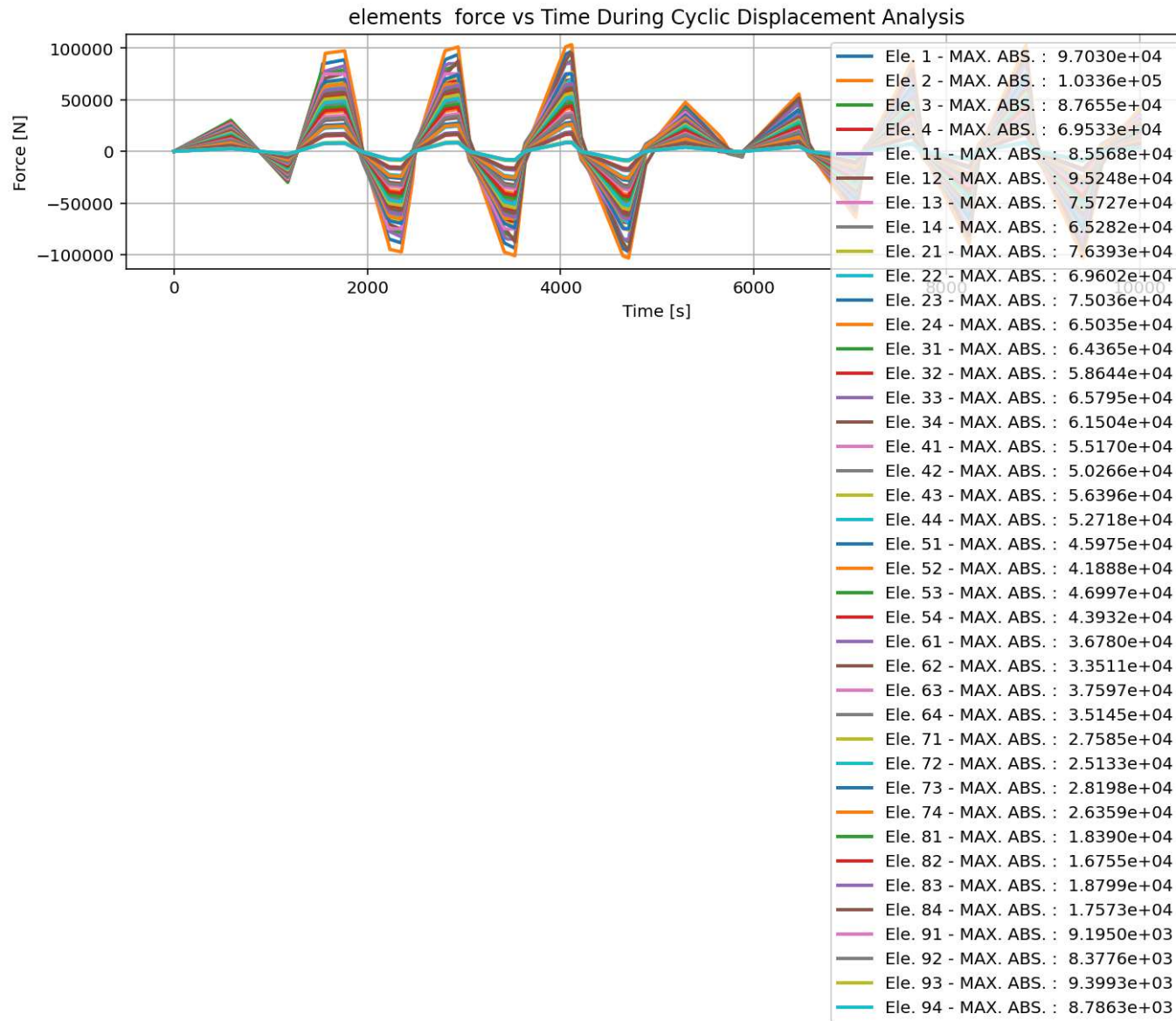
CYCLIC DISPLACEMENT ANALYSIS



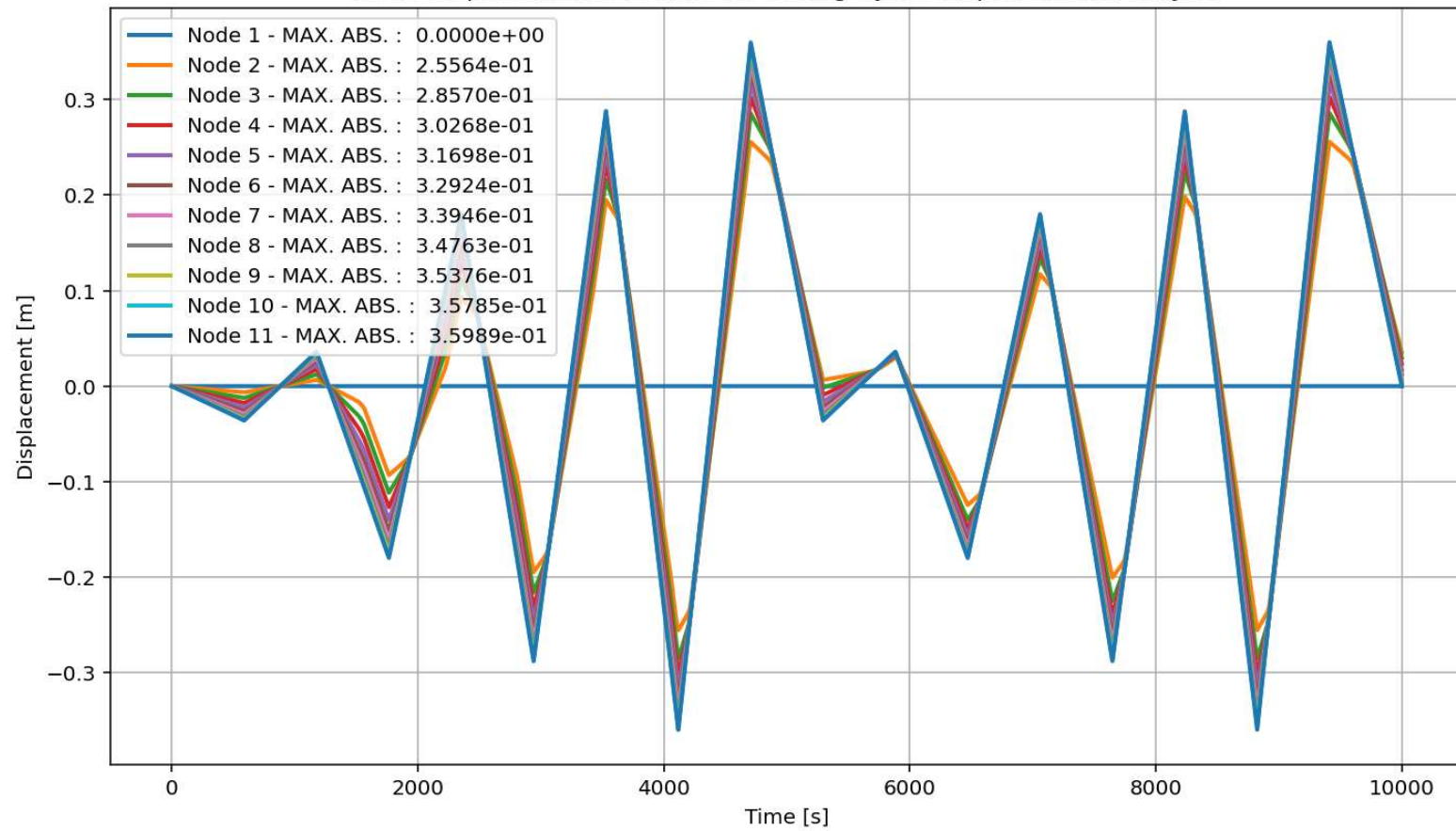
Base Reaction and Displacement of Structure During Cyclic-Displacement Analysis





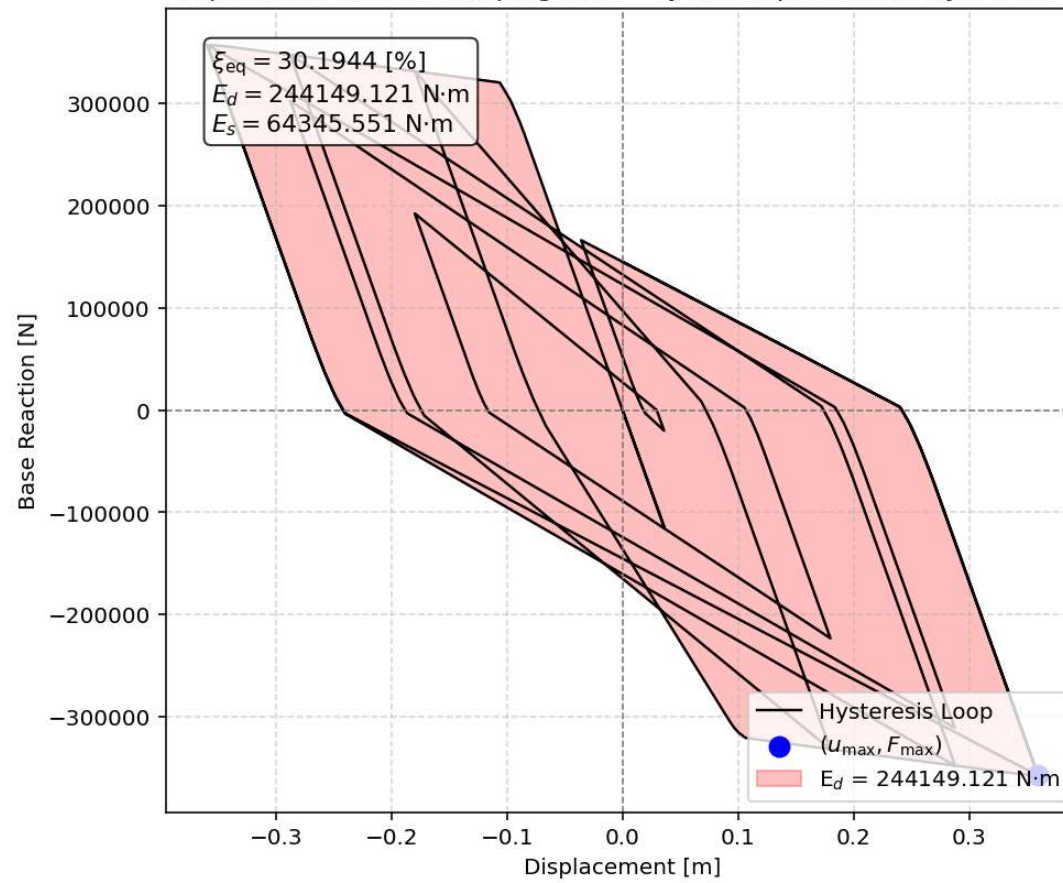


Node Displacements vs Time for During Cyclic Displacement Analysis





Equivalent viscous damping ratio - Cyclic Displacement Hysteresis



STATIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS

Spyder (Python 3.12)

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C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...MNS_8_ANA\p(t)_MDOF_10_FLOORS_04_COLUMNS_8_ANA.py

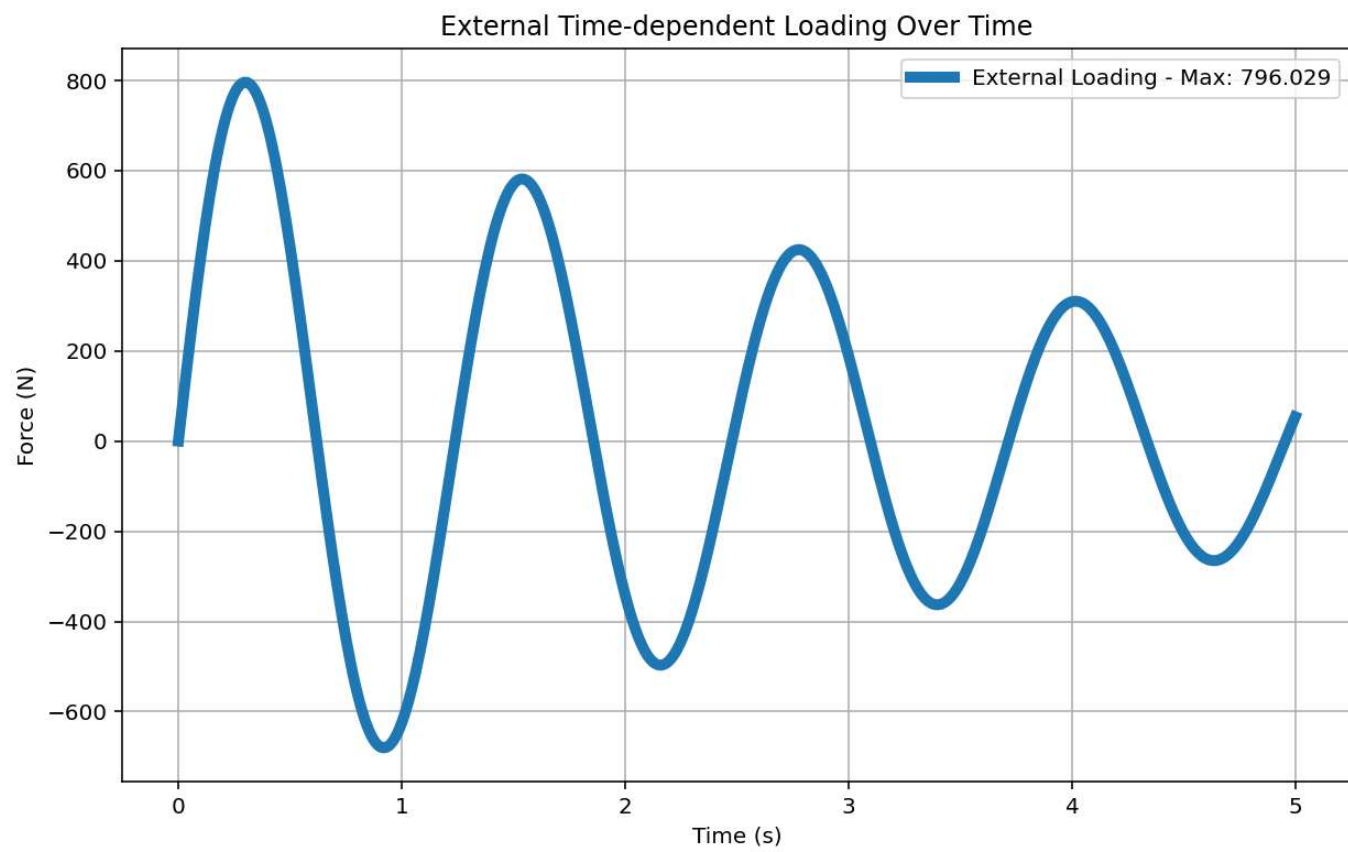
P(t)_MDOF_10_FLOORS_04_COLUMNS_8_ANA.py X

```
1139 # EXTERNAL TIME-DEPENDENT LOADING ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
1140 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
1141 ANAL_TYPE = 'STATIC_EXTERNAL_TIME-DEPENDENT_LOADING'
1142
1143 DATA = MDOF(MAT_TYPE, TOTAL_MASS, ANAL_TYPE)
1144
1145 (reaction_ETDLS, disp_ETDLS, DI_ETDLS,
1146  ele_force_ETDLS, node_displacements_ETDLS,
1147  PERIOD_MIN_ETDLS, PERIOD_MAX_ETDLS) = DATA
1148
1149
1150 XDATA = disp_ETDLS
1151 YDATA = reaction_ETDLS
1152 XLABEL = 'Displacement [m]'
1153 YLABEL = 'Base Reaction [N]'
1154 TITLE = 'Base Reaction and Displacement of Structure During Static External Time-dependent Loading
1155 COLOR = 'black'
1156 SEMIOLOGY = False
1157 PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMIOLOGY)
1158
1159 # PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
1160 plt.figure(0, figsize=(12, 8))
1161 plt.plot(disp_ETDLS, PERIOD_MIN_ETDLS, linewidth=3)
1162 plt.plot(disp_ETDLS, PERIOD_MAX_ETDLS, linewidth=3)
1163 plt.title('Period of Structure During Static External Time-dependent Loading Analysis')
1164 plt.ylabel('Structural Period [s]')
1165 plt.xlabel('Displacement [m]')
1166 #plt.semilogy()
1167 plt.grid()
1168 plt.legend([f'PERIOD - MIN VALUES: Min: {np.min(PERIOD_MIN_ETDLS):.3f} (s) - Mean: {np.mean(PERIOD
1169 f'PERIOD - MAX VALUES: Min: {np.min(PERIOD_MAX_ETDLS):.3f} (s) - Mean: {np.mean(PERIC
1170 ]])
1171 plt.show()
1172
```

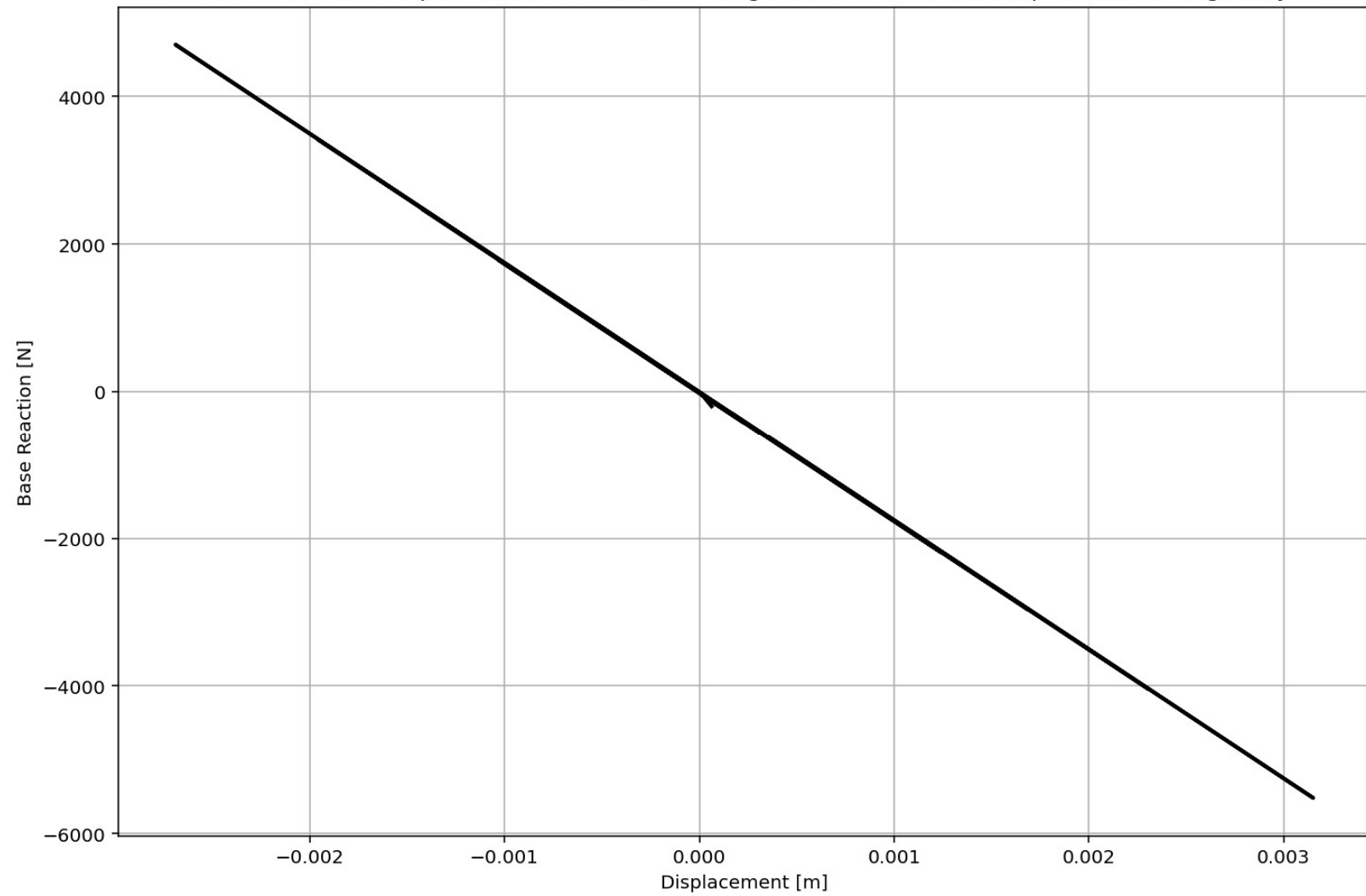
Base Reaction and Displacement of Structure During Static External Time-dependent Loading Analysis

IPython Console Plots Variable Explorer Help Debugger History Files

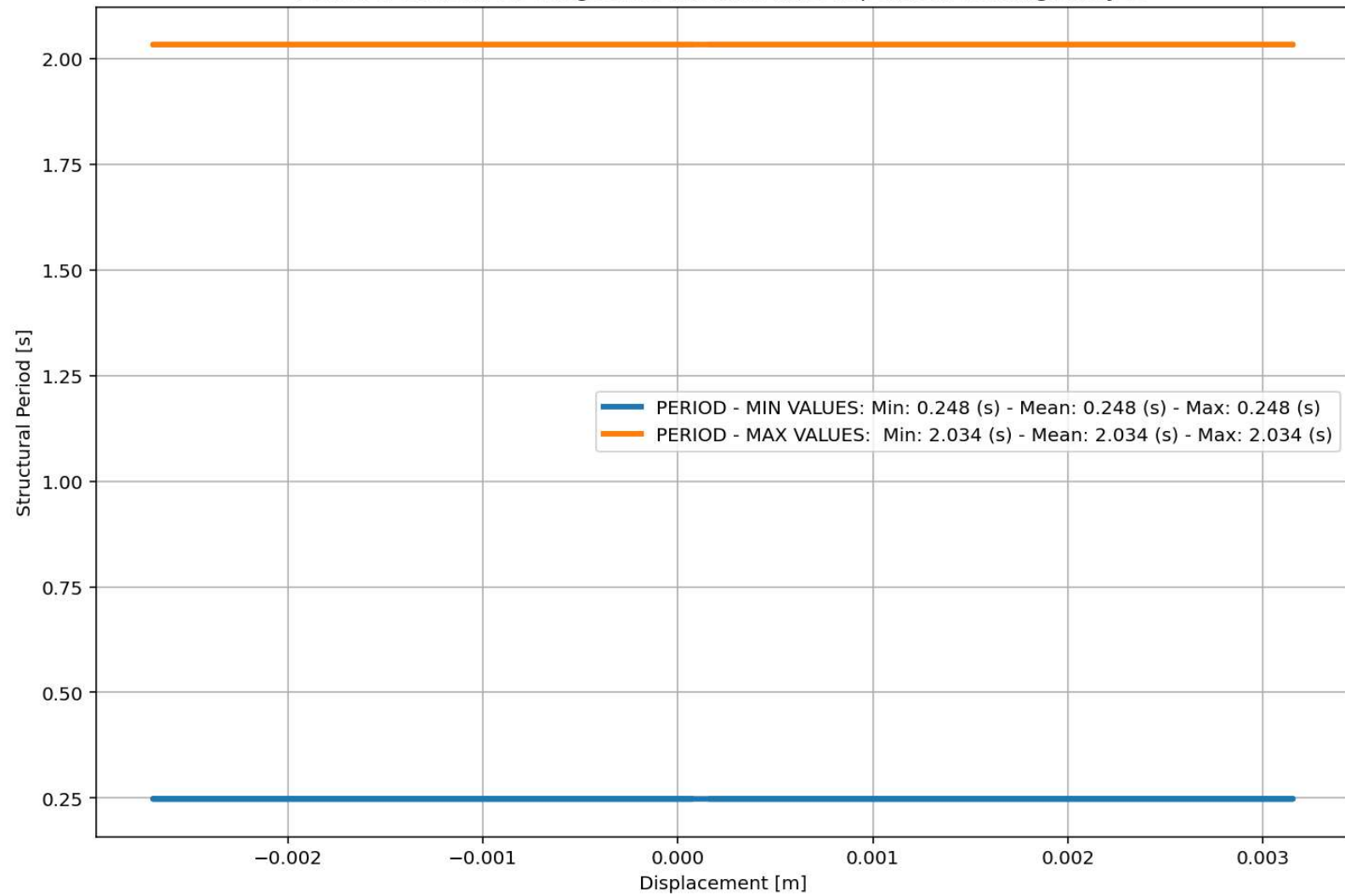
Inline Conda: anaconda3 (Python 3.12.7) ✓ LSP: Python Line 1, Col 1 UTF-8 CRLF RW Mem 44%

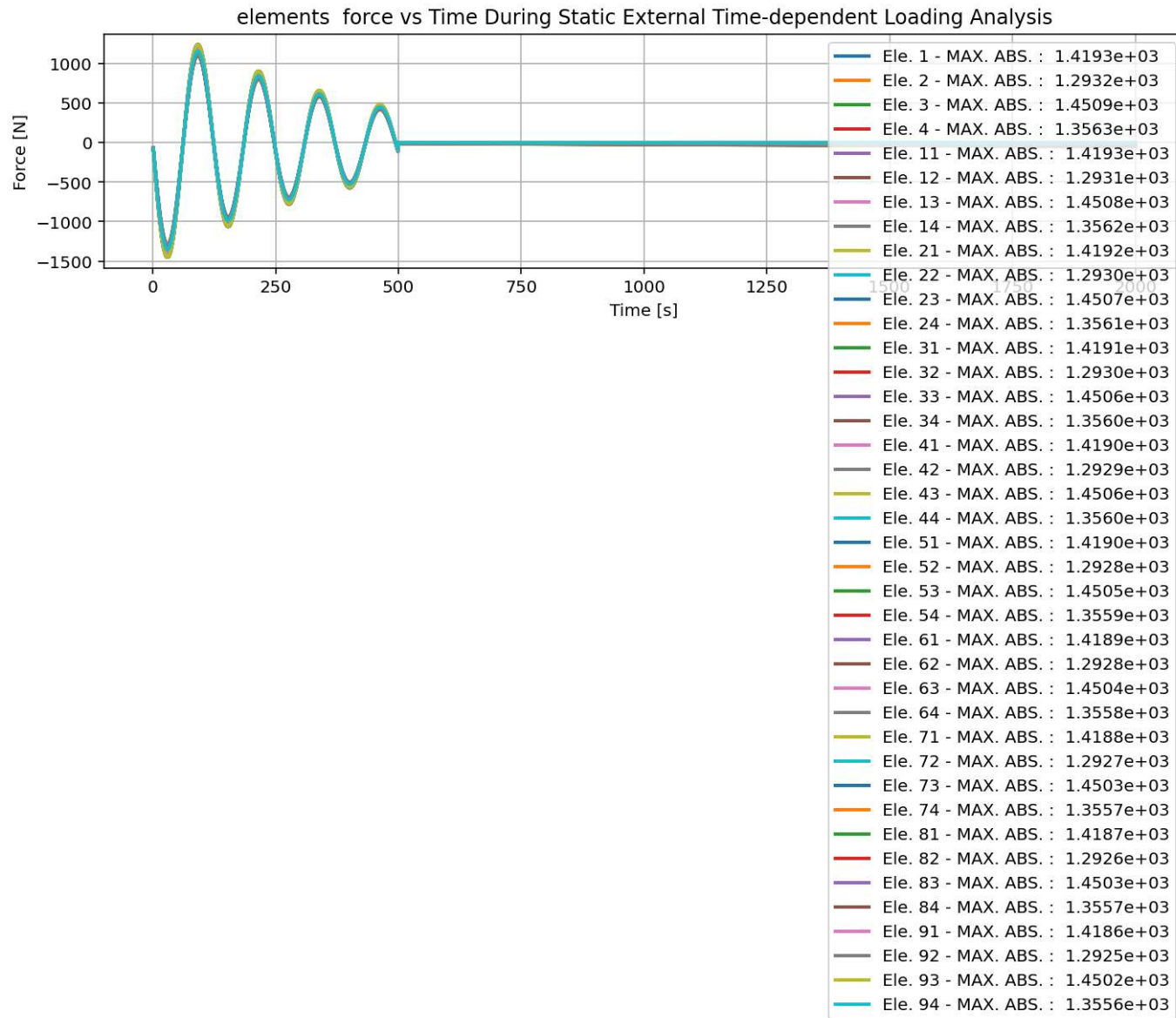


Base Reaction and Displacement of Structure During Static External Time-dependent Loading Analysis

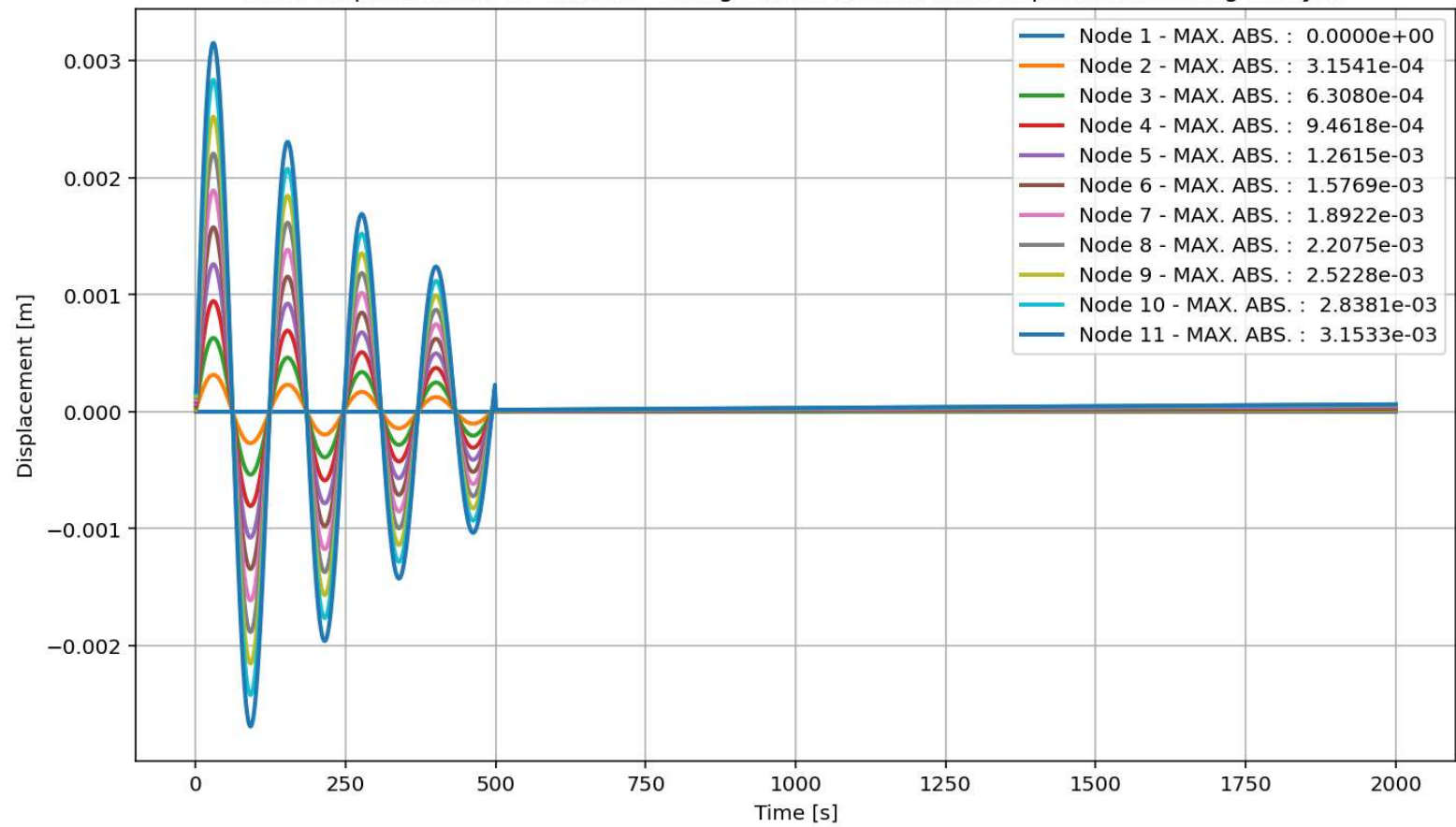


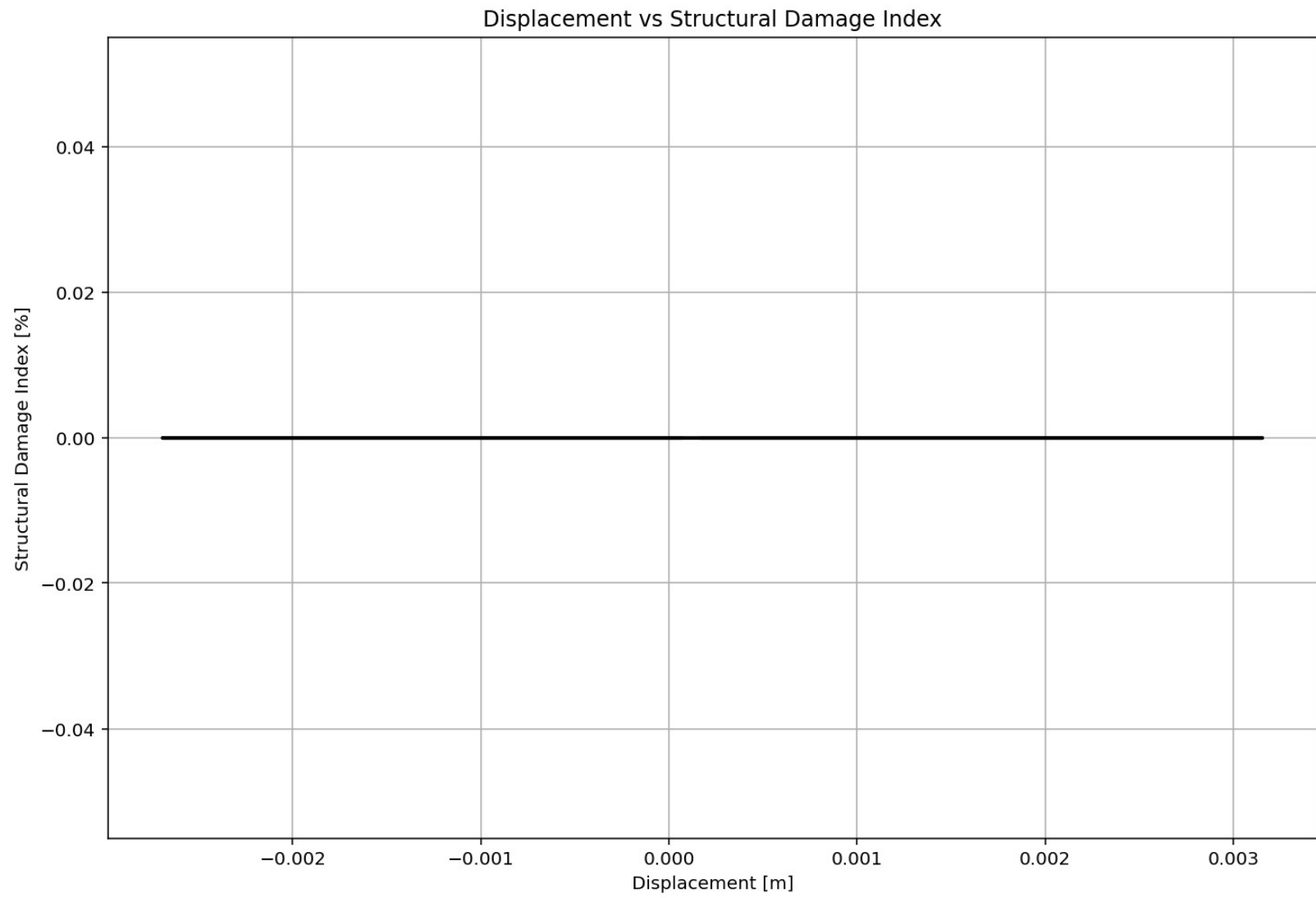
Period of Structure During Static External Time-dependent Loading Analysis



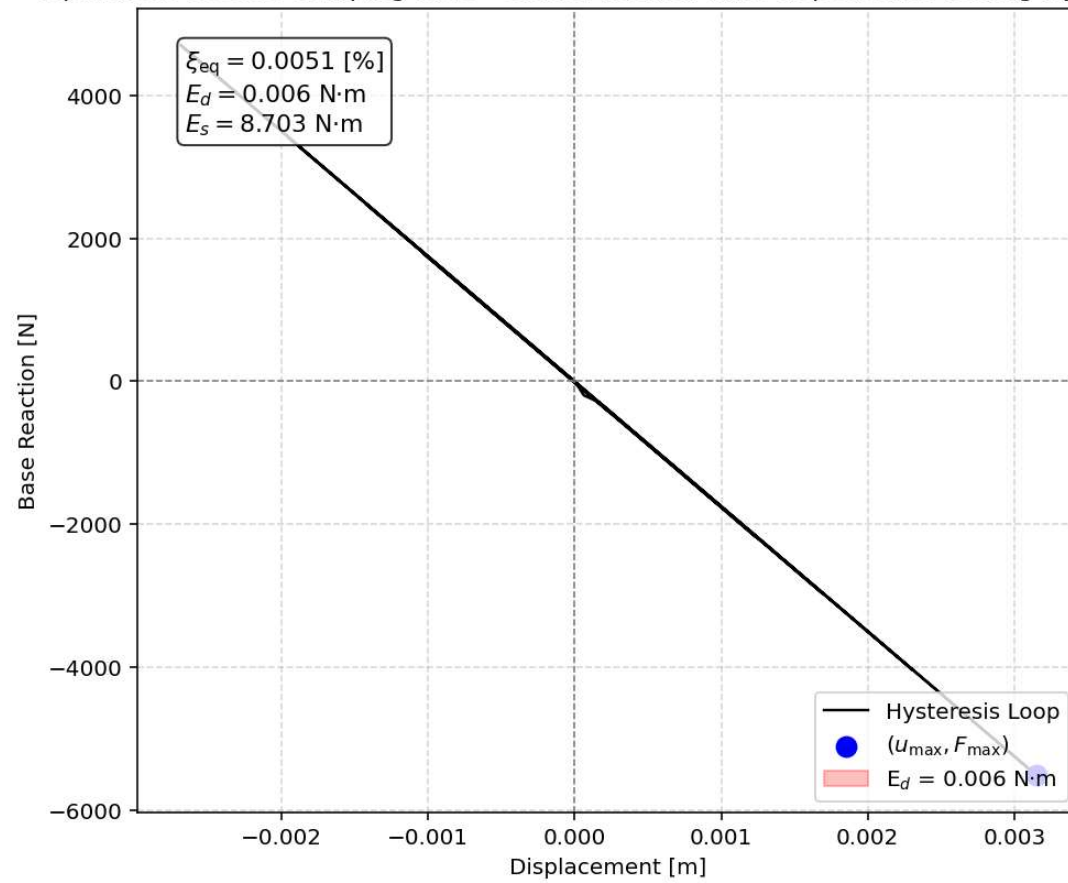


Node Displacements vs Time for During Static External Time-dependent Loading Analysis





Equivalent viscous damping ratio - Static External Time-dependent Loading Hysteresis



DYNAMIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS

Spyder (Python 3.12)

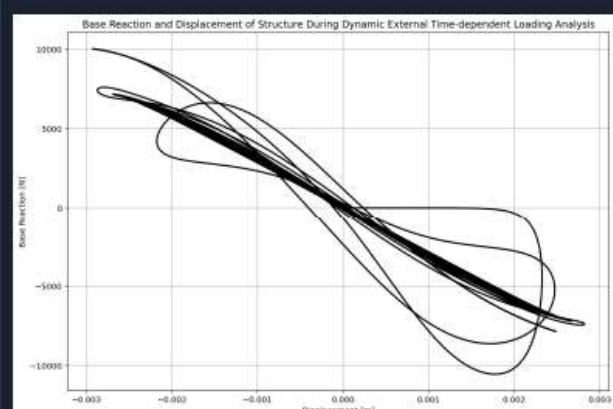
File Edit Search Source Run Debug Consoles Projects Tools View Help

C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...MNS_8_ANA\P(t)_MDOF_10_FLOORS_04_COLUMNS_8_ANA.py

P(t)_MDOF_10_FLOORS_04_COLUMNS_8_ANA.py X

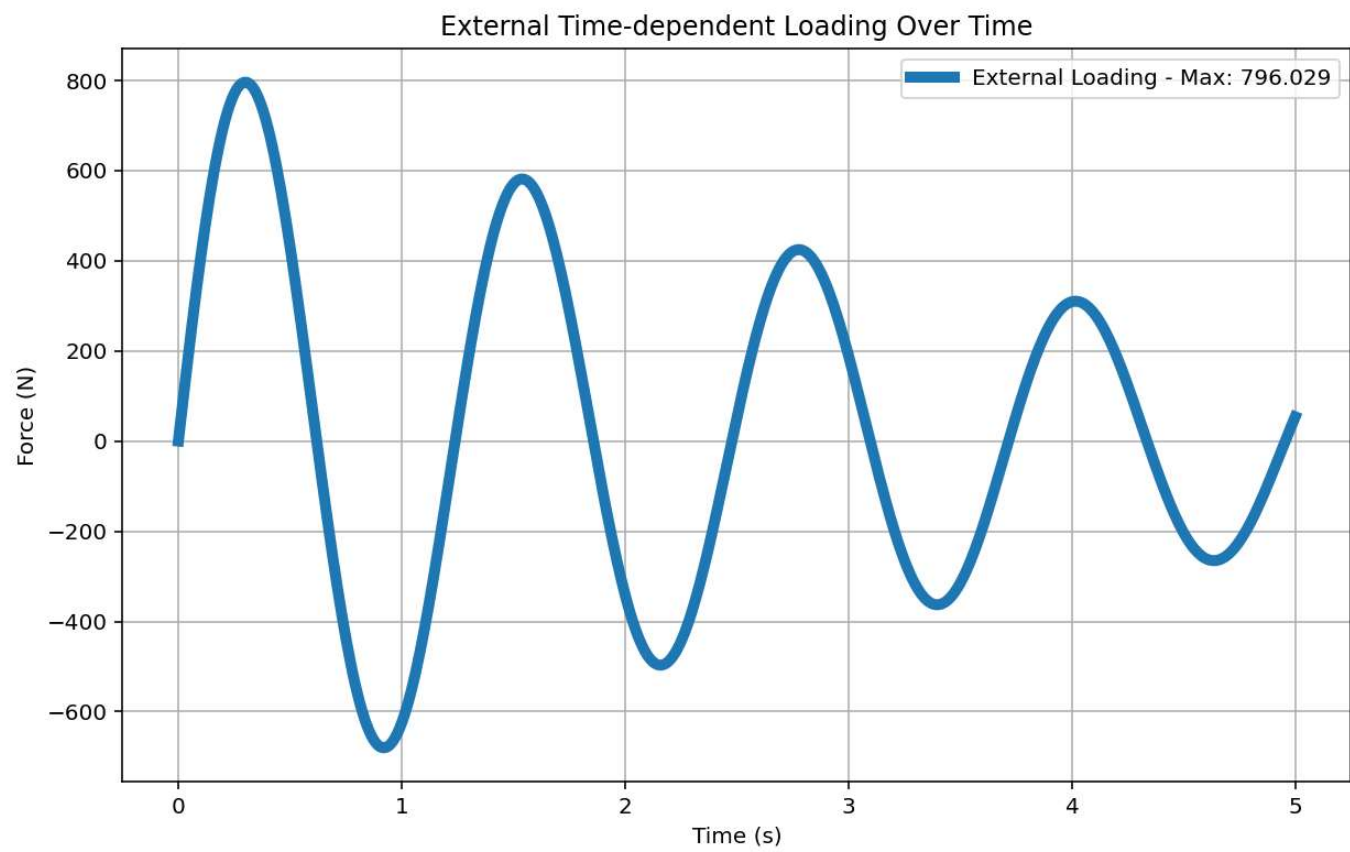
```
1204 # EXTERNAL TIME-DEPENDENT LOADING ANALYSIS (DYNAMIC TIME-HISTORY ANALYSIS)
1205 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
1206 ANAL_TYPE = 'DYNAMIC_EXTERNAL_TIME-DEPENDENT_LOADING'
1207
1208 DATA = MDOF(MAT_TYPE, TOTAL_MASS, ANAL_TYPE)
1209
1210 (time_ETDLD, reaction_ETDLD, disp_ETDLD, velo_ETDLD, acc_ETDLD, DI_ETDLD,
1211  ele_force_ETDLD, node_displacements_ETDLD,
1212  stiffness, PERIOD, damping_ratio,
1213  PERIOD_MIN_ETDLD, PERIOD_MAX_ETDLD) = DATA
1214
1215
1216
1217 XDATA = disp_ETDLD
1218 YDATA = reaction_ETDLD
1219 XLABEL = 'Displacement [m]'
1220 YLABEL = 'Base Reaction [N]'
1221 TITLE = 'Base Reaction and Displacement of Structure During Dynamic External Time-dependent Loading Analysis'
1222 COLOR = 'black'
1223 SEMILOGY = False
1224 PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
1225
1226 PLOT_TIME_HISTORY(time_ETDLD, reaction_ETDLD, disp_ETDLD, velo_ETDLD, acc_ETDLD)
1227
1228 # PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
1229 plt.figure(0, figsize=(12, 8))
1230 plt.plot(disp_ETDLD, PERIOD_MIN_ETDLD, linewidth=3)
1231 plt.plot(disp_ETDLD, PERIOD_MAX_ETDLD, linewidth=3)
1232 plt.title('Period of Structure During Dynamic External Time-dependent Loading Analysis')
1233 plt.ylabel('Structural Period [s]')
1234 plt.xlabel('Displacement [m]')
1235 #plt.semilogy()
1236 plt.grid()
1237 plt.legend([f'PERIOD - MIN VALUES: Min: {np.min(PERIOD_MIN_ETDLD):.3f} (s) - Mean: {np.mean(PERIOD
```

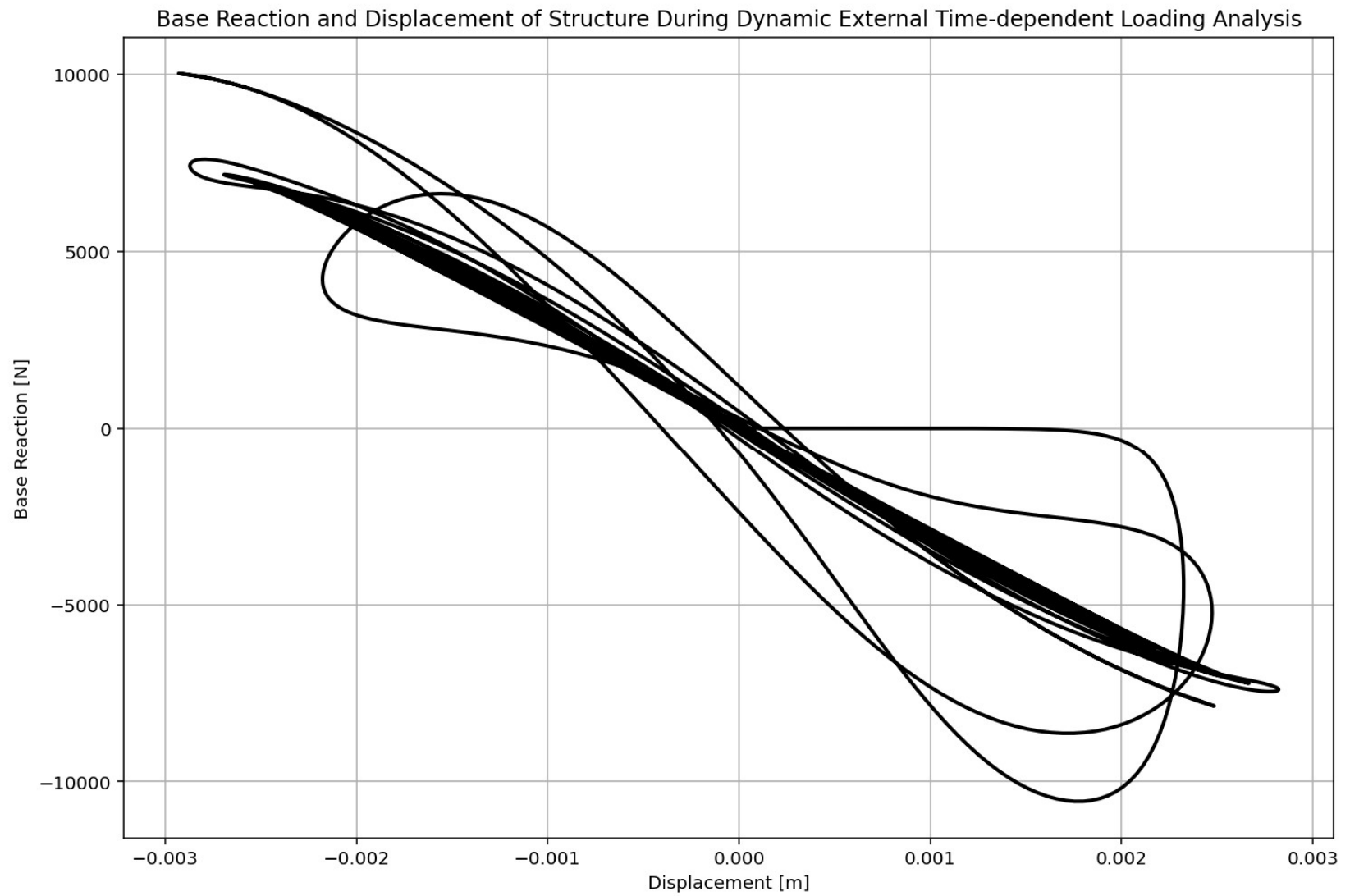
Base Reaction and Displacement of Structure During Dynamic External Time-dependent Loading Analysis



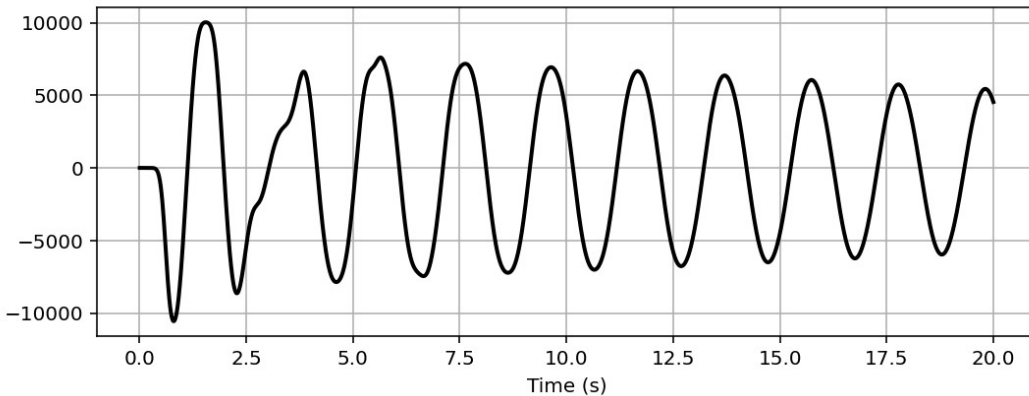
IPython Console Plots Variable Explorer Help Debugger History Files

Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 1, Col 1 UTF-8 CRLF RW Mem 45%

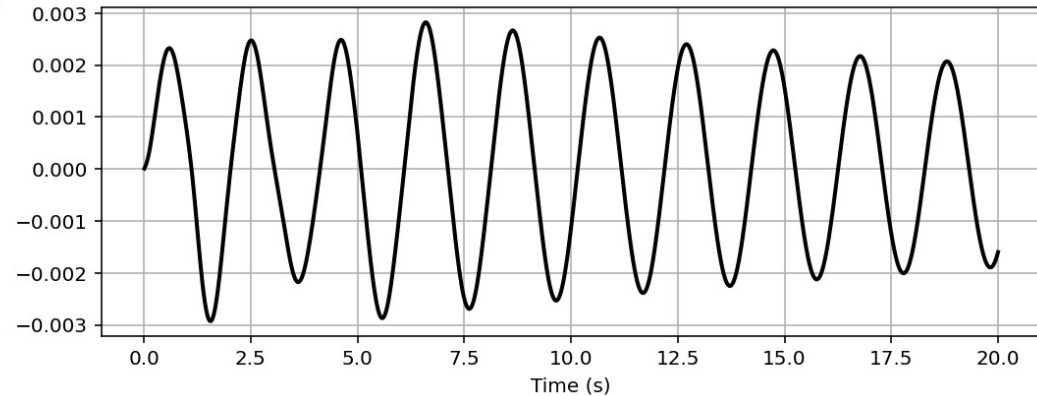




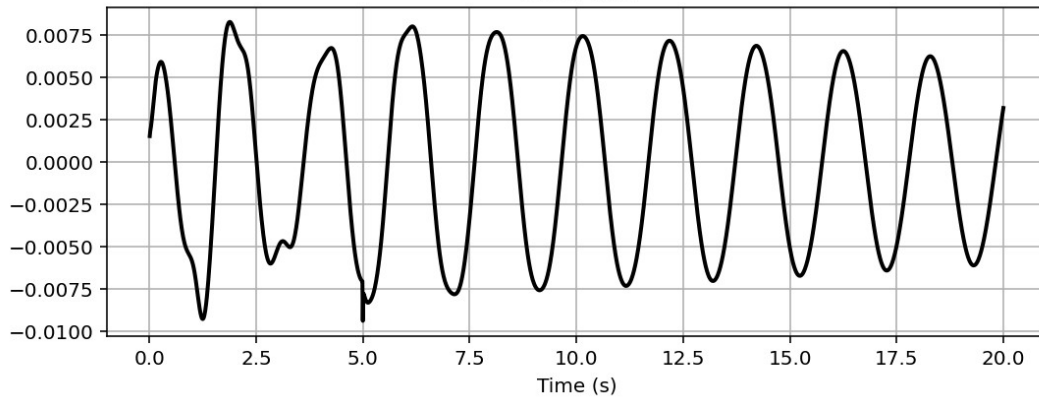
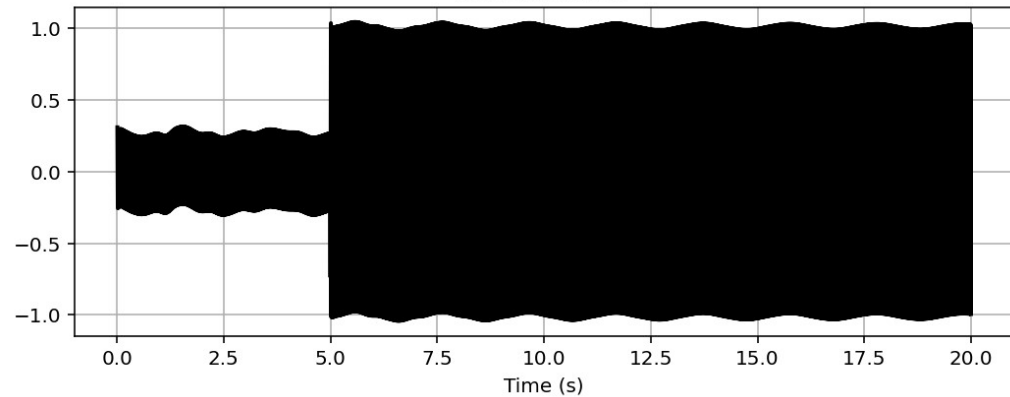
Reaction [N]



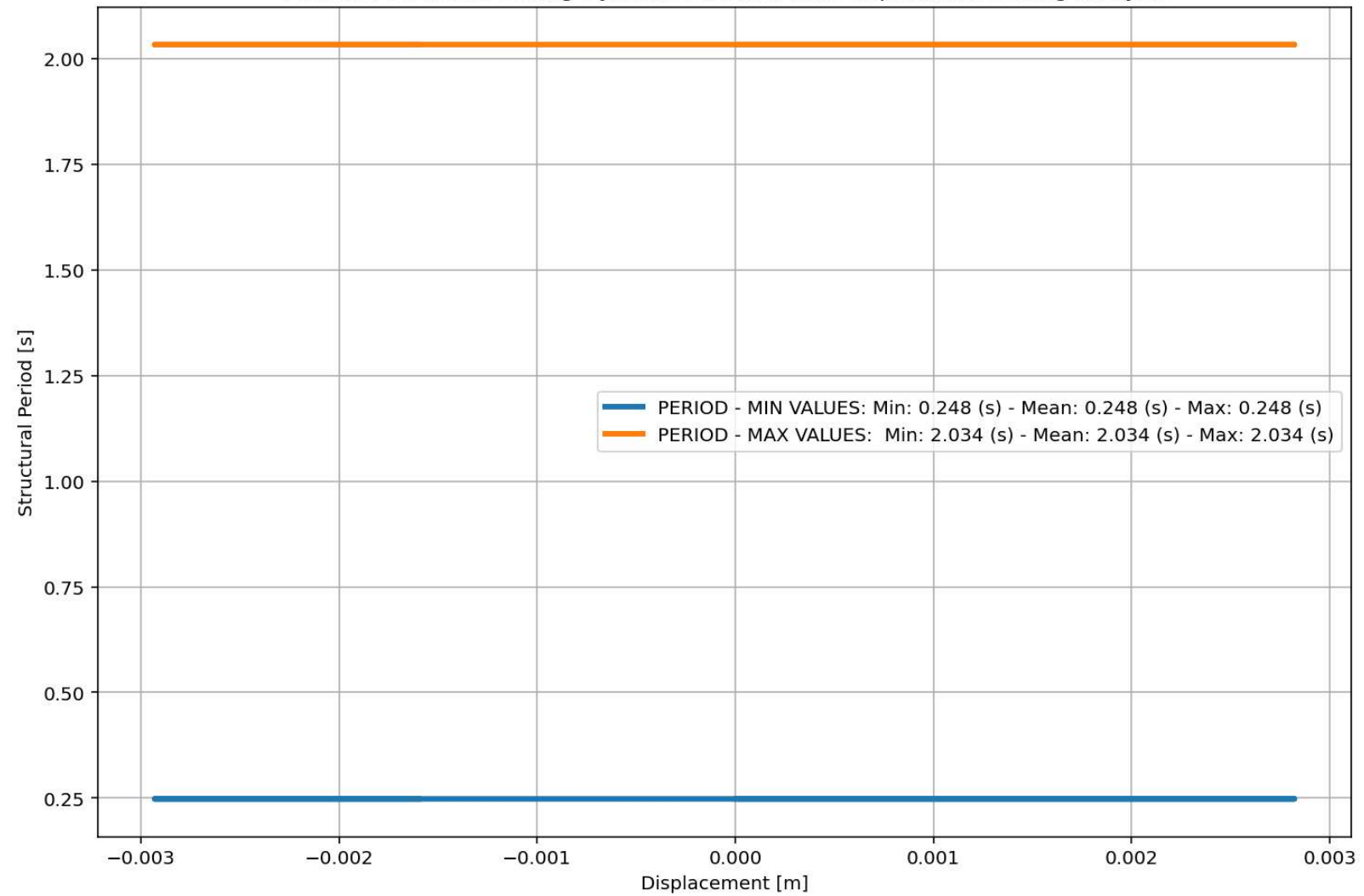
Displacement [m]

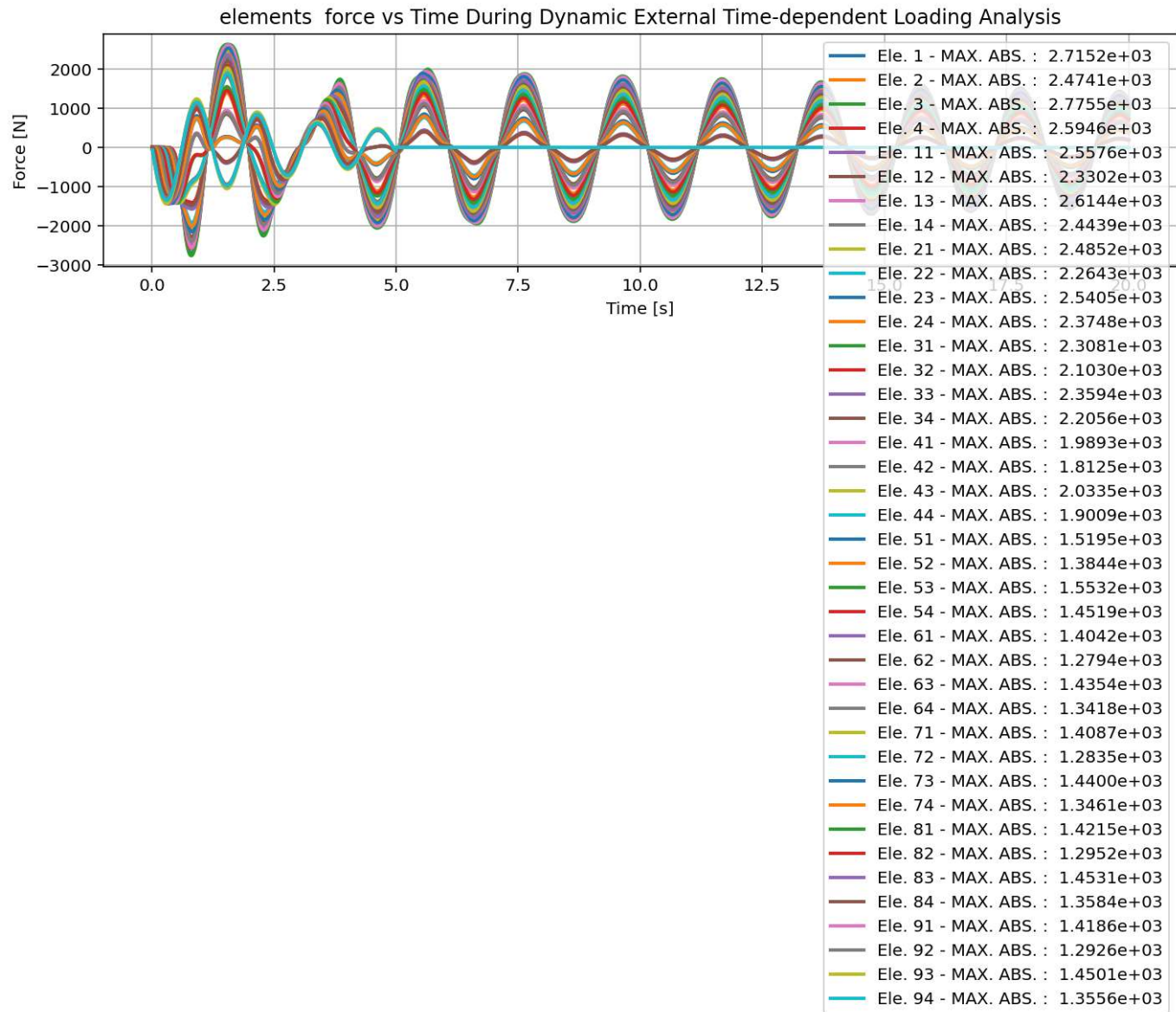


Velocity [m/s]

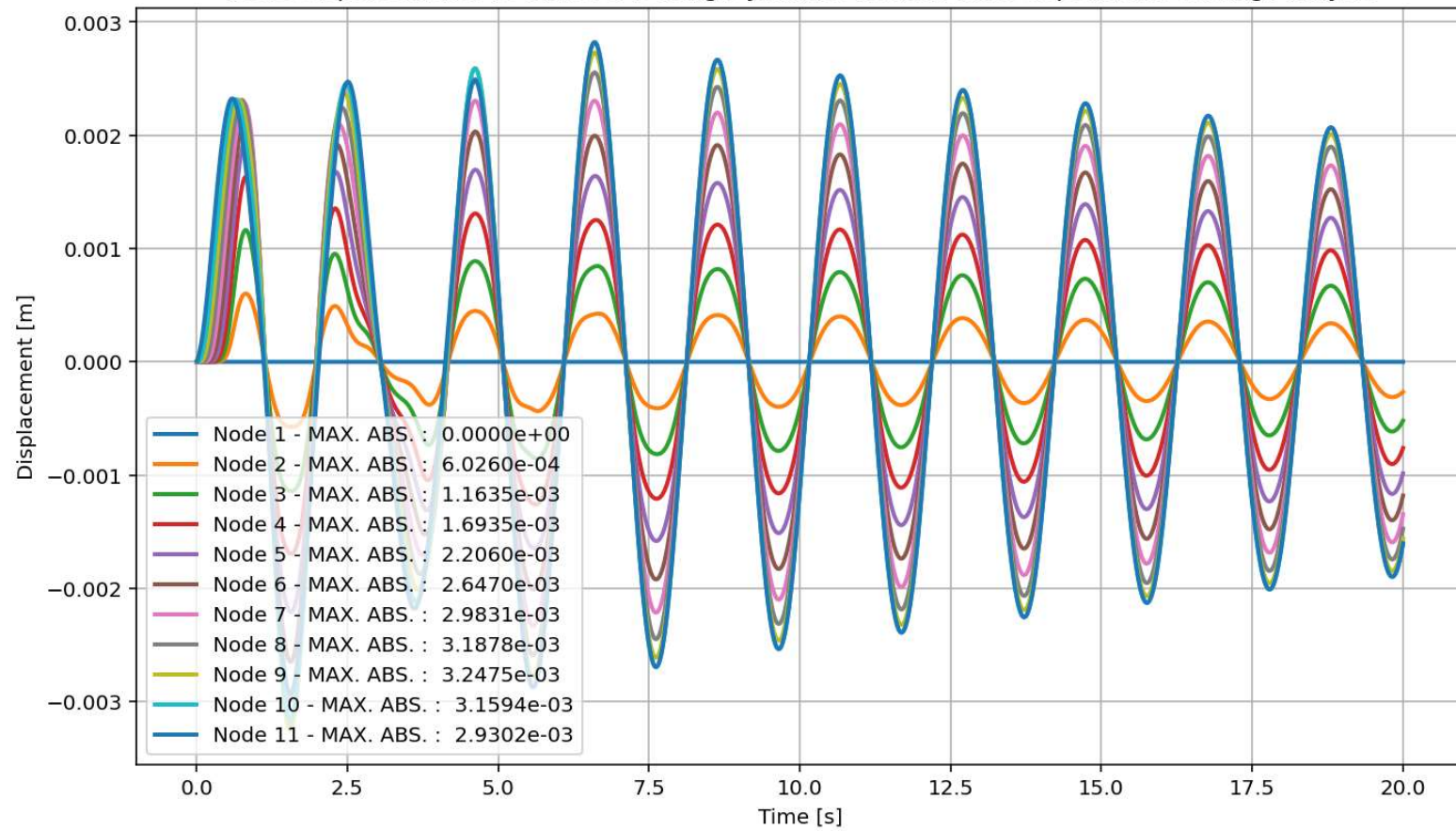
Acceleration [m/s²]

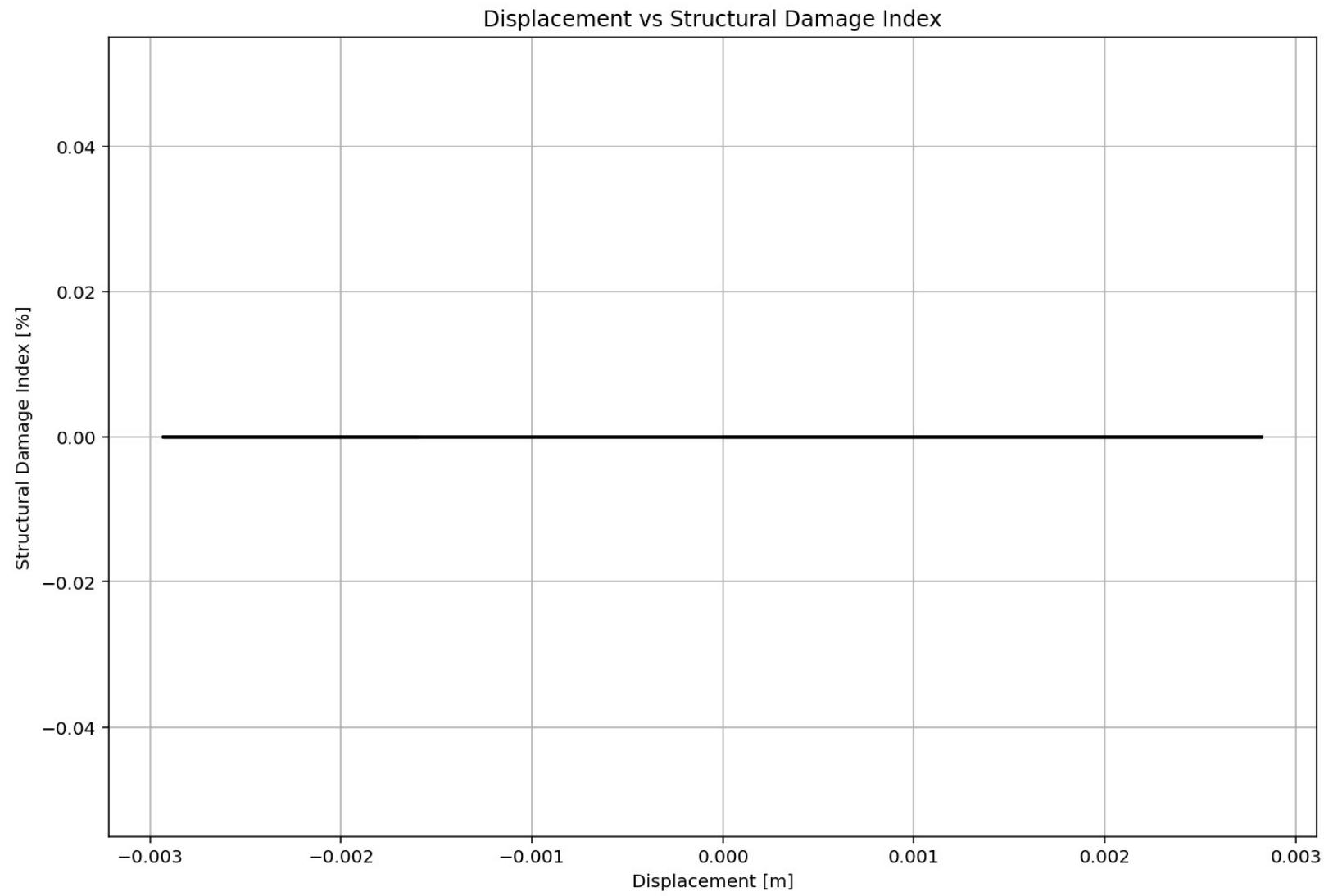
Period of Structure During Dynamic External Time-dependent Loading Analysis



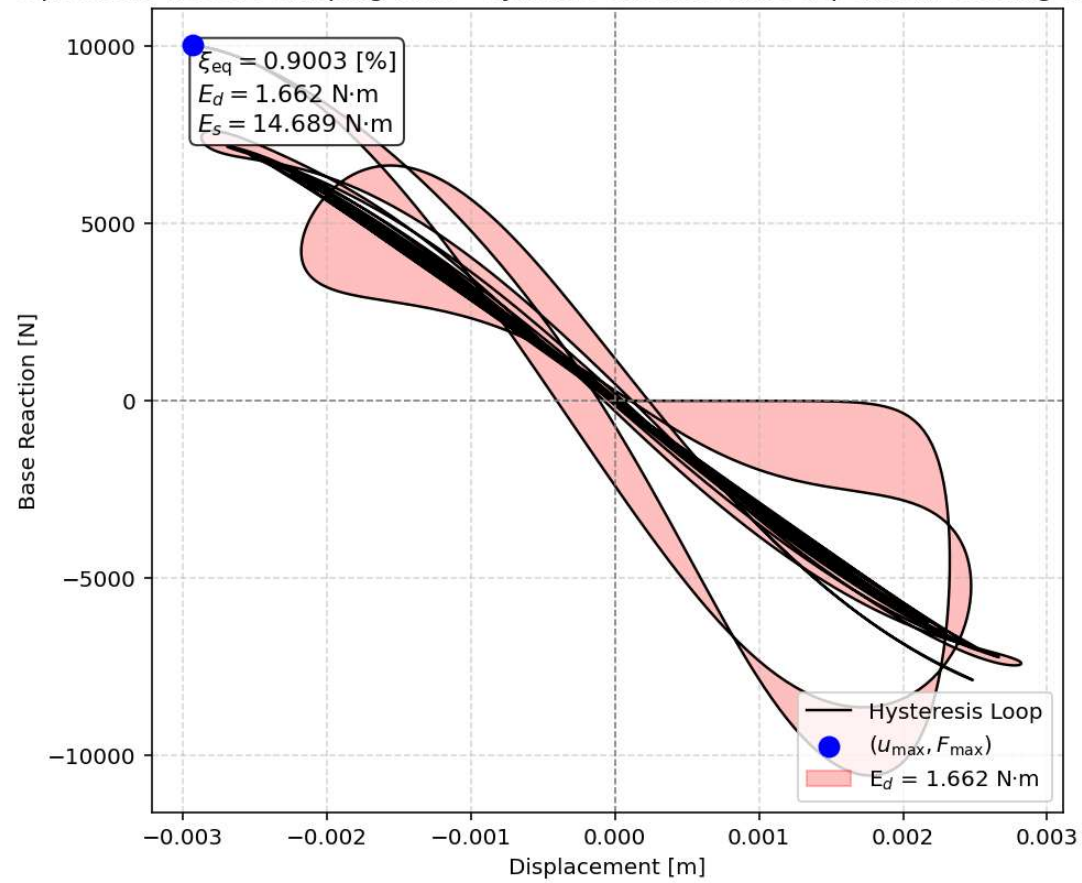


Node Displacements vs Time for During Dynamic External Time-dependent Loading Analysis

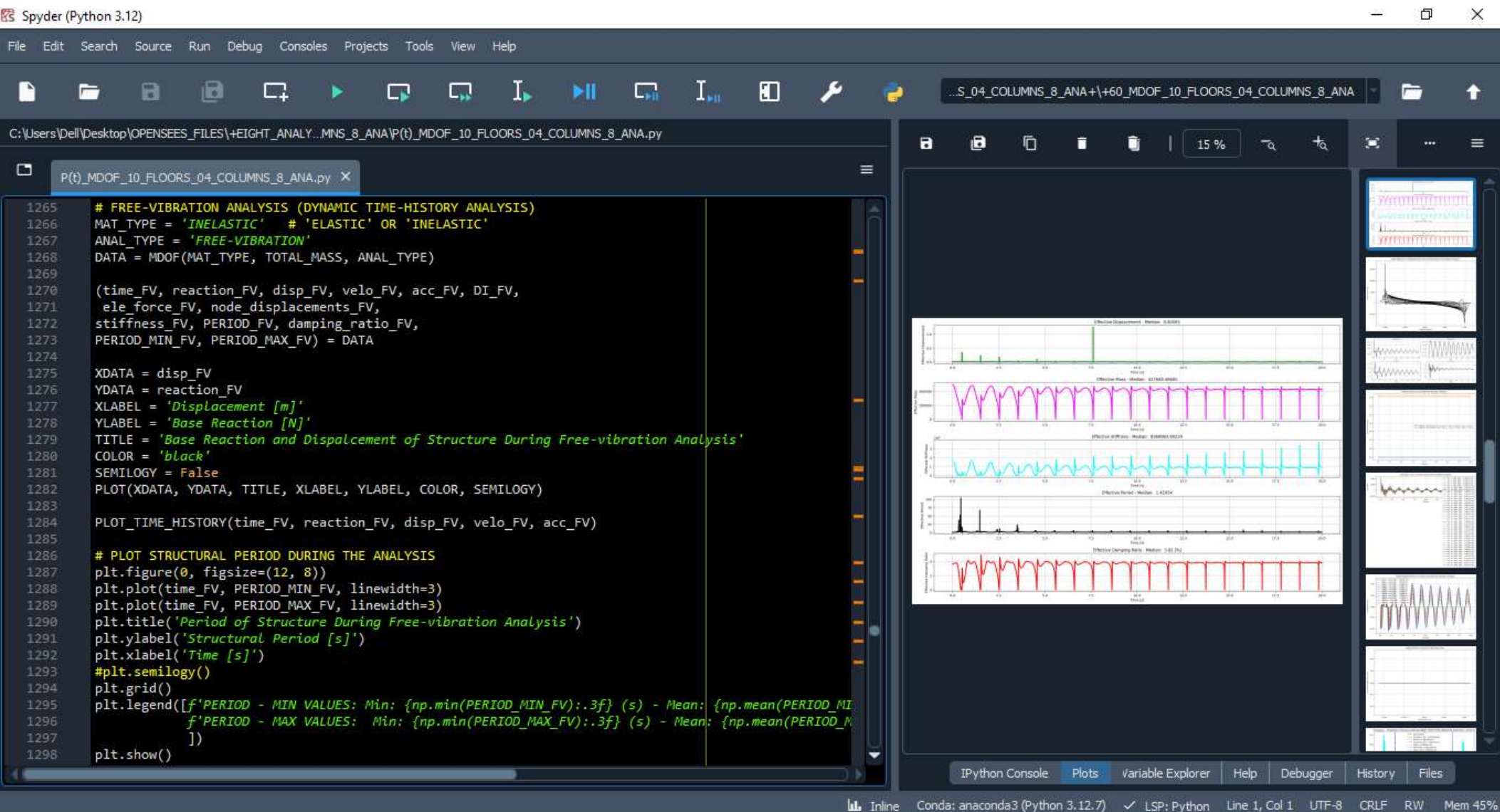




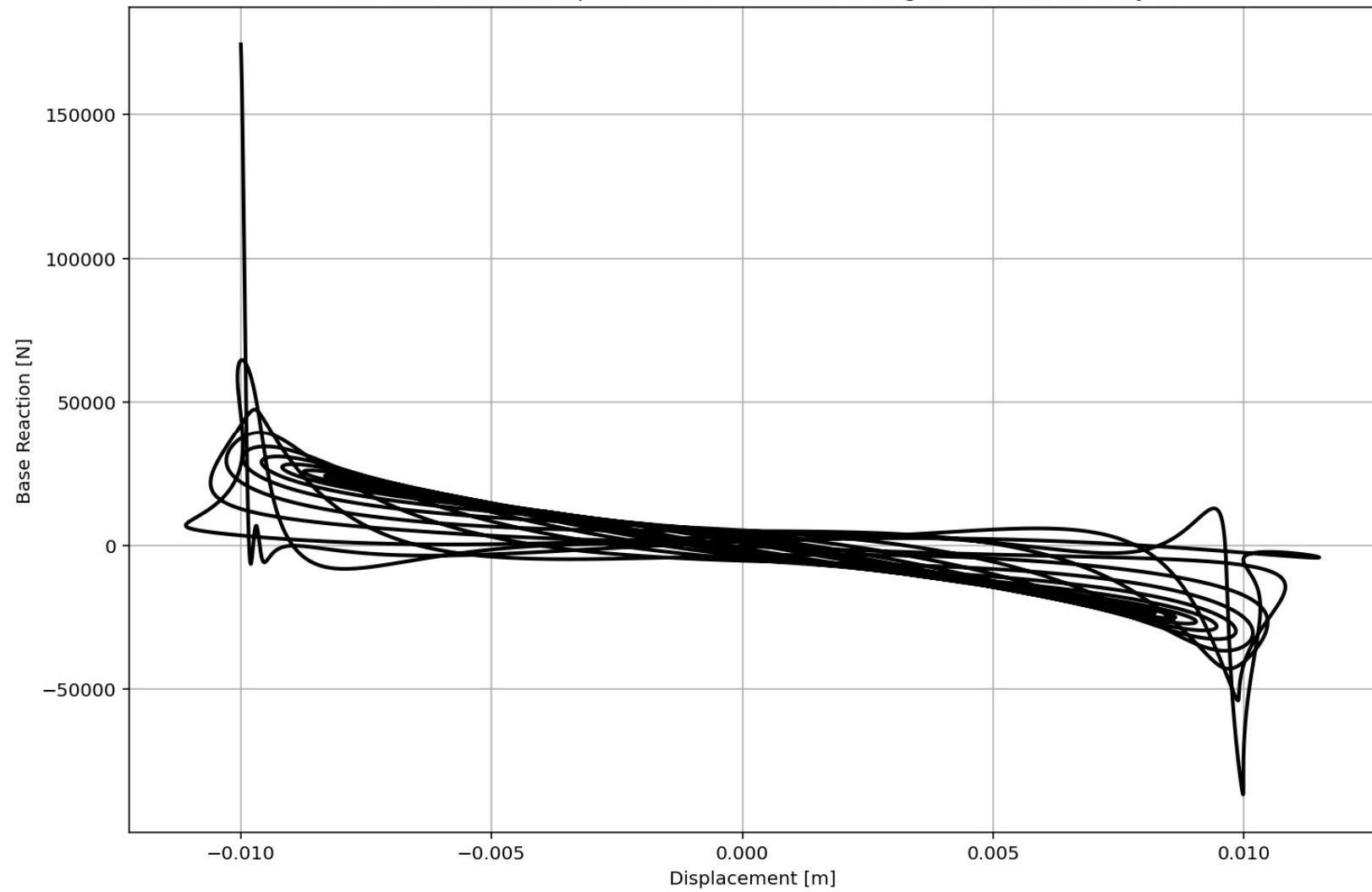
Equivalent viscous damping ratio - Dynamic External Time-dependent Loading Hysteresis

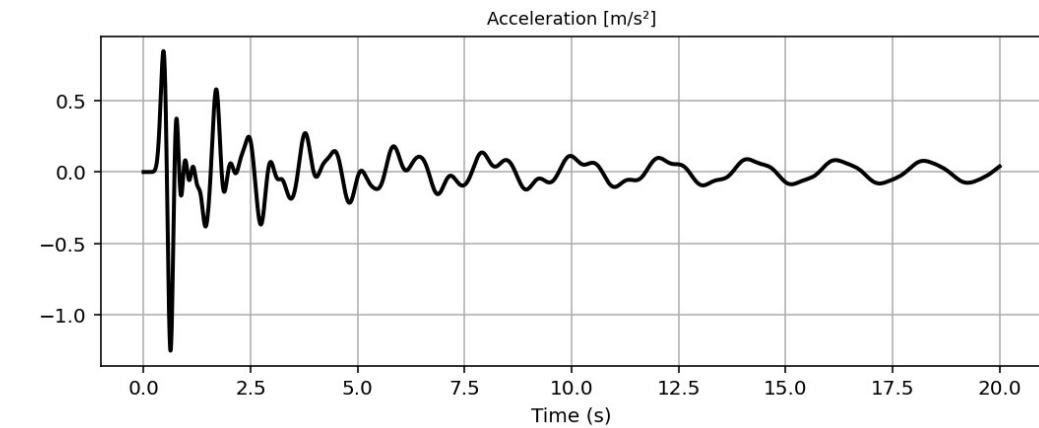
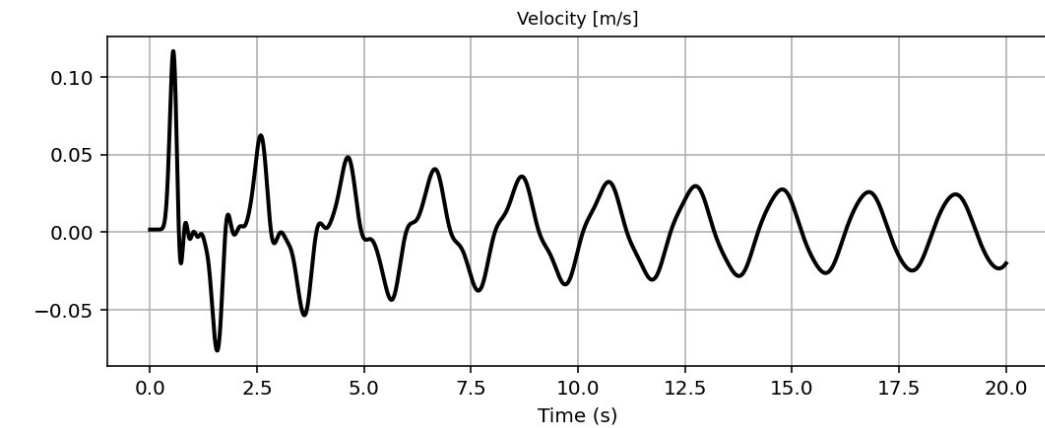
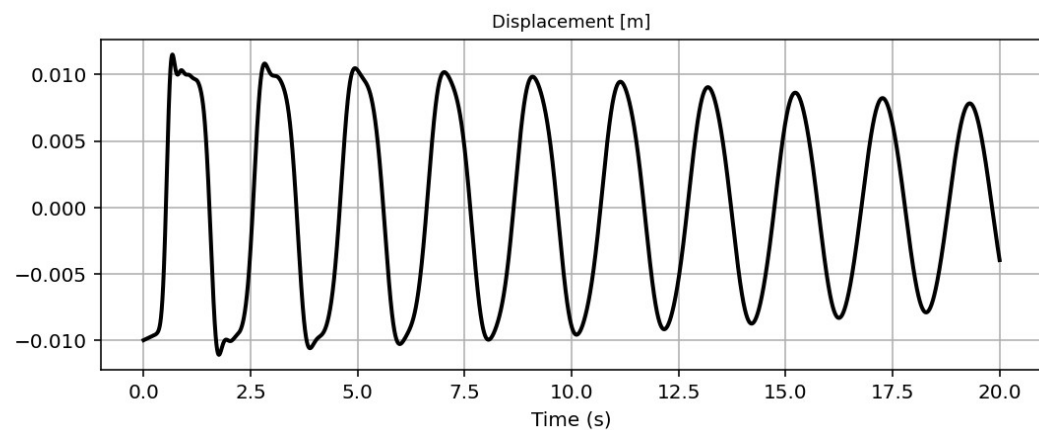
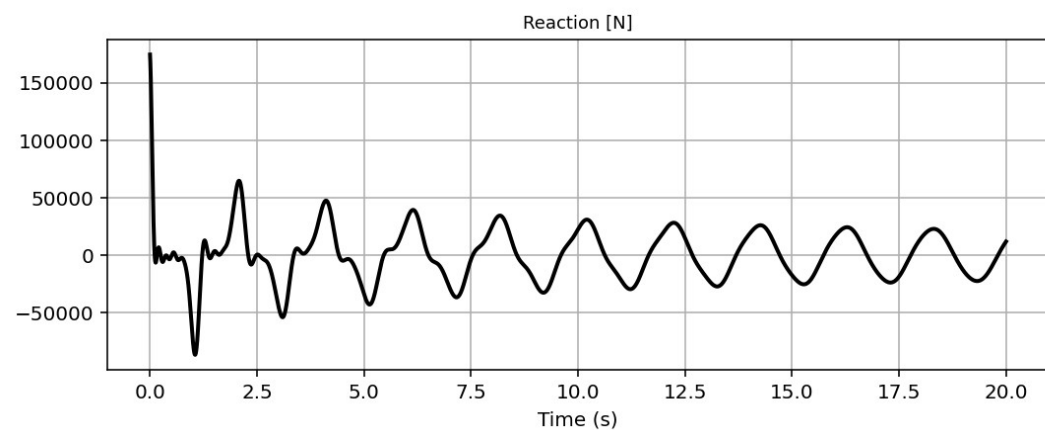


FREE-VIBRATION ANALYSIS

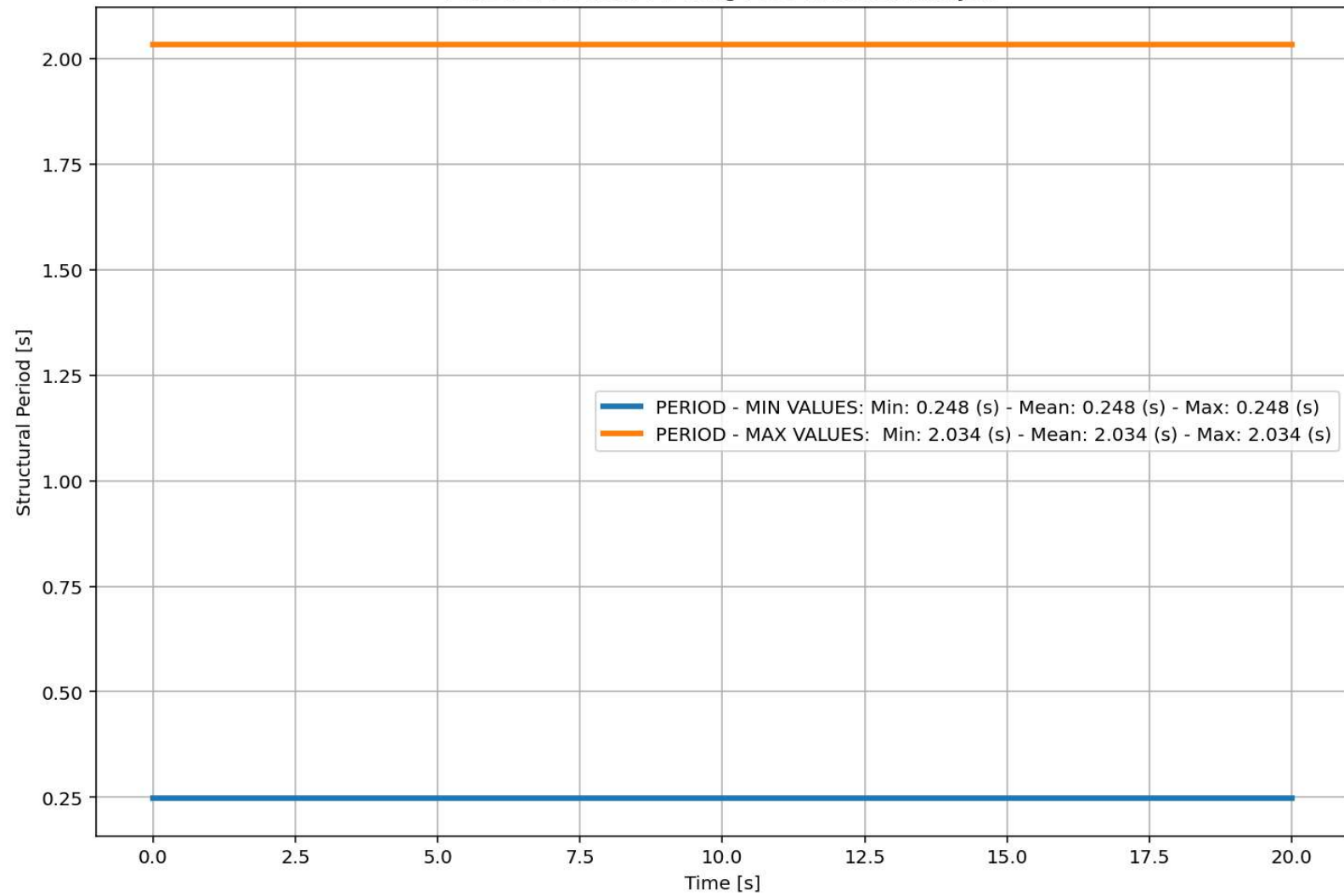


Base Reaction and Dispalcement of Structure During Free-vibration Analysis

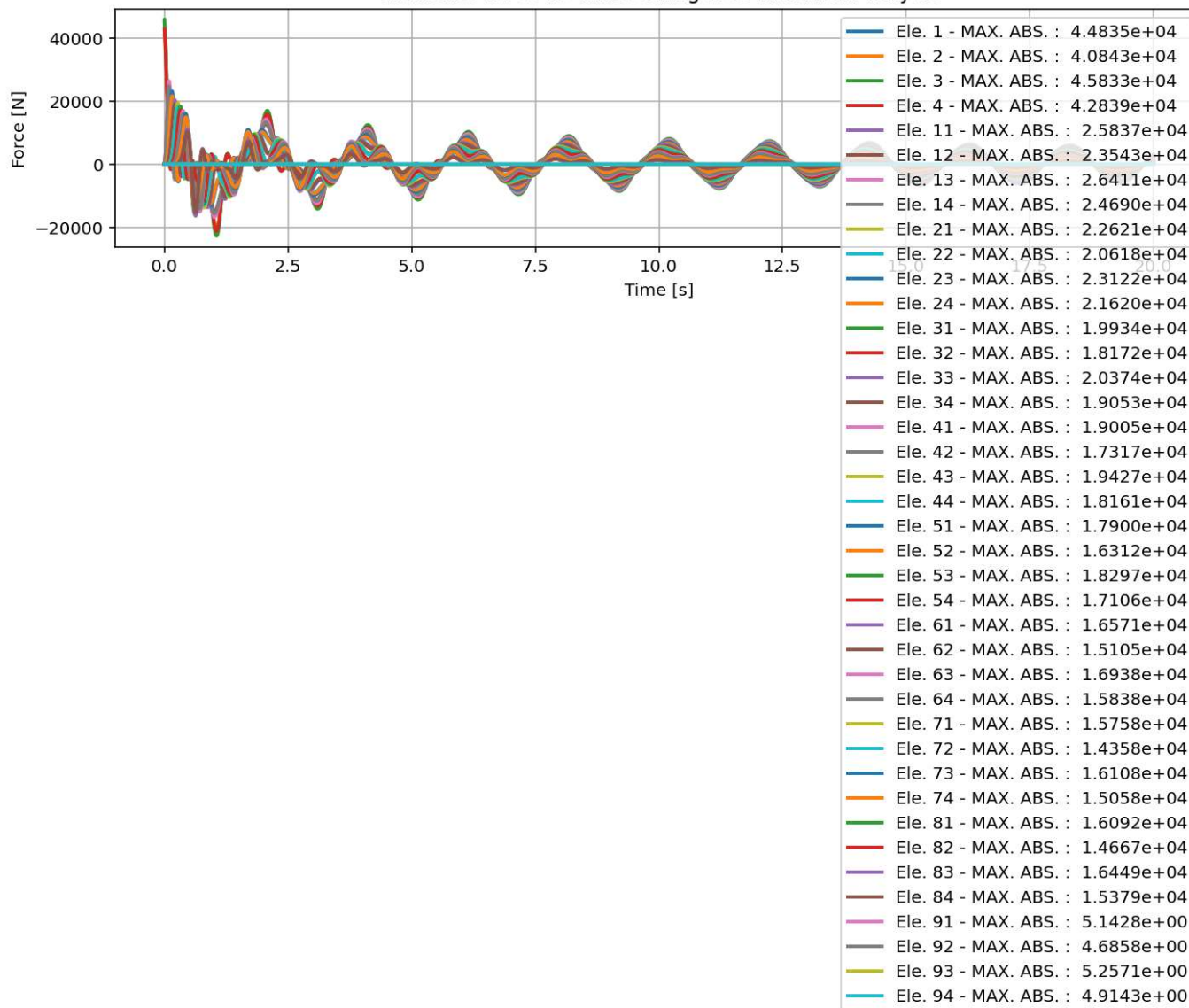




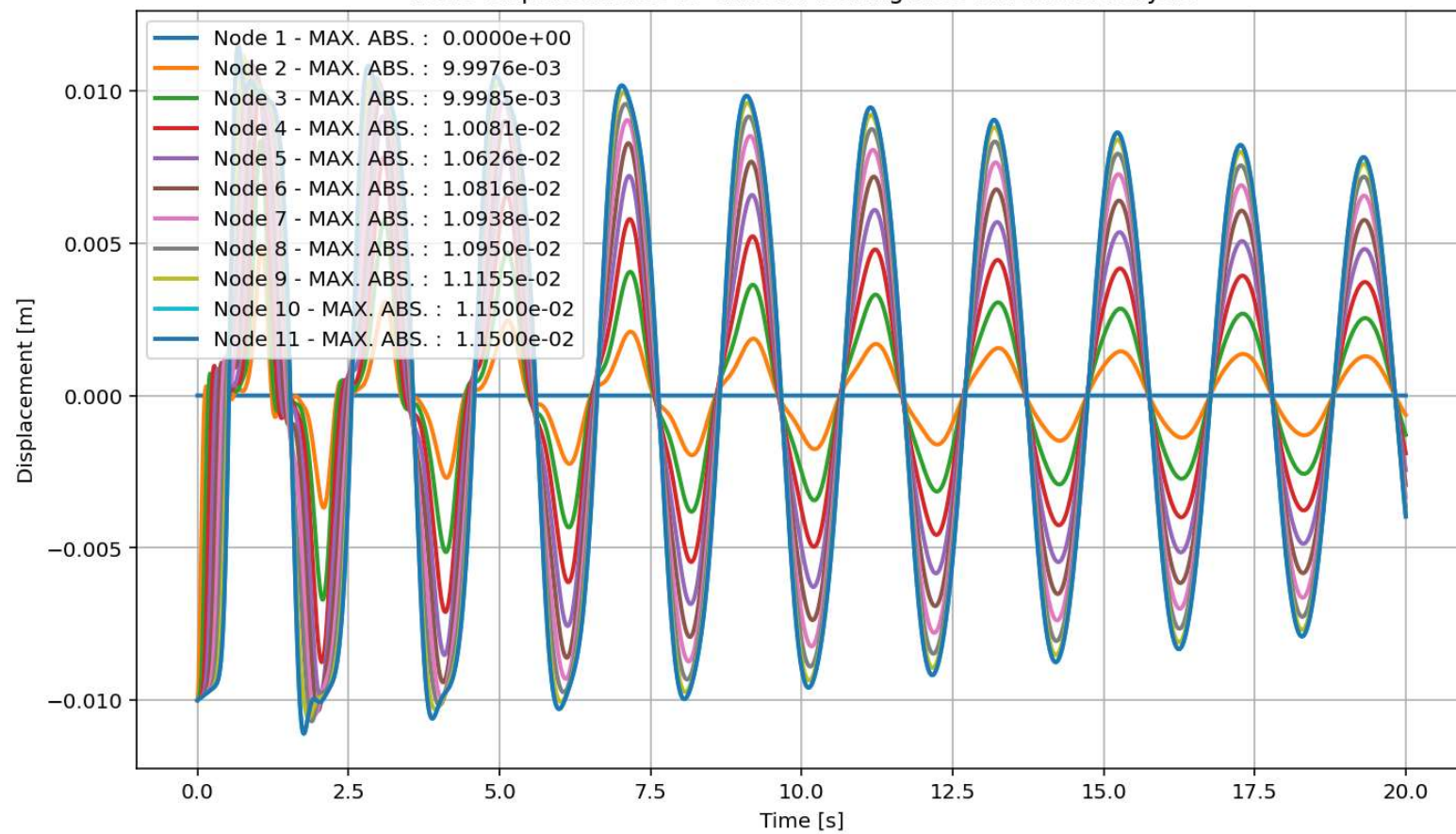
Period of Structure During Free-vibration Analysis

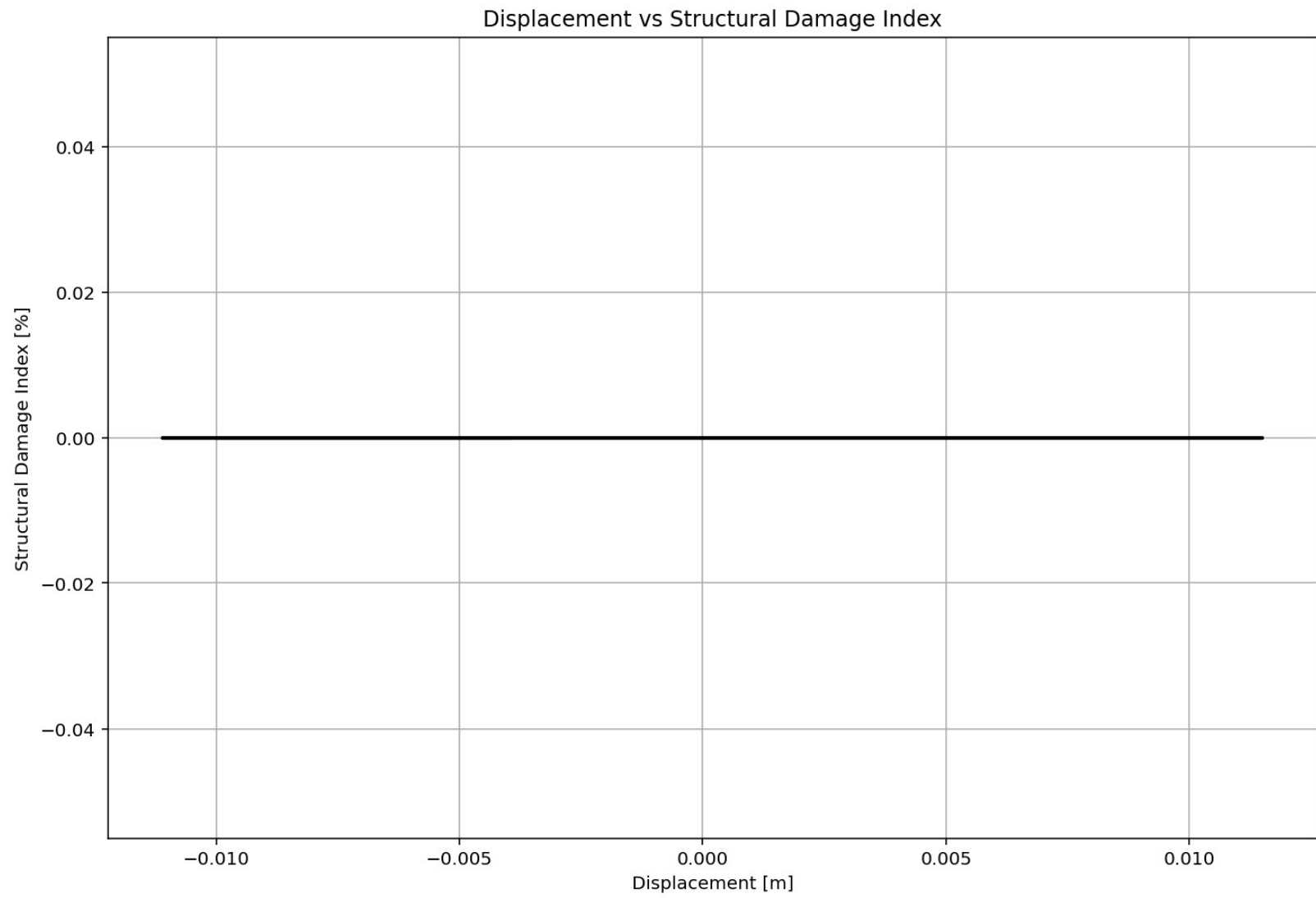


elements force vs Time During Free-vibration Analysis



Node Displacements vs Time for During Free-vibration Analysis





Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...OMPUTE_EFFECTIVE_PROPERTIES_FUN_FREE_VIBRATION.py

P(t)_MDOF_10_FLOORS_04_COLUMNS_8_ANA.py X COMPUTE_EFFECTIVE_..._FREE_VIBRATION.py X

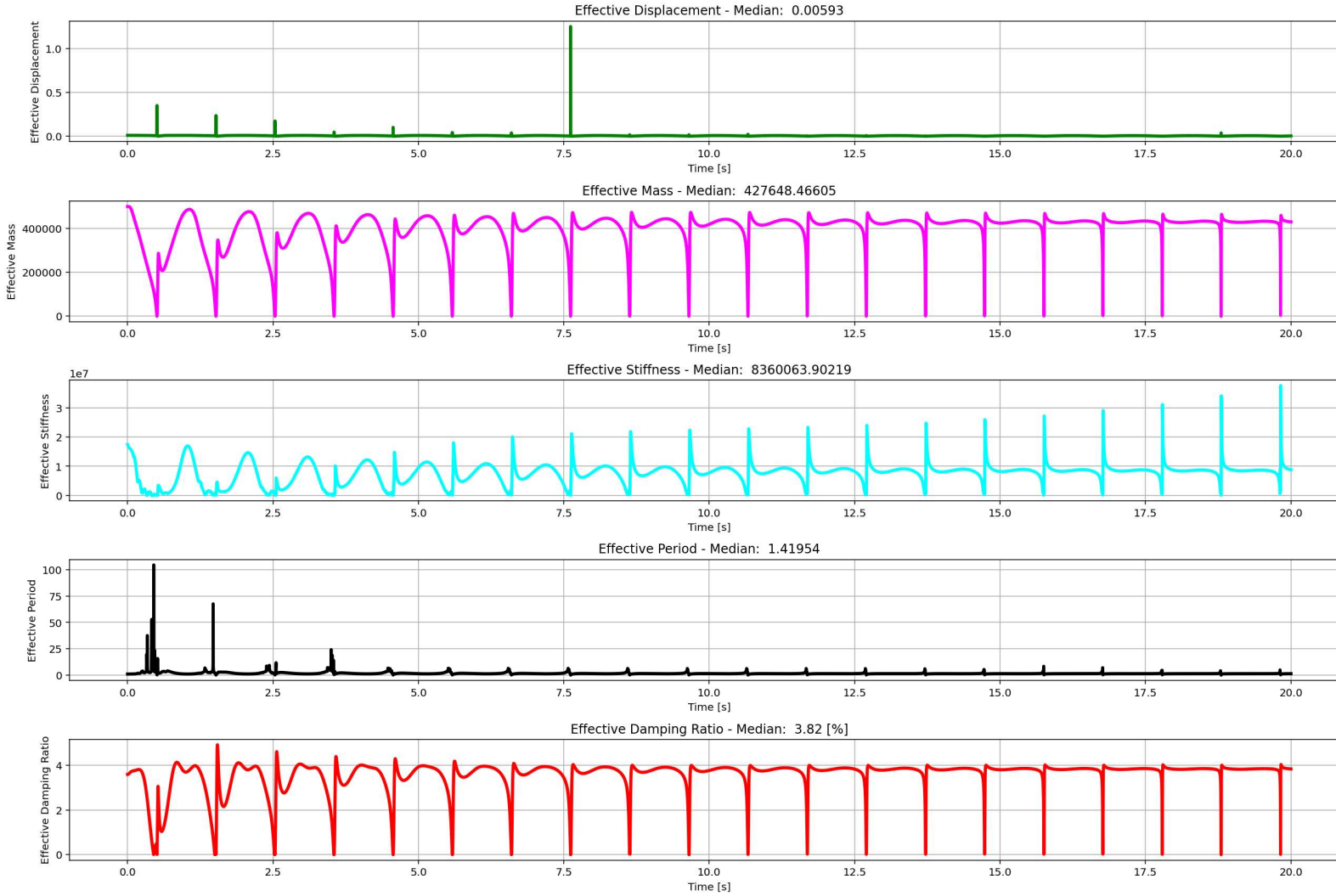
```
1  """ EQUIVALENT SDOF SYSTEM DERIVATION VIA DISPLACEMENT-BASED SEISMIC DESIGN PROCEDURE WITH PUSHOVER
2  # Change MDOF to SDOF System with Displacement Based Design Concept
3  # THIS PYTHON SCRIPT IS WRITTEN BY SALAR DELAVAR GHASHGHAEE (QASHQAI)
4  """
5  This script implements a displacement-based pushover transformation,
6  converting a multi-degree-of-freedom (MDOF) system into an equivalent
7  single-degree-of-freedom (SDOF) system for seismic assessment.
8  It calculates effective modal properties-displacement, mass, and
9  stiffness-by weighting element forces and nodal displacements according
10 to a presumed deformed shape.
11 The derived effective period provides a simplified dynamic characteristic
12 for performance-based engineering. The visualizations effectively track the
13 evolution of these equivalent parameters throughout the nonlinear analysis steps.
14 """
15 #-----
16 import numpy as np
17
18 displacement_X_1 = np.array(list(node_displacements.values())[1]) # DOF 02
19 displacement_X_2 = np.array(list(node_displacements.values())[2]) # DOF 03
20 displacement_X_3 = np.array(list(node_displacements.values())[3]) # DOF 04
21 displacement_X_4 = np.array(list(node_displacements.values())[4]) # DOF 05
22 displacement_X_5 = np.array(list(node_displacements.values())[5]) # DOF 06
23 displacement_X_6 = np.array(list(node_displacements.values())[6]) # DOF 07
24 displacement_X_7 = np.array(list(node_displacements.values())[7]) # DOF 08
25 displacement_X_8 = np.array(list(node_displacements.values())[8]) # DOF 09
26 displacement_X_9 = np.array(list(node_displacements.values())[9]) # DOF 10
27 displacement_X_10 = np.array(list(node_displacements.values())[10]) # DOF 11
28
29 ele_force_01 = np.array(list(ele_force.values())[0]) # ELEMENT 01
30 ele_force_02 = np.array(list(ele_force.values())[1]) # ELEMENT 02
31 ele_force_03 = np.array(list(ele_force.values())[2]) # ELEMENT 03
32 ele_force_04 = np.array(list(ele_force.values())[3]) # ELEMENT 04
33 ele_force_05 = np.array(list(ele_force.values())[4]) # ELEMENT 05
34 ele_force_06 = np.array(list(ele_force.values())[5]) # ELEMENT 06
```

...S_04_COLUMNS_8_ANA+)+60_MDOF_10_FLOORS_04_COLUMNS_8_ANA

15 %

IPython Console Plots Variable Explorer Help Debugger History Files

Inline Conda: anaconda3 (Python 3.12.7) ✓ LSP: Python Line 1, Col 1 UTF-8 CRLF RW Mem 46%



$$\Delta_d = \frac{\sum_{i=1}^n m_i \Delta_i^2}{\sum_{i=1}^n m_i \Delta_i}$$

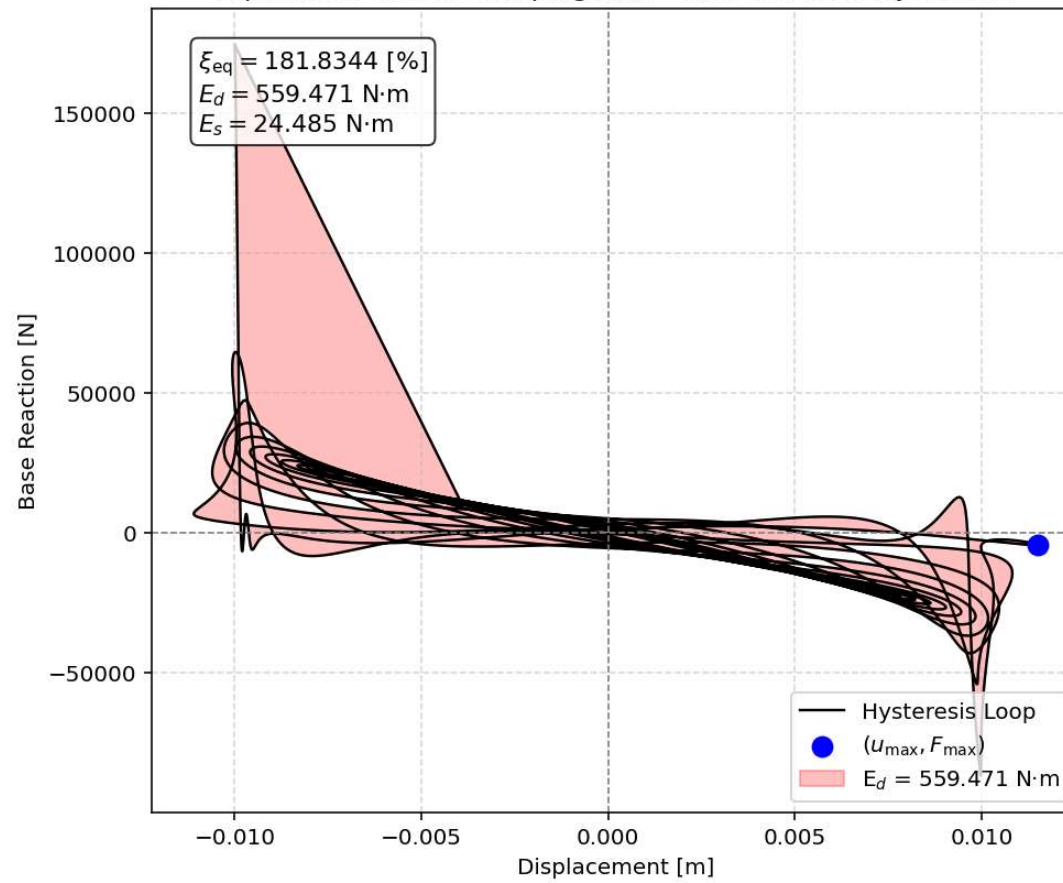
$$m_{eff} = \frac{\sum_{i=1}^n m_i \Delta_i}{\Delta_d}$$

$$k_{eff} = \frac{\sum_{i=1}^n k_i \Delta_i}{\Delta_d}$$

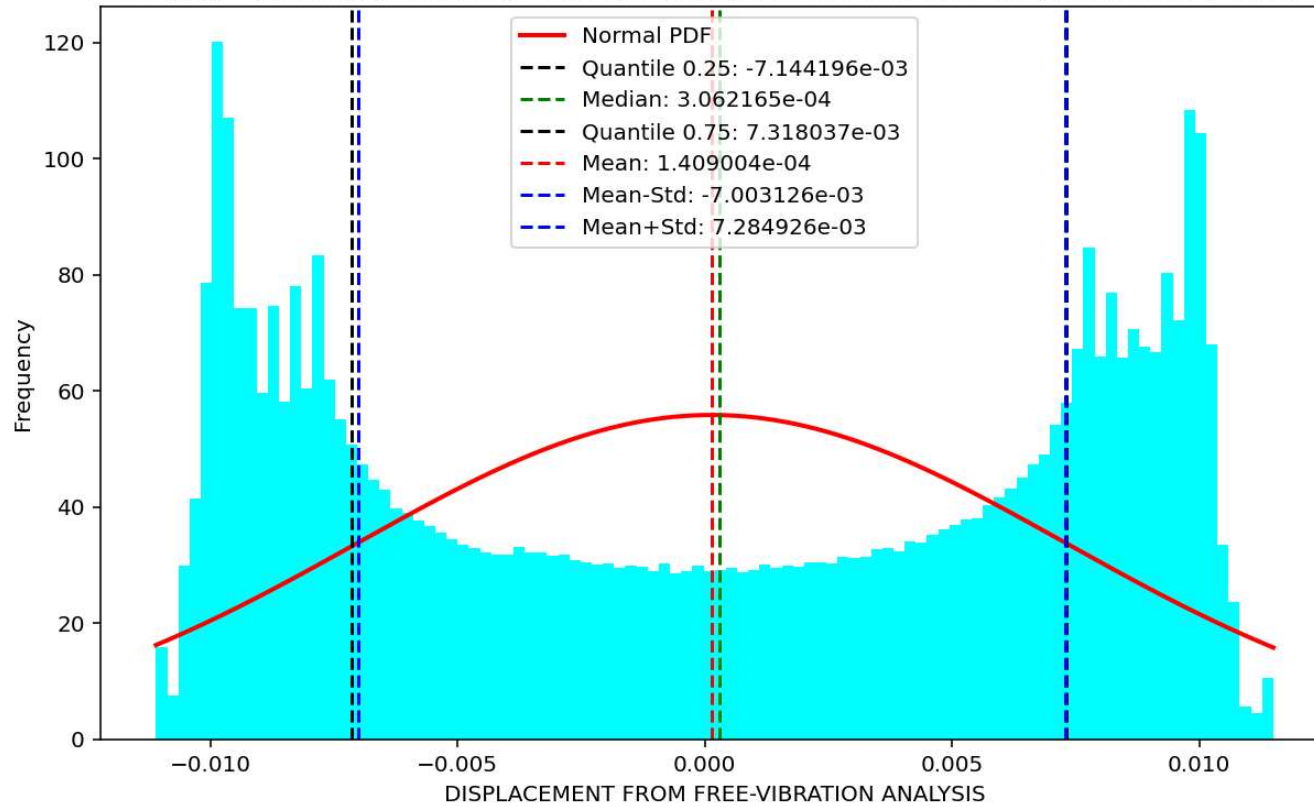
$$T_{eff} = \frac{2\pi}{\sqrt{\frac{k_{eff}}{m_{eff}}}}$$

$$\xi_{eff} = \frac{\sum_{i=1}^n \xi_i \Delta_i}{\Delta_d}$$

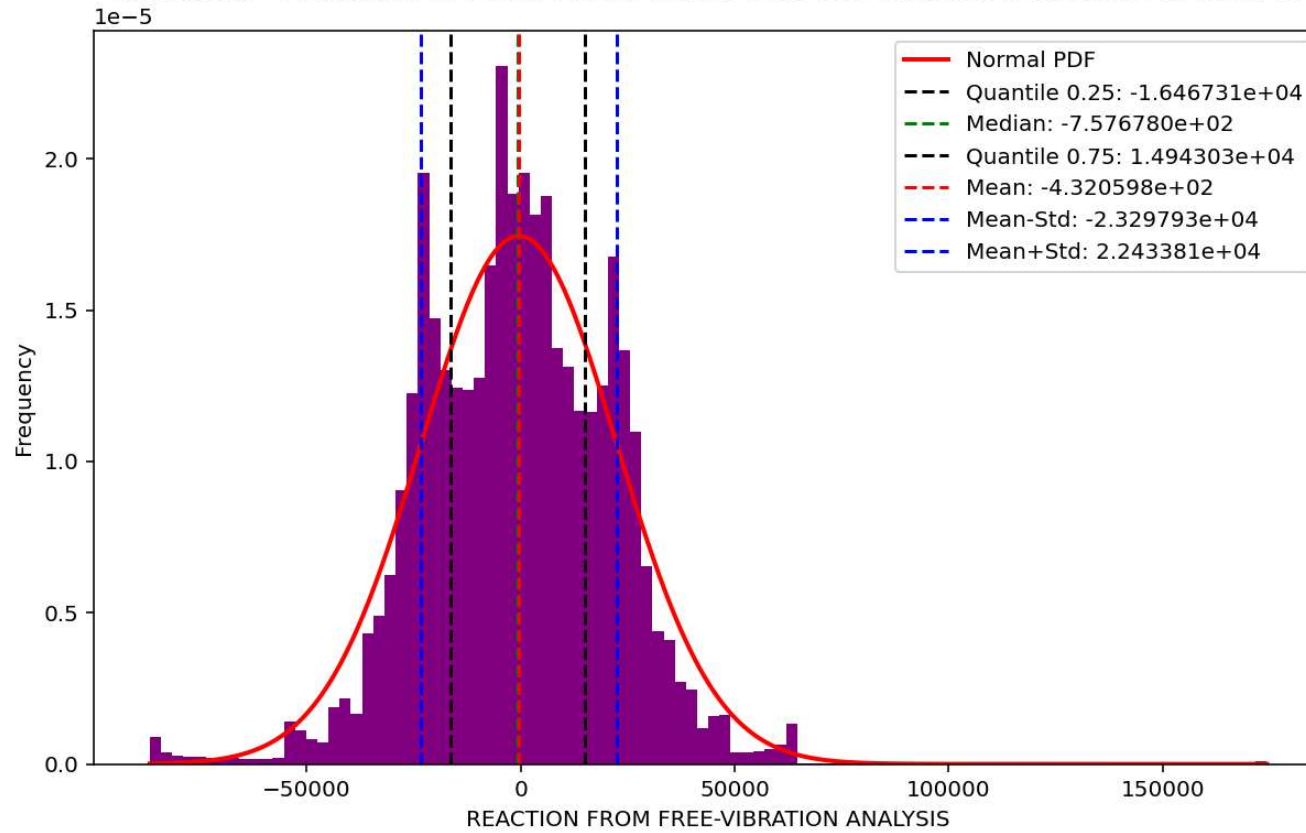
Equivalent viscous damping ratio - Free-vibration Hysteresis



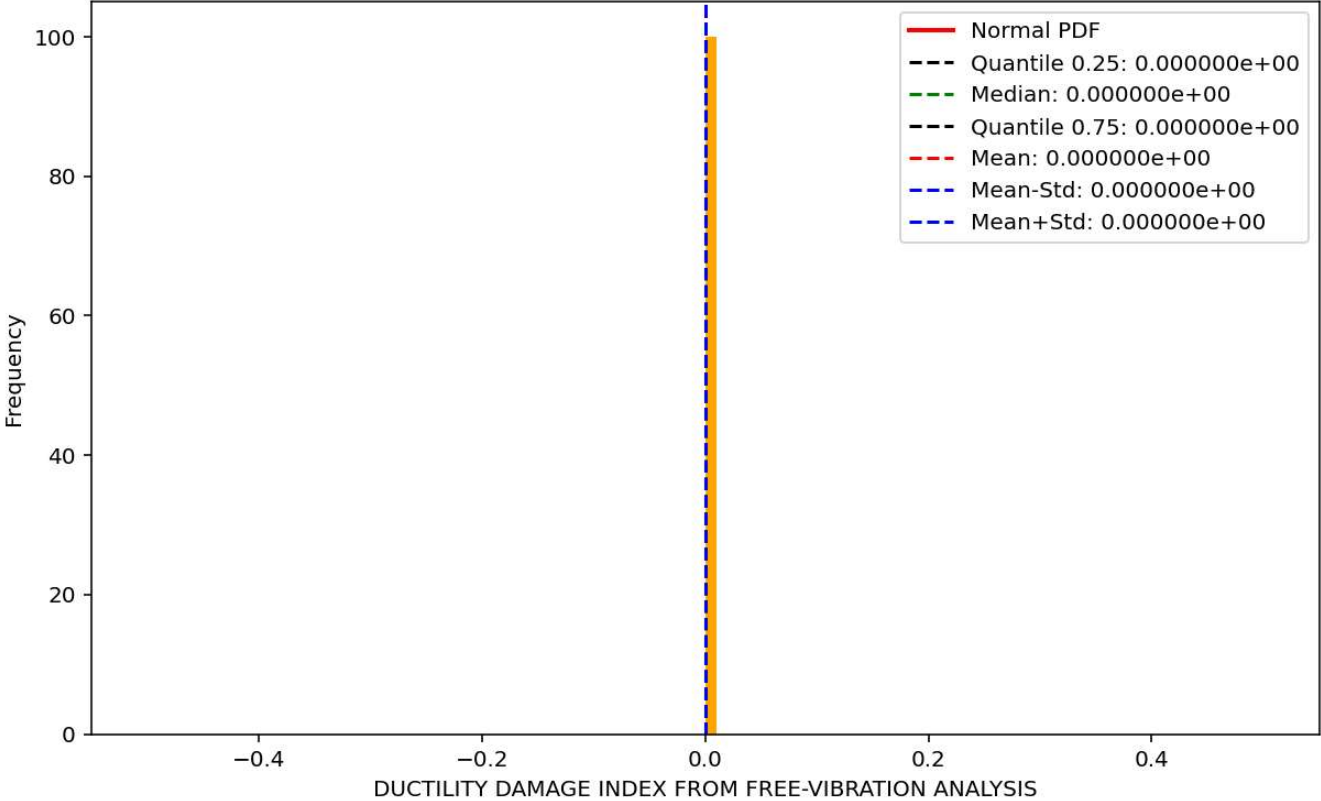
Histogram - Probability of Positive DISPLACEMENT FROM FREE-VIBRATION ANALYSIS is 50.90 %



Histogram - Probability of Positive REACTION FROM FREE-VIBRATION ANALYSIS is 48.62 %



Histogram - Probability of Positive DUCTILITY DAMAGE INDEX FROM FREE-VIBRATION ANALYSIS is 0.00 %



SEISMIC ANALYSIS

Spyder (Python 3.12)

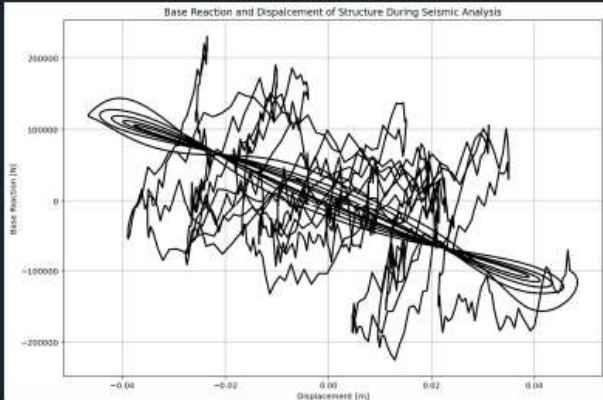
File Edit Search Source Run Debug Consoles Projects Tools View Help

C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...MNS_8_ANA\p(t)_MDOF_10_FLOORS_04_COLUMNS_8_ANA.py

P(t)_MDOF_10_FLOORS_04_COLUMNS_8_ANA.py COMPUTE_EFFECTIVE_..._FREE_VIBRATION.py

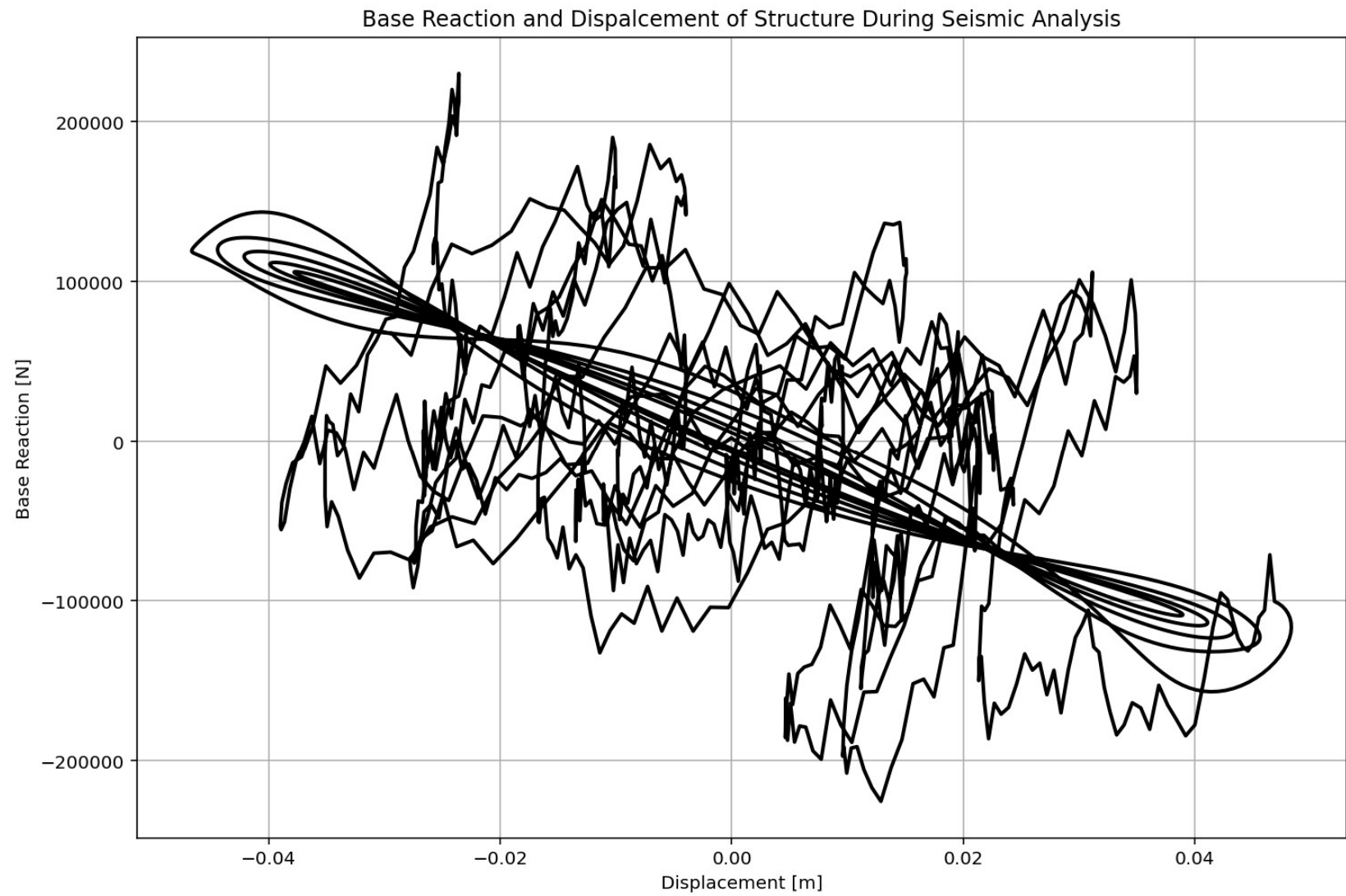
```
1330 # SEISMIC ANALYSIS (DYNAMIC TIME-HISTORY ANALYSIS)
1331 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
1332 ANAL_TYPE = 'SEISMIC'
1333
1334 DATA = MDOF(MAT_TYPE, TOTAL_MASS, ANAL_TYPE)
1335
1336 (time_SEI, reaction_SEI, disp_SEI, velo_SEI, acc_SEI, DI_SEI,
1337  ele_force_SEI, node_displacements_SEI,
1338  stiffness_SEI, PERIOD_SEI, damping_ratio_SEI,
1339  PERIOD_MIN_SEI, PERIOD_MAX_SEI) = DATA
1340
1341
1342 XDATA = disp_SEI
1343 YDATA = reaction_SEI
1344 XLABEL = 'Displacement [m]'
1345 YLABEL = 'Base Reaction [N]'
1346 TITLE = 'Base Reaction and Displacment of Structure During Seismic Analysis'
1347 COLOR = 'black'
1348 SEMIOLOGY = False
1349 PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMIOLOGY)
1350
1351 PLOT_TIME_HISTORY(time_SEI, reaction_SEI, disp_SEI, velo_SEI, acc_SEI)
1352
1353 # PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
1354 plt.figure(0, figsize=(12, 8))
1355 plt.plot(time_SEI, PERIOD_MIN_SEI, linewidth=3)
1356 plt.plot(time_SEI, PERIOD_MAX_SEI, linewidth=3)
1357 plt.title('Period of Structure During Seismic Analysis')
1358 plt.ylabel('Structural Period [s]')
1359 plt.xlabel('Time [s]')
1360 #plt.semilogy()
1361 plt.grid()
1362 plt.legend([f'PERIOD - MIN VALUES: Min: {np.min(PERIOD_MIN_SEI):.3f} (s) - Mean: {np.mean(PERIOD_MIN_SEI):.3f} (s)',
1363            f'PERIOD - MAX VALUES: Min: {np.min(PERIOD_MAX_SEI):.3f} (s) - Mean: {np.mean(PERIOD_MAX_SEI):.3f} (s)'])
```

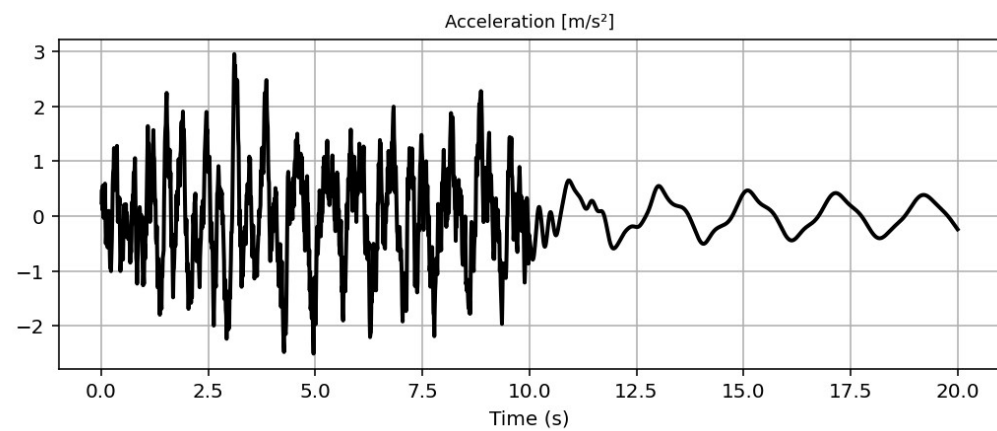
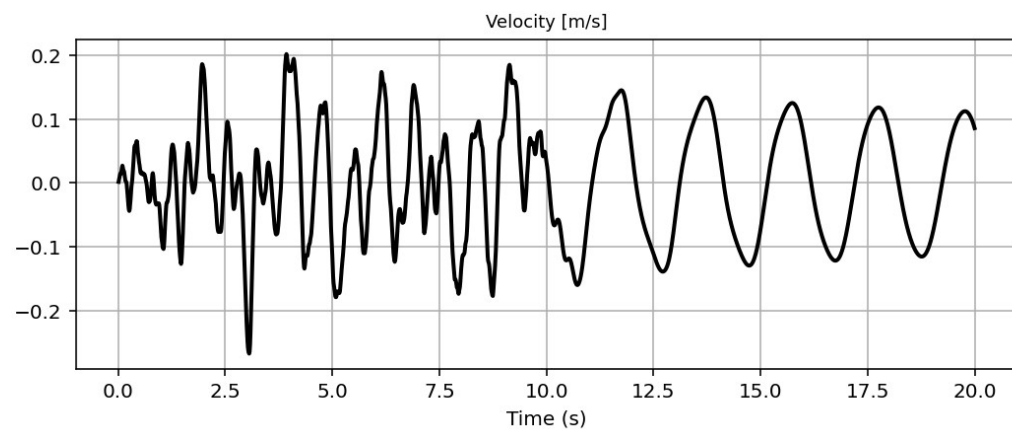
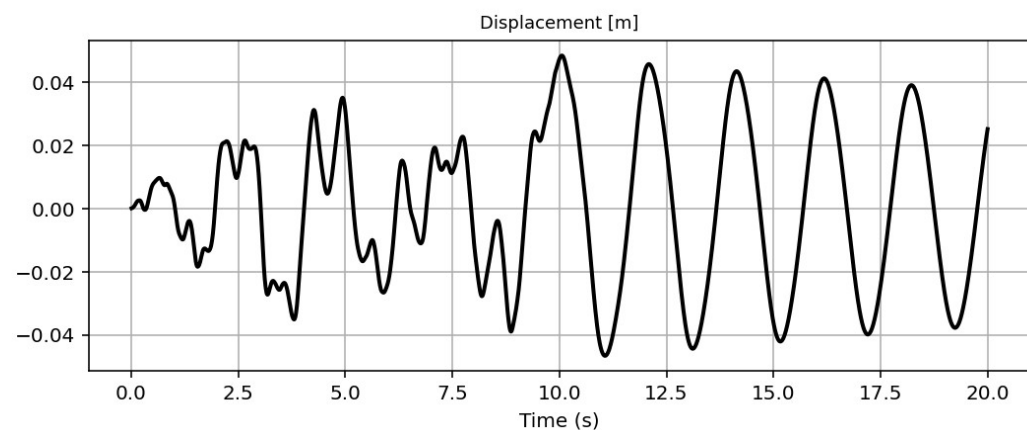
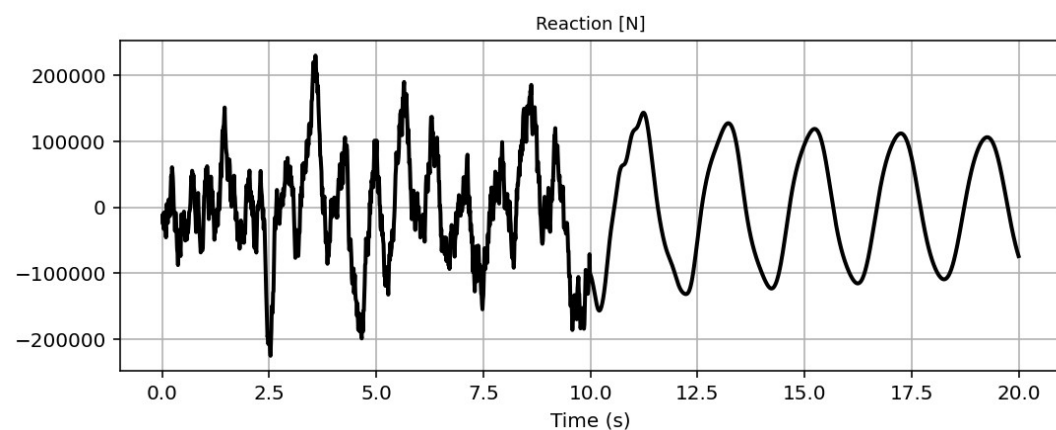
Base Reaction and Displacment of Structure During Seismic Analysis



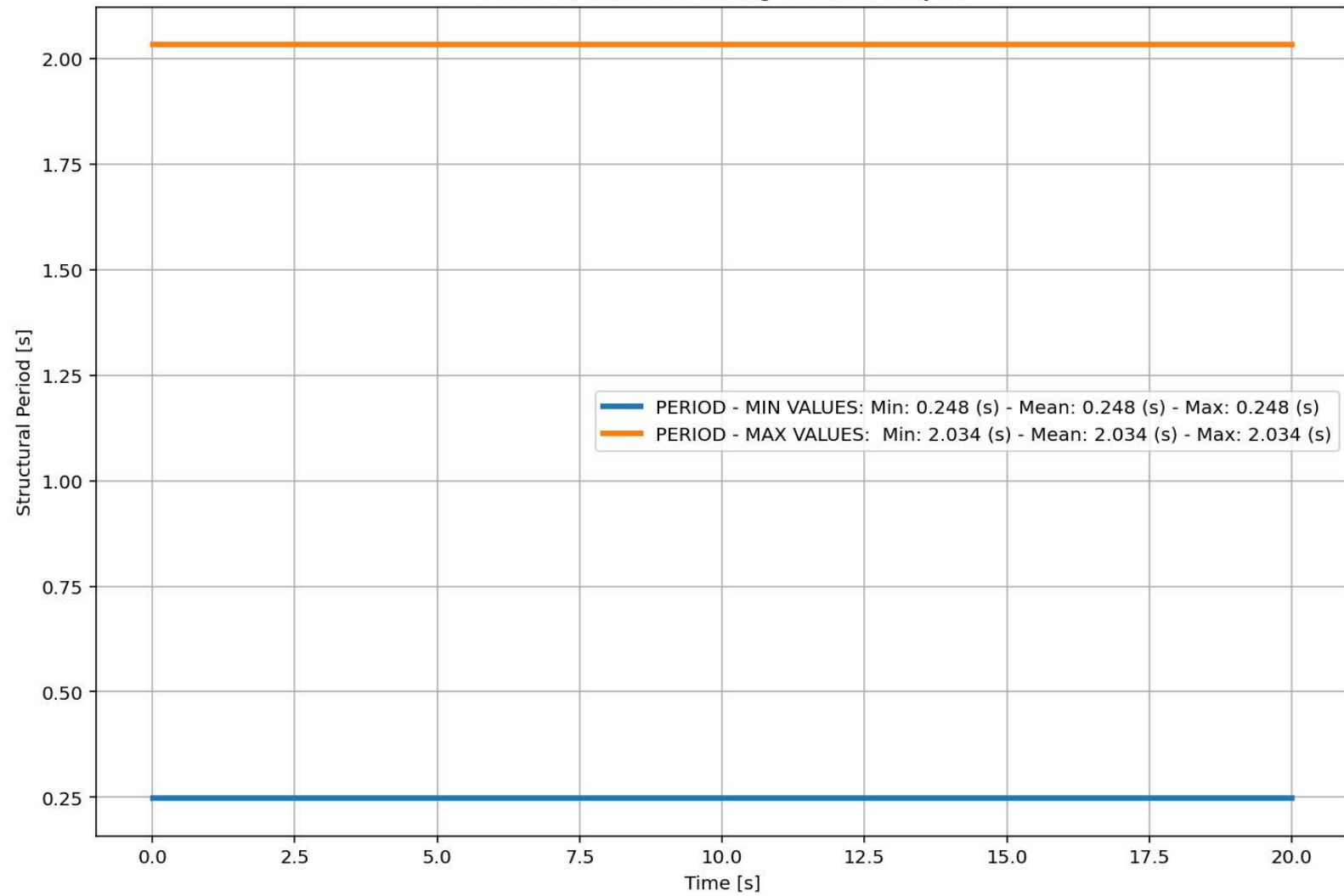
IPython Console Plots Variable Explorer Help Debugger History Files

Inline Conda: anaconda3 (Python 3.12.7) ✓ LSP: Python Line 1, Col 1 UTF-8 CRLF RW Mem 43%

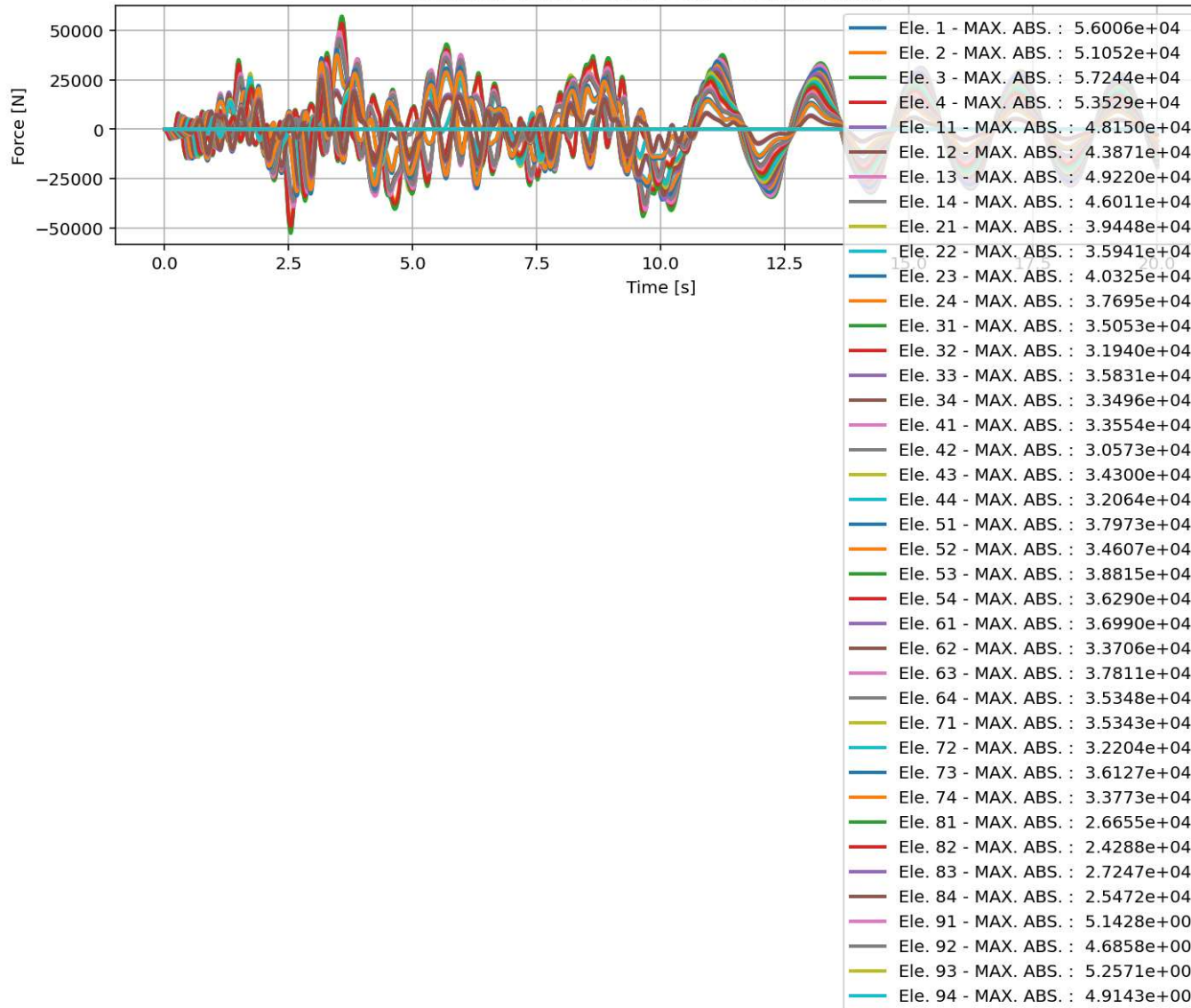




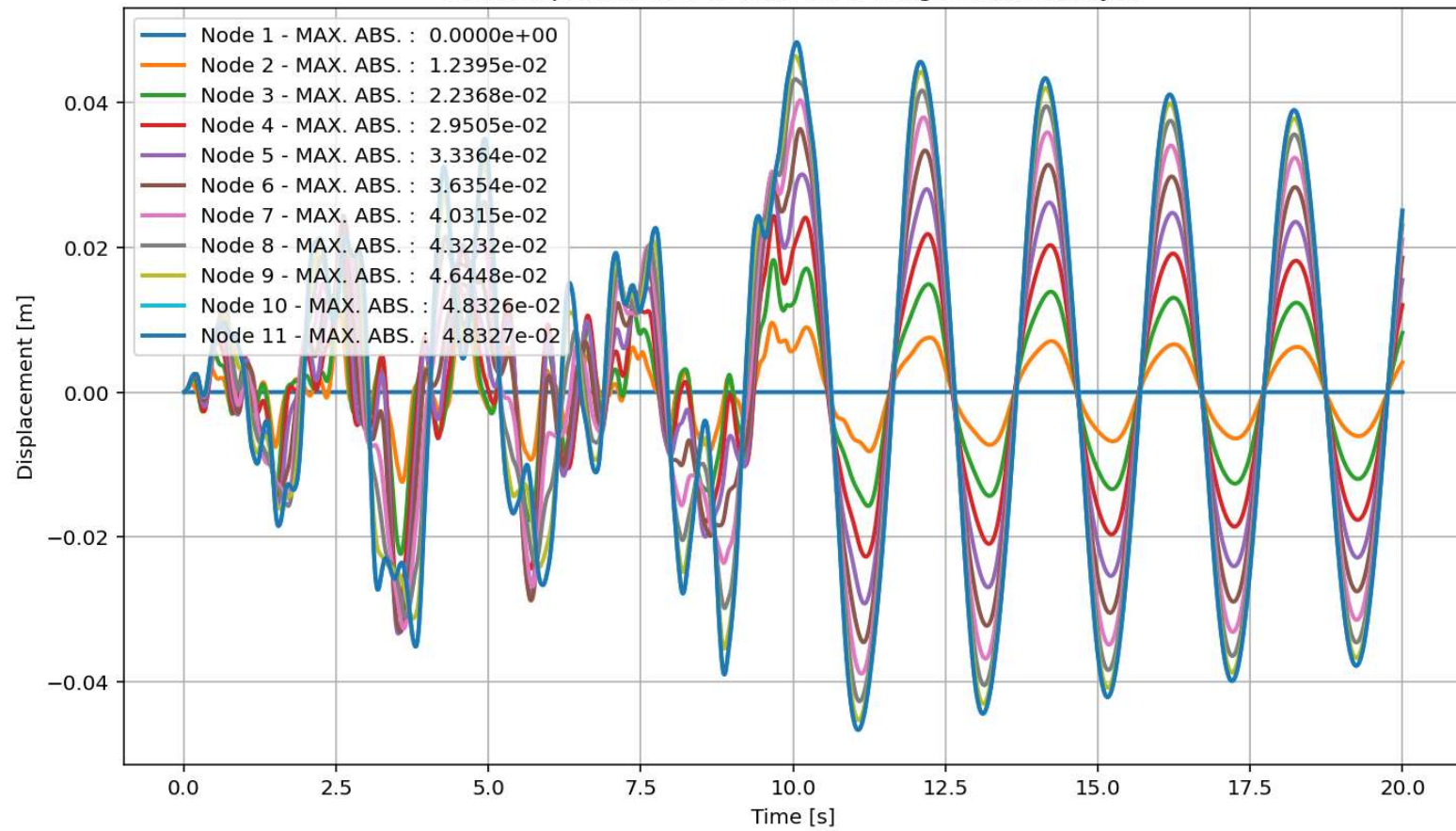
Period of Structure During Seismic Analysis

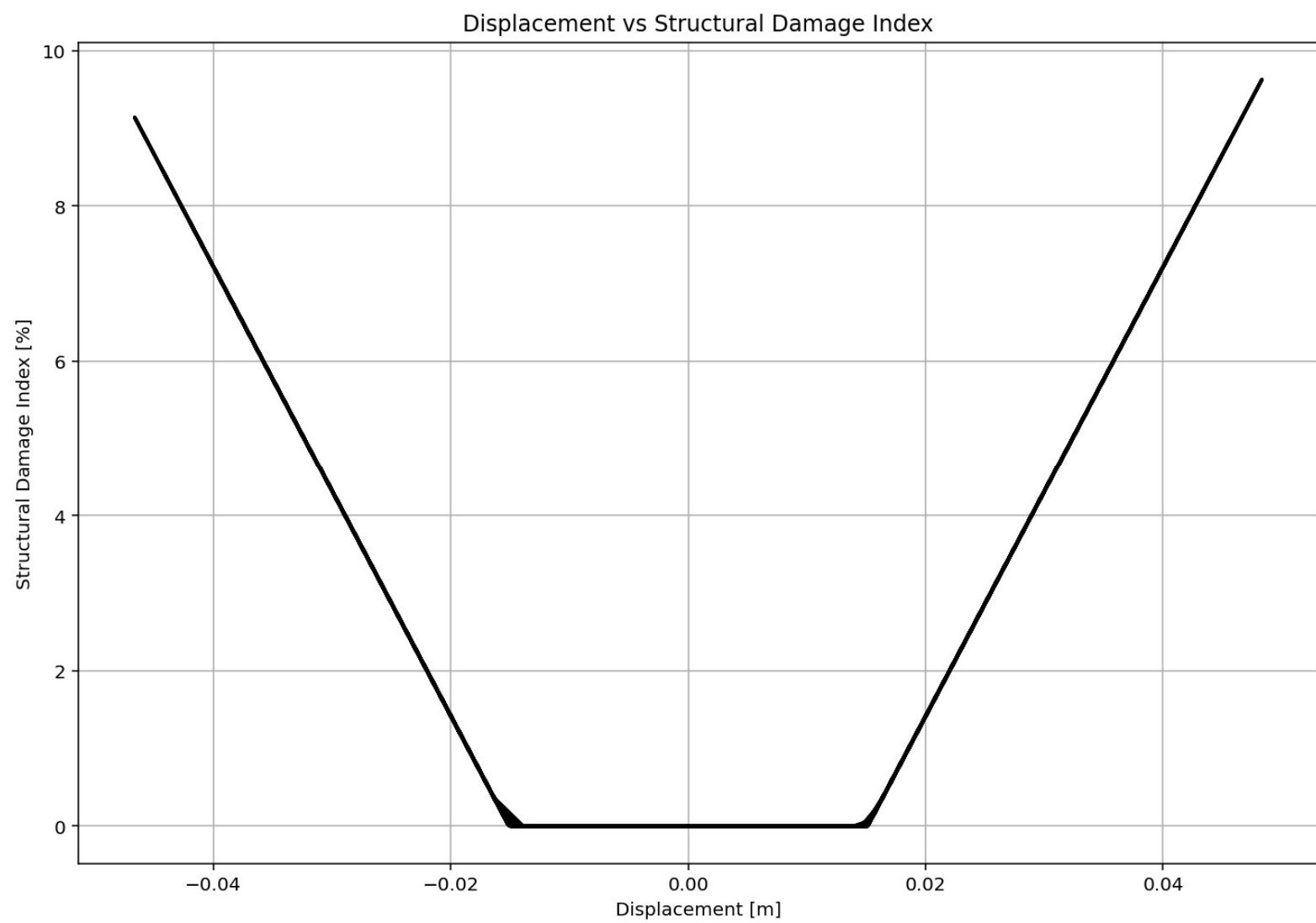


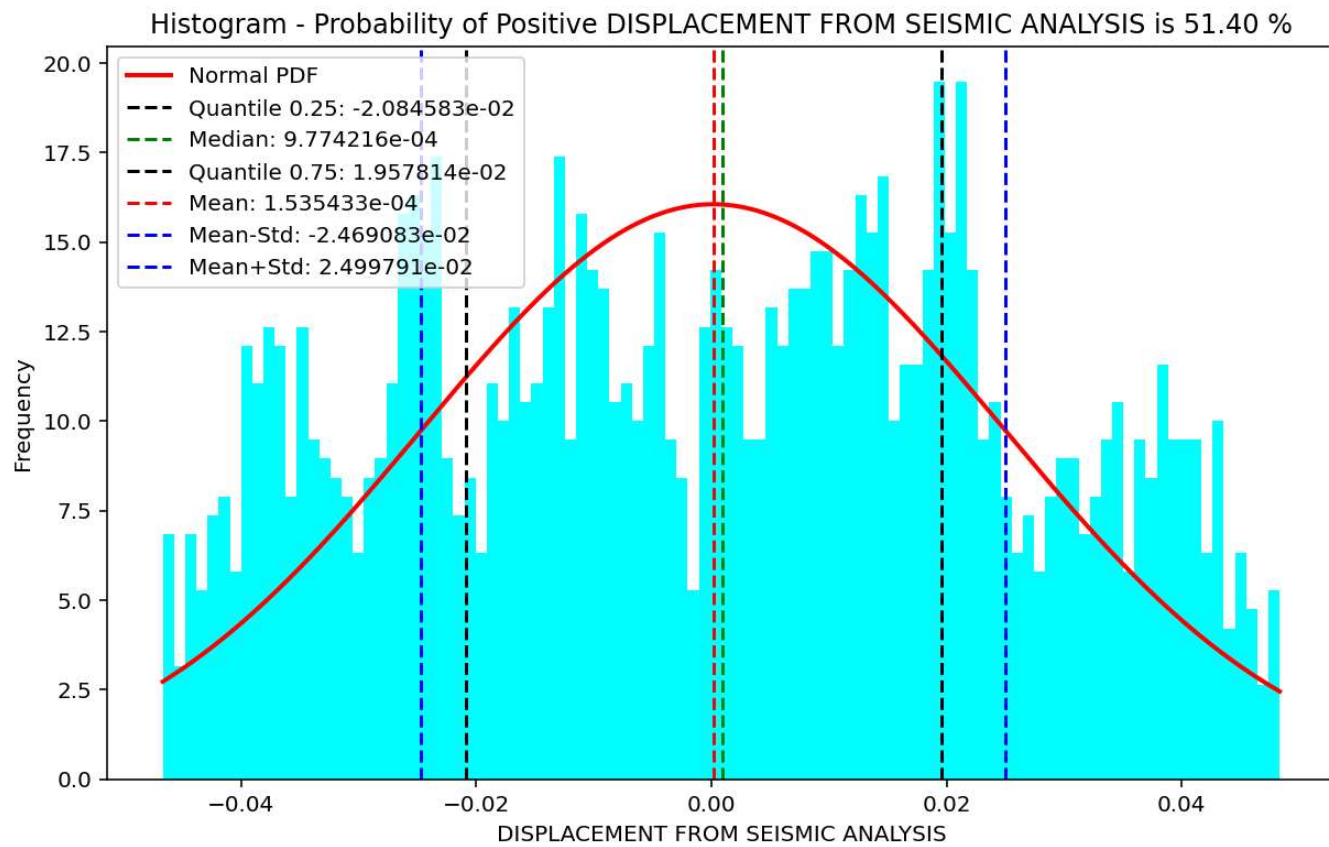
elements force vs Time During Seismic Analysis

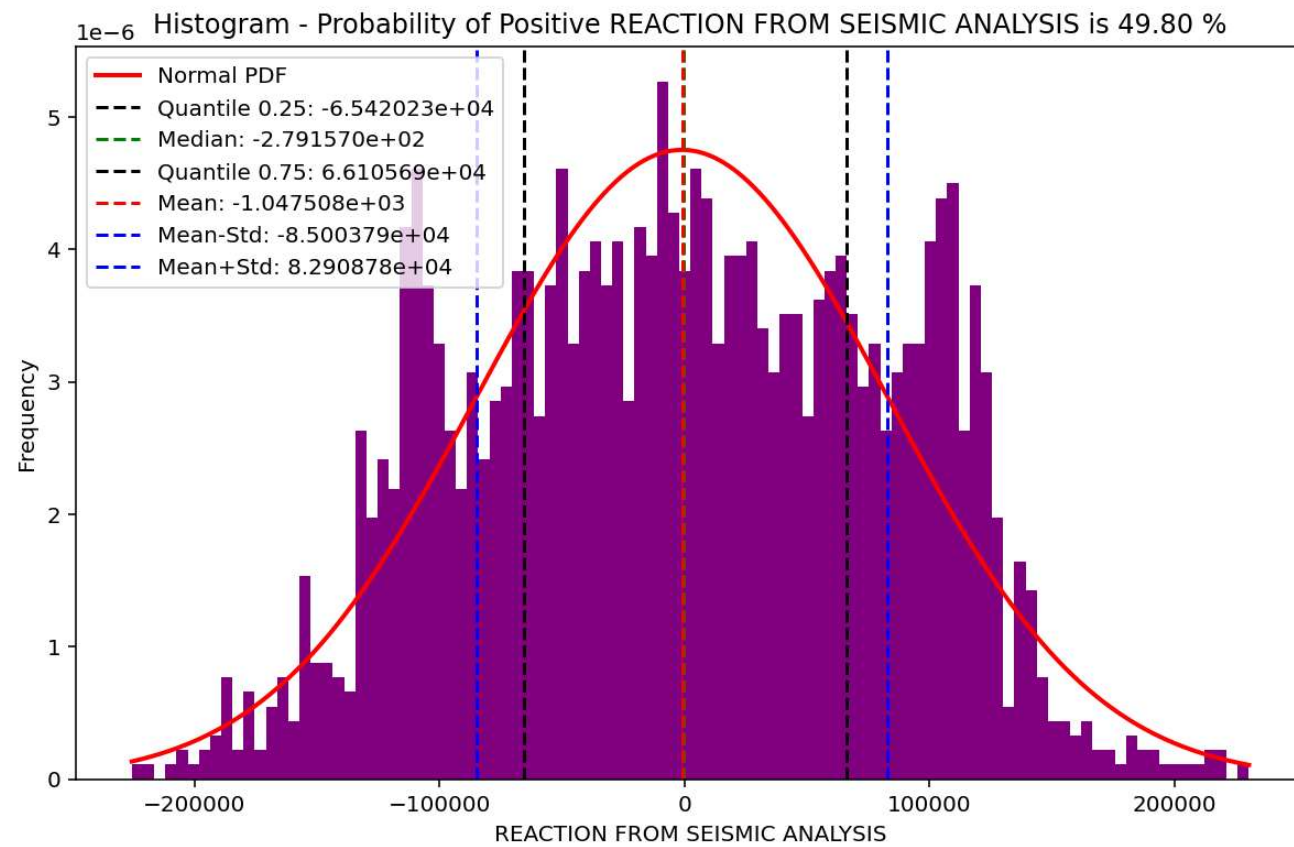


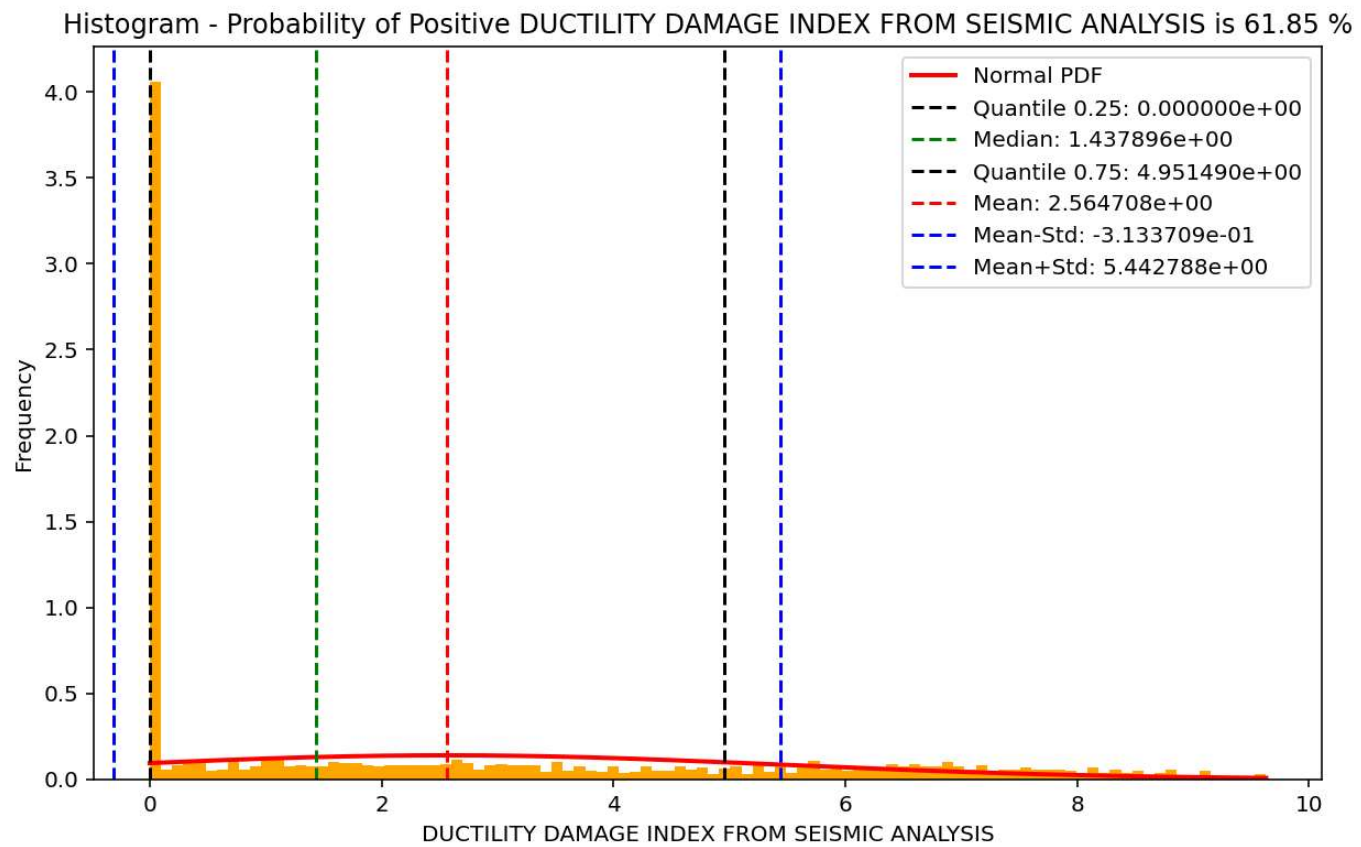
Node Displacements vs Time for During Seismic Analysis



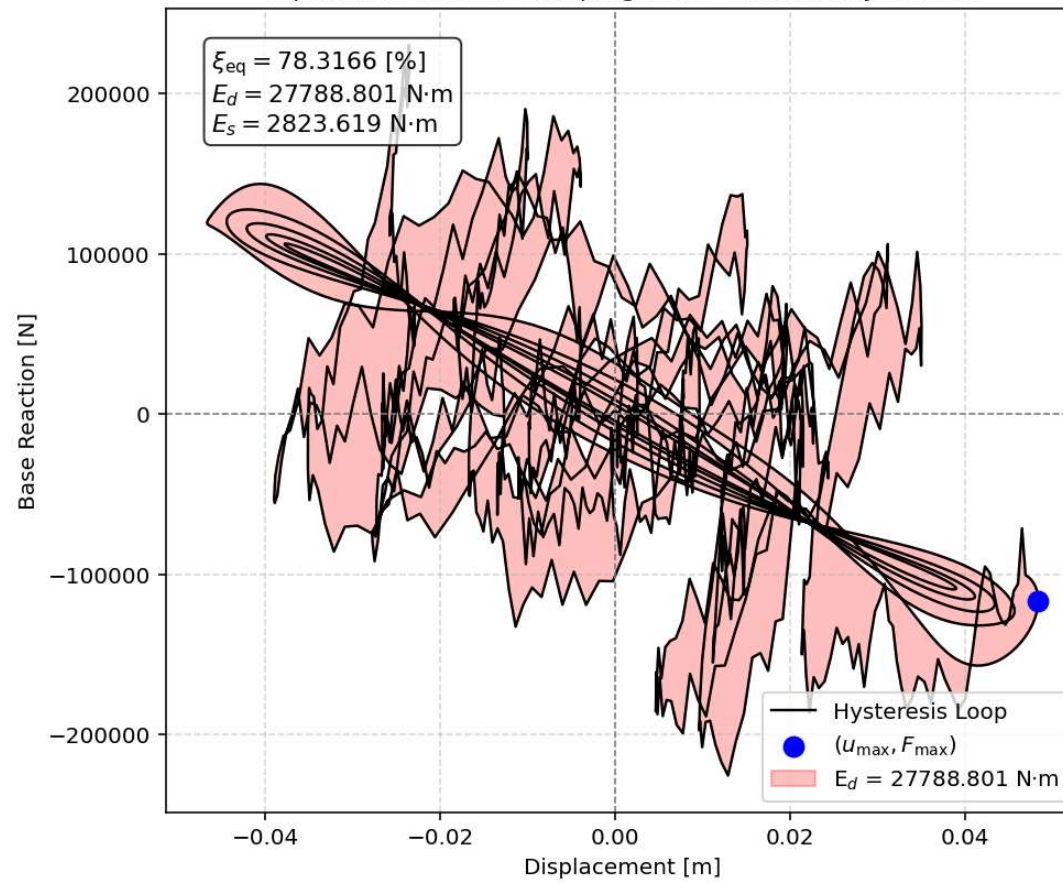


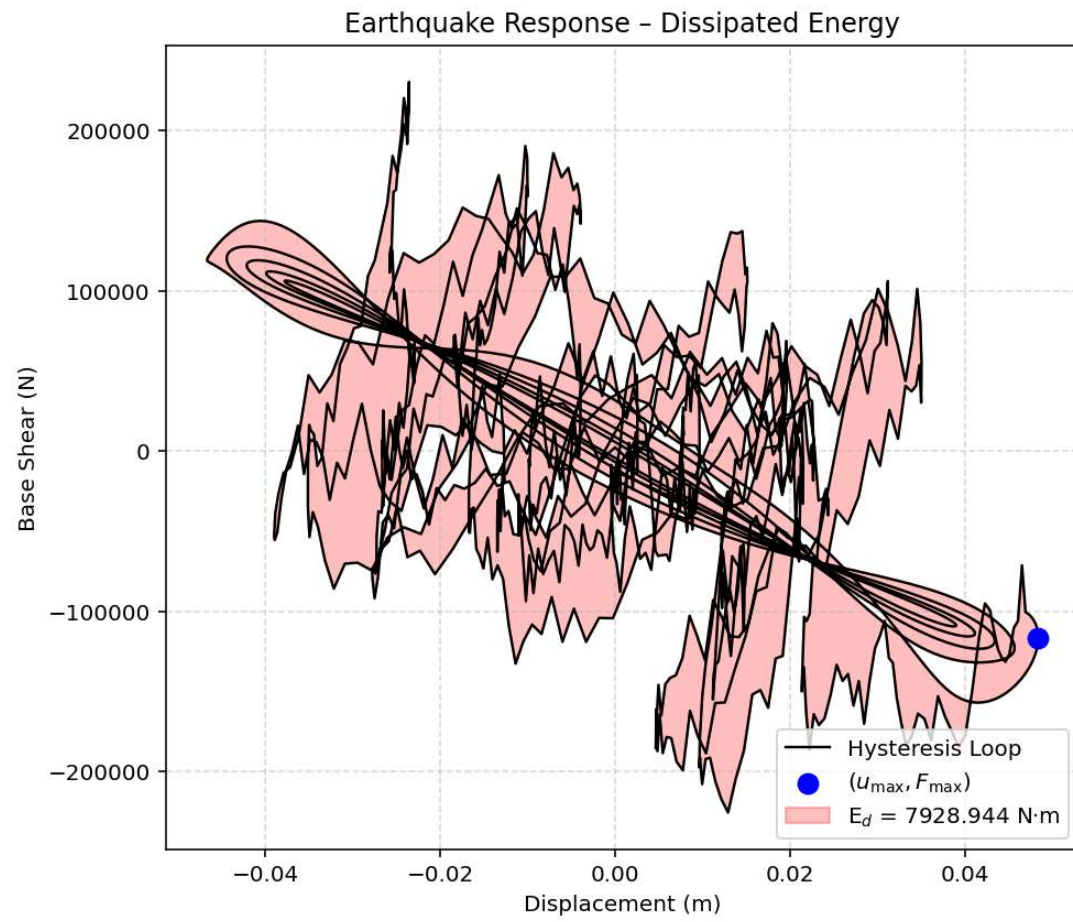


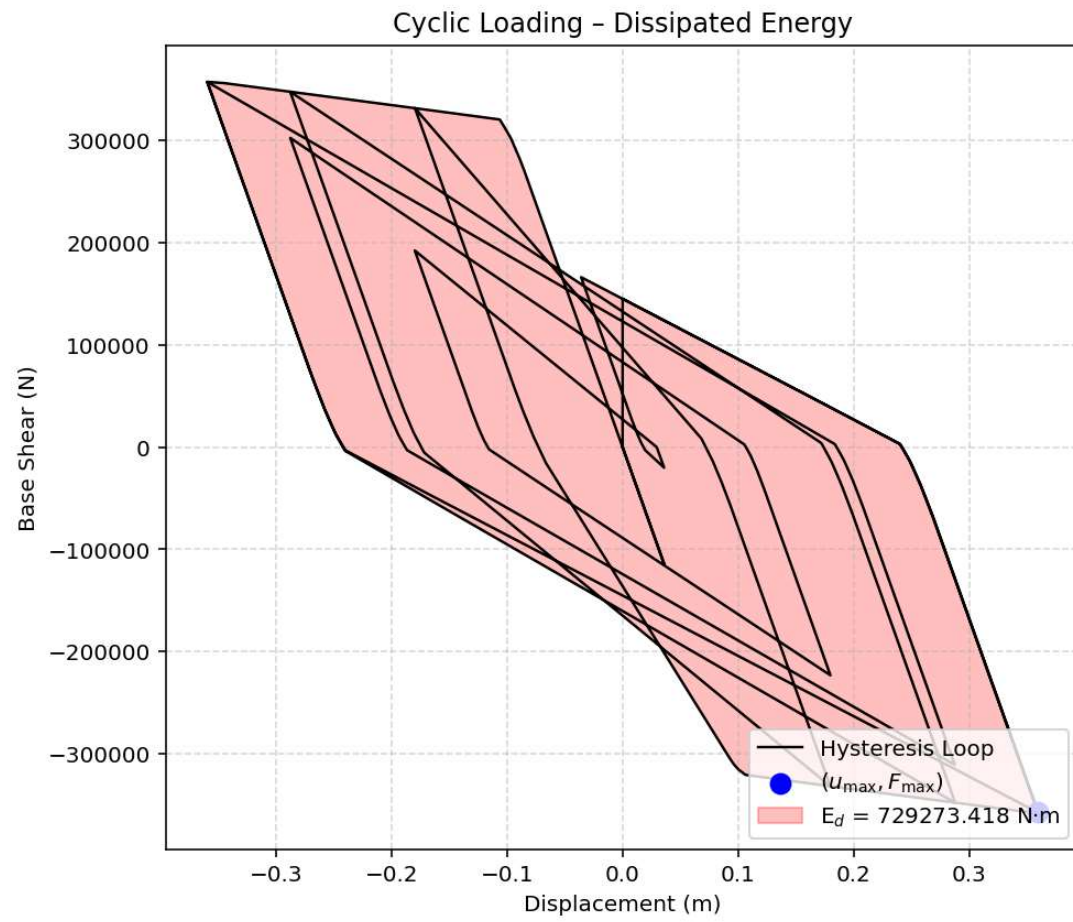




Equivalent viscous damping ratio - Seismic Hysteresis







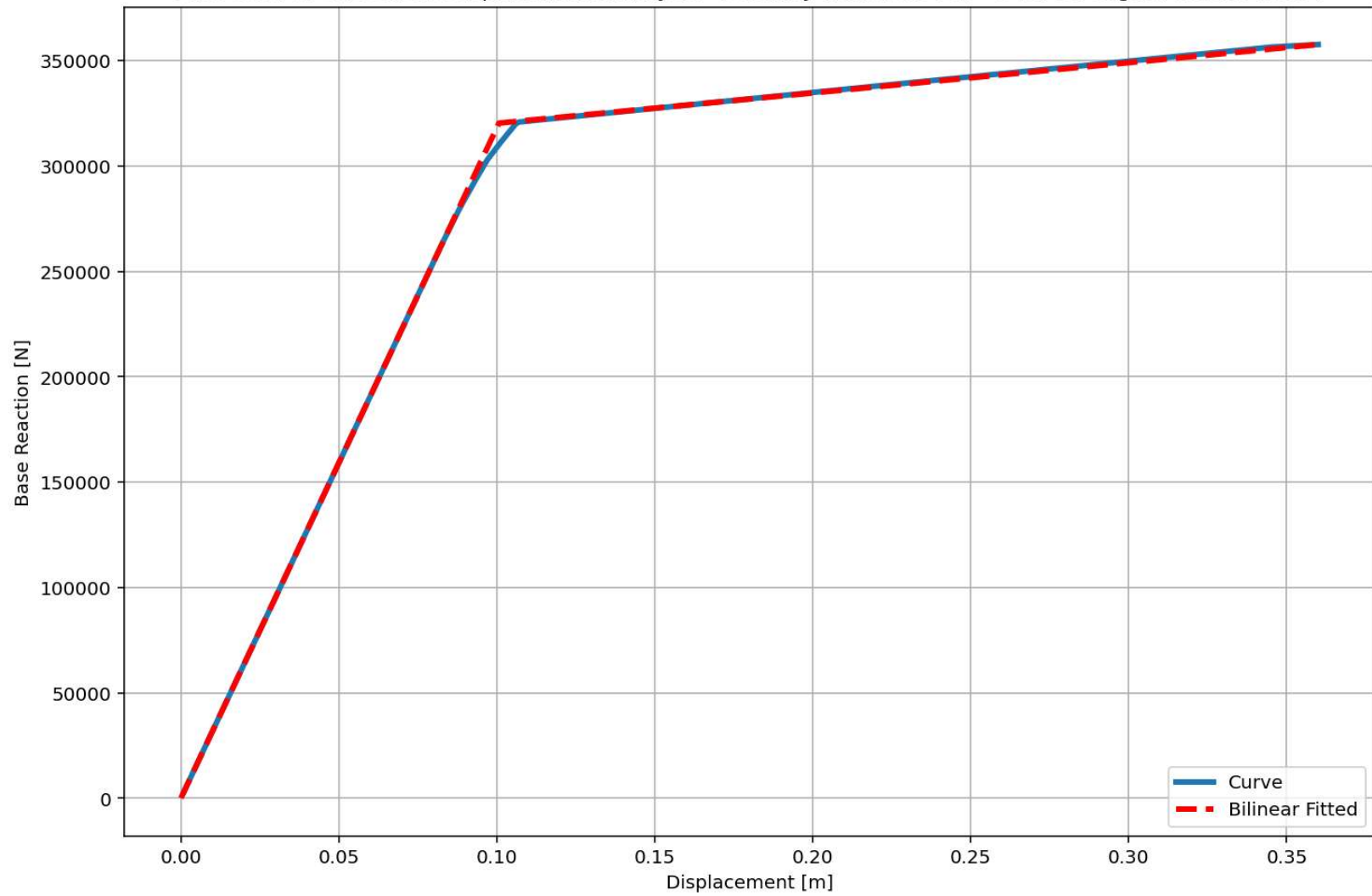
Dissipated Energy from Earthquake= 7928.94 N·m

Dissipated Energy from Cyclic Displacement= 729273.42 N·m

DISSIPATED ENERGY CAPACITY INDEX: 1.087 [%]

ZONE 1: VERY MINOR DAMAGE

Last Data of BaseShear-Displacement Analysis - Ductility Ratio: 3.5761 - Over Strength Factor: 1.1164



+=====+

= Analysis curve fitted =

Disp Base Shear

[[0.00000000e+00 0.00000000e+00]

[1.00669128e-01 3.20310863e+05]

[3.60000000e-01 3.57591396e+05]]

+=====+

+-----+

Structure Elastic Stiffness : 3181818.18

Structure Plastic Stiffness : 993309.43

Structure Tangent Stiffness : 143756.63

Structure Ductility Ratio : 3.58

Structure Over Strength Factor: 1.12

+-----+

Over Strength Coefficient (Ω_0): 1.1164

Displacement Ductility Ratio (μ): 3.5761

Ductility Coefficient (R_μ): 3.5761

Structural Behavior Coefficient (R): 3.9923

Structural Ductility Damage Index in X Direction: -20.1835 (%)

Seismic Fragility Curves by Damage State from DYN. TIME-HISTORY ANA. DUCTILITY DAMAGE INDEX [%]

